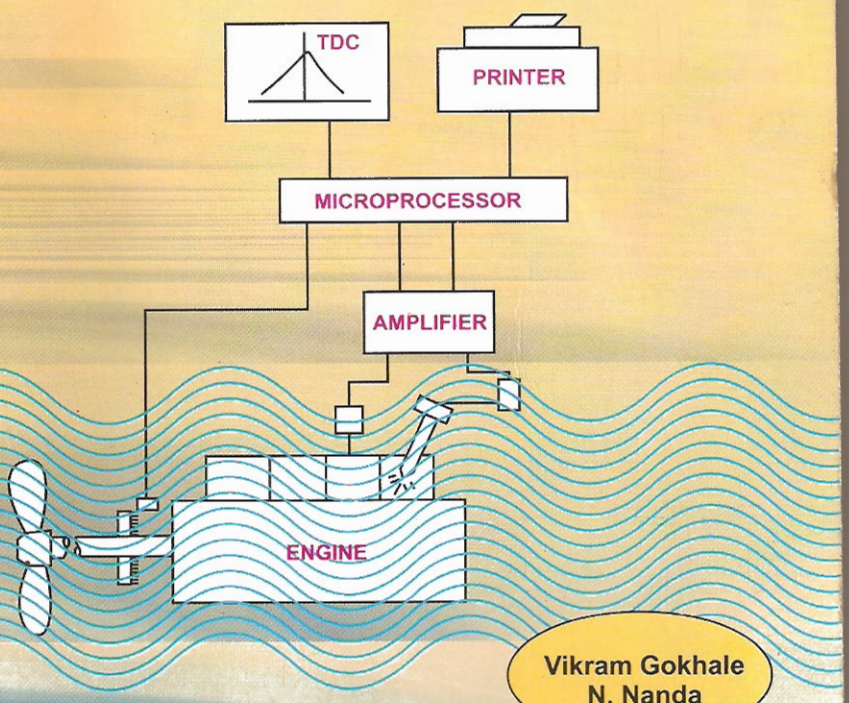
MARINE AUTOMATION AND CONTROL SYSTEMS



Vikram Gokhale
N. Nanda

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PREFACE

We have great pleasure in releasing this latest Edition – **Marine Automation and Control Systems**, which has been written to confirm to the STCW 95 standard.

We would like to thank Capt. Navin Passey, for his support of our endeavours, as also our colleague, Shri Vinod Kumar Chawla, for his painstaking efforts in helping us literally re-make this book, in its new avatar.

We would also like to thank Capt. Srirang S Manjeshwar and Capt. Anand M Shingatgeri of Wallem Maritime Training Centre, Mumbai, for their inputs for the new Chapter on Electronic Navigation Aids.

Last, but not the least, it is primarily written for all our students, appearing for their Certificate of Competency examinations, by providing answers to typical questions, along with simplified diagrams. This is also ideally suited as training material for the new value-added Automation courses.

Several new Chapters have been added, specifically on Motor controls, Programmable Logic Controllers, Control Diagram Fundamentals and Electronic Navigational Aids.

For today's high-tech mariners, 'hands-on' experience, on the latest types of control systems and microprocessor-controlled engines, is becoming an increasingly significant factor. for landing lucrative jobs and improving one's long-term prospects, besides enhancing skills. We sincerely hope, that this will address all their needs.

Vikram Gokhale

N. Nanda

Authors of the NG Series of Marine text-books

Contents

Chapter 1 1
AUTOMATION
Advantages of automation 2
Unattended Machinery Spaces 3
Remote control of machinery 6

APPLICATION OF COMPUTERS IN SHIPS
Basics 8
RAM 9
DRAM Controller 9
I/O 10
Disk storage 11
Programs 10
Power supply 11
Bus 11
History of the PC 12
Serial ports 13
Parallel ports 13
Display adaptor 14
PCMCIA cards 14
Keyboard, mouse controllers 14
ROM Bios 15
Real time clock 15
Integrated circuit 16
Bits, bytes and characters 16
Applications of computers in ships 17
Basics of Automation - Need for industrial standards 19

Chapter 2 21
SENSORS
Pressure measurement 22
Well type manometer 23
Bourdon type gauge 24
Diaphragm gauge 26
Capsule type gauge 27
Differential pressure cell 28
Aneroid barometer 29
Testing of gauges 29
Temperature measurement 31
Liquid-in-glass thermometer 31
Bimetallic thermometer 32
Filled system thermometers 32
Liquid filled thermometer 33
Vapour pressure thermometer 33
Gas filled thermometer 34
Resistance thermometer 34
Thermocouple 35
Thermister 37
Level measurement 38
Gauge glass 38
Float gauge 38
Displacement gauges 40
Hydrostatic type 40
Pneumatic bubbler system 41
Pneumercator gauge 43
Igema gauge 44
Electrical capacitance gauge 45

Proximity Sensors 46
Optical Sensors 47

Chapter 3 48
CONTROL FUNDAMENTALS 48
Terminology 49
Open loop 50
Closed loop 51
Discontinuous, on-off or two step control 52
Differential gap 53
Transfer lag 53
Feedback 53
Ratio control 53
Cascade control 54
Condition monitoring 55
Vibration and noise monitoring 55
Temperature monitoring 57
Ferrography 58
Spectrometric Analysers 58
Atomic absorption spectrometer 59
Particle Counters 59
Corrosion Monitoring 60
Performance Trend Monitoring 61
Piezo Electrical Accelerometer 62
Two step action 63
Proportional action 64
Proportional band 65
Integral action 67
Proportional plus Integral action 68
Derivative action 69
Three term control 71
Closed loop systems using P+I+D Control 72

Tuning of PID Controllers 72
'Adjust and Observe' Method 74
Ziegler-Nichols Method 75
Oscillation Method 75
Open Loop Method 75

Chapter 4 76
PLC CONTROL DIAGRAM FUNDAMENTALS 76
Control Terminology 77
Components in Control circuits 78
Switches 78
Indicator Lamps 81
Time delay relays or timesrs 82
Control transformers 83
Boolean logic 84

Chapter 5 87
PROGRAMMABLE LOGIC CONTROLLERS 87
Limitations of older control systems 87
PLC Input Systems 89
Output 90
Digital control 91
Drawing the ladder diagram 93
Final step 94
Adding rungs, drawing connections 95
Creating a simple ladder logic diagram 96
Latch Instructions 97
DC inputs 98
AC inputs 100

Relay Outputs 102
Transistor outputs 104
Communication 106

Chapter 6 107
MARINE EQUIPMENT 107
Electromagnetic log 108
Echo sounder 110
Mechanical Tachometer 111
Electrical Tachometer 112
Salinometer 113
Dionic water purity meter 114
pH meter 115
Oxygen analyser 116
Dynamic viscosity meter 117
Viscotherm 118

Chapter 7 119
PNEUMATIC CONTROL SYSTEMS
Pneumatics 119
Flapper and Nozzle 120
Proportional action 123
Split range control 124
Pneumatic system air treatment 125
Air dryers 126
Pressure reducing valve 127
Compressed air system 128
Intercooling 128

The need for Dryers 130
Lubricator 131

Chapter 8 132
CONTROL SYSTEM COMPONENTS
Poppet valves 133
Spool valves 134
Rotary valves 135
Actuators 136
Valve positioner 138
Valve characteristics 139
Equal percentage 140
Quick opening 140
Piston and rotary actuators 141
Materials 142
Boiler water level control 143

Chapter 9 144
STARTING AND REVERSING
Starting and Reversing 144
Direct adminission of Compressed air 145
Starting air period 146
Rules for starting air systems 147
Main Engina Interlocks 148
Typical contrl system of a modern marine two stroke engine 149
Electronic governor 152
Overspread protection device 155
Indicator diagrams 156

Chapter 10 158
MOTOR CONTROLS
Motor Starter 158
Overload protection for AC motors 160
Specifiying motor starter 162
DC Motor controller 164
Variable speed AC motor drive 171

Chapter 11 174
ELECTRONIC NAVIGATIONAL AIDS
Advanced radar plotting aids 175
Global positioning system 176
Electronic charts display and information system 177

CHAPTER 1

Automation

Upon completion of this chapter, you will learn :

- Need for automation
- Requirements of Unattended Machinery Spaces
- Development of the PC and PLC
- Bits, Bytes and Characters
- Applications of Computers
- Basics of Automation

From ancient times, man has learnt how to survive, by controlling his environment. Whereas animals and plants adapted to suit the environmental conditions, man has developed to a stage, where he can actually control his environment, such as maintaining of ambient temperature through the invention of air-conditioning.

In order to succeed, man has learnt to continually improvise and modify systems, so as to achieve better and better control. Thus, we can say that Automation has always existed, ever since humans moved out of caves and started taking charge of their lives.

After the introduction of machinery, during the Industrial Revolution, man learnt to manufacture products using machines. Soon there was a need to control these machines, without having to manually operate them. In the 1960s / early 1970s, electro-mechanical relays, timers, counters and other devices were introduced.

As time went on, the age of the microprocessor arrived, with a consequent change in the systems, leading to the Programmable Logic Controllers (PLC), which were initially introduced in the automobile industry, by companies like General Motors.

Early PLCs were nothing more than replacements for Relays, limited in their applications and programming. Then came the Internet and the break-through in satellite communication systems, which has

led to the present day SCADA systems, which can now be remotely controlled and monitored.

Advantages of automation

There are several factors to be considered when deciding to go in for Automation.

1. Very accurate monitoring and operation of equipment. Efficient monitoring ensures that equipment always runs at the correct parameters, reducing the wear and tear associated with running at constantly varying parameters.

2. The human operator is relieved of repetitive and tiring tasks, such as watch-keeping, so that he is able to concentrate on matters which require attention, such as preventive maintenance, trouble-shooting of faults or analysis of data to predict future failure and take proper preventive action.

3. Automatic control will give a very consistent result, while human operator's output will vary among individuals.The consistency is important in analysis of data, where any variation is considered as a departure from normal parameters, and often signals the starting of problems. These can be tackled easily at an earlier stage. By tackling the problem at an early stage, the damage due to deteriorating conditions is avoided.

4. There will be a significant saving in fuel / maintenance costs, since Automation enables the machine to always run at its optimum conditions. Fuel consumption can vary considerably, always increasing if parameters (such as viscosity) are not maintained within optimum limits.

5. Complex machinery operations can be coordinated easily using automated means.

6. Operating costs may be less, (less man-power), however higher cost of maintenance must also be considered.

Unattended Machinery Spaces

There are various factors to be considered for running a vessel with Unmanned Machinery Spaces.(U.M.S.)

1] There should be periodic inspections or rounds, before and after the unmanning of the machinery spaces.

2] All documents, concerning fitness for U.M.S. operation, should be carefully prepared and properly maintained, to ensure that all the possible precautions are taken, equivalent to running the engine room manned.

3] Fire Protection. There should be an approved Fire detection and Alarm system, which should be capable of giving an early warning of fire in the machinery spaces, especially for a) Boiler, b) Scavenge air belt of the Main Engine, c) Crankcase of the Main Engine - Provision of an Oil mist detector where Engines of 2250 kW (and above) power output are provided, or the engine bore exceeds 300 mm.

4] Protection against Flooding. Bilge wells to be monitored for excessive accumulations under normal angles of heel and trim. For auto bilge pumps, a 'long -run' alarm should be provided. This indicates excessive filling of the bilges, since the pump is not able to cope with the ingress of water.
The controls for the bilge pump and remote valves should be easily approachable, and above the possible level of flooding.

5] Bridge control of the Main Engine. Under all sailing conditions, the speed, rudder and (in case of controllable pitch propellers) the pitch must be controllable from the Bridge. The Bridge telegraph orders are to be duplicated in the Engine control room, as well as the Emergency or Local station near the Main engine. Indicators are to be provided, on the Bridge, for giving the Main engine's speed (R.P.M.) and the Ahead/ Astern direction, and (in case of controllable pitch propeller) pitch. Additionally, a means of remote Trip of the Main Engine, is to be provided. Also, there should be a gauge showing available Starting air pressure and a Low starting air pressure alarm. Air compressors, for Starting air, should be of the automatic type, which are capable of meeting the

requirements, relating to the time taken to fill the Main air Bottles. The system should be arranged to give three consecutive start attempts, failure of which should transfer controls to the Engine control room, where the fault can be rectified, and the Main engine started.

6] Remote control of the Main engine is to be possible from only one location at any time, and there should be an indication, e.g. a light, showing which location (Wheel-house, Engine control room or Emergency station) is in control, at any time. There should be a change-over system, to transfer control of the Main engine, from one location to another. It should be ensured that both the stations involved in transferring of controls (e.g. from Bridge to Engine control room), should show the same speed (fuel lever setting) and direction (Ahead or Astern), before the transfer of controls is possible. This ensures that there can be no conflicting instructions given, during any change -over of control, and prevent accidents. There must be an approved means of communication, e.g. telephone, at all three locations, such that it must be possible for all three stations to be in contact at all times. The local control station / Emergency station, at the Main engine, should be equipped with all the necessary Gauges and controls, to ensure safe running of the engine, in case of failure of the remote control system. If possible, it should be arranged that Speed and Direction (Ahead/ Astern) of the Main engine be kept constant, in case of remote control failure, till Local controls are taken over.

7] A comprehensive Alarm system is to be provided, for indication of machinery faults, which should be distinct from the Fire alarm system. This system should be capable of displaying (by means of a panel of lights) any abnormality of the machinery, both on the Bridge, as well as in the Accommodation spaces, including the Duty Engineer's cabin and the Chief Engineer's cabin, all public rooms. The power supply, for this alarm system, should have a back- up, from the Emergency switchboard, in case of a Black- out, and there should be an alarm, to indicate this condition.

The alarm system should be capable of indicating more than one alarm, at the same time.

For convenience, the alarms may be grouped, as e.g. Main engine, Aux. Machinery, Bilge levels , and so on. Non- acceptance of the alarm, by the Duty Engineer, should further relay the alarm to the Chief Engineer's cabin.

8] Adequately designed safety systems are to be provided, for the automatic shut- down of Main engine, Aux. Engines or Boiler, in case of any serious malfunction. Aux. Engines used for power generation, must be capable of automatic start and coming on Load, in case of failure of the running machine. This can be done by means of a Stand- by feature.

In case of electrical power Generators, means should be provided for shedding excess load, e.g. a Preferential Trip, which will shed the load that is considered non- essential for the immediate running of the Main Engine.(e.g. Galley power, deck machinery power, accommodation Blowers and A/C).

All pumps, essential for the running of the Main engine, must be provided in duplicate, and a auto stand- by facility provided, so that the stand- by pump must be capable of automatically starting, on detection of failure of the running pump, e.g. by means of a pressure switch. Each pump must be capable of performing the duty independently.

9] Steering gear for U.M.S. operation must be of the type, such that it can automatically isolate and regain Steering control, in case of single failure, by automatic operation of the Isolating valves, e.g. safe-matic steering system. There should be two separate Steering gear pumps, which can independently steer the vessel, and one of them must have their power supply through the Emergency switch-board, such that it will continue to work, inspite of a Black- out, by drawing its power from the emergency supply.

10] There must be an approved system for ensuring safety of the personnel entering the Machinery spaces, during U.M.S. operation, for answering Alarms or carrying out essential repairs. E.g. Dead man alarm system.

This operates by means of a timer, which is activated, when the Duty engineer enters the Machinery space. It must be manually reset, after a pre-determined time interval, by means of reset switches located in the engine room.

In case of incapacitation of the Duty engineer, failure to reset the timer will sound the alarm on the Bridge, and someone can be deputed to investigate and take requisite action.

Remote control of machinery

Microprocessor controllers are the principle devices used to control various processes. Each controller is able to receive inputs from the engine room, perform processing and alarm checking, perform control algorithms, and output to motor controllers, valves and other types of actuators.

Besides direct monitoring and control of engine room systems, the microprocessor can perform trend analysis to give an early warning of possible break-downs.

One such system is the SIPWA system used to detect piston ring wear. Piston ring wear is one of the main reasons for having a unit overhaul every 8000 hours. Since the actual wear can be monitored, the Time Between Overhaul (TBO) can be safely extended, with savings in costs- spare parts, man hours as well as reduced downtime, allowing more time for other maintenance. This also helps to plan the intervals between unit overhauls, so that work can be easily carried out, at a convenient place, e.g. when the ship is alongside for longer time.

The Control Room usually contains various panels for monitoring and running of the machinery - Main engine, various pumps, compressors, boilers, alternators and other equipment.

Automated control rooms usually have a CRT display, which gives various parameters, mimic diagrams, bar- charts, performance graphs and other data. Engine combustion can easily be monitored by means of electronic indicator and draw cards. The advantage of using an electronic indicator is -

1. There is a consistancy of results, since many readings can be stored and the average taken.

2. The readings can be superimposed on each other, so that there can be a comparison of performance, eg readings taken on successive Loaded voyages can be compared on the screen / analysed by the microprocessor.

3. Since the mechanical indicator depended on a spring, it was only accurate for slow speed engines. The electronic indicator uses sensors like peizo-electric transducers, which can work with any speed engine - from 50 rpm to 2000 rpm, so that the same instrument can be used for the Main engine as well as the auxiliary engines.

4. The electronic indicator can be used on any engine having a standard indicator cock and is also portable.

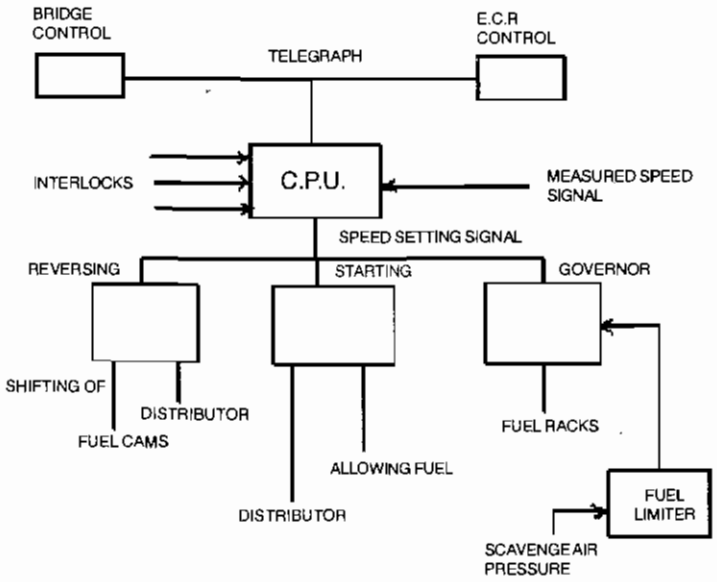
An on-line printer can print out the log 'on-demand' or at regular intervals. An Alarm printer will provide the time, type and numerical values of various alarm conditions.

The control systems for Main engine, Boilers and Alternators contain numerous interlocks, to safeguard against possible emergency conditions, including remote shut-downs / trips and indications.

There is an alarm panel provided, which is duplicated in various remote locations, viz. Navigation Bridge, Chief Engineer's and Duty Engineer's cabins, and public spaces in the Accomodation.

With the ship in UMS mode, the Navigating watchkeeper can control the speed and direction from the Bridge by means of the remote control system, and the Main engine control system is able to automatically bypass the Critical Speed range.

Fig. 1



Application of Computers in ships

Basics

There are several components in a computer, but we can divide them into five key divisions :

1. The **Processor** (CPU),
2. The **Memory** ,
3. The **Input / Output** circuitry (I/O),
4. **Disc storage** and
5. **Programs.**

Besides this, there are other components, such as the Power **supply, the** Motherboard and other peripheral cards.

The majority of the components that make up the computer are **housed on a single printed circuit board** called as the system board **or Motherboard. The usual components** on the motherboard include **the main processor chip and its support** circuits, memory, and I/O **interfaces (serial port, parallel port, keyboard,** and so on).

The Processor or the CPU, the Central Processing Unit, is the **brain of the computer..It executes programs.**

There are currently two main types of chips:

1. **Those made by Intel and others,** which make IBM compatible chips **using the CISC or Complicated** Instruction Set Computer architecture.

2. Those **made by Motorola and others,** which make MacIntosh / Apple compatible **chips using RISC or** Reduced Instruction Set Computer architecture.

There are still others which **emulate** the main types, and are referred to as clones.

The speed of the chip is in **MHz or GHz** and refers to the speed of operation of the chip, and not **necessarily that** of the whole system. Thus, upgrading the chip, to a higher speed one, need not always improve the system speed significantly, because of the other speed 'bottle-necks' , which need to be reduced.

The **Memory** is the computer's work-place, like the work bench of a carpenter. Its physical capacity decides the speed and limit of its operations. When IBM first came out with its PS/2 model, it came with only 8 Megabytes or MB of memory. Today we talk of Gigabytes and Terabytes of memory.

RAM or Random Access Memory is the volatile storage area where the computer carries out its work. It is volatile, because it depends on being constantly refreshed by a memory controller chip. This can only be done while the machine is On, hence the contents of the RAM are lost when the system is shut down, and any data is usually stored in the Hard Disk before shutting down.

RAM is of different types -
SIMM or Single Inline Memory Module
DIMM or Double Inline Memory Module
DRAM or Dynamic Random Access Memory
SRAM or Static Random Access Memory
VRAM or Video Random Access Memory

The RAM is the work area, and often contains the Operating system software, the data entered for work, the space allocated for calculations, and is directly affecting the speed of the system, as a whole.

The speed of the RAM could not keep up with the requirements of the microprocessor, so another bottle-neck was created, which hindered performance. To overcome this, a second smaller memory, called Cache memory, was placed in between the Chip and the DRAM.

This smaller cache memory being much faster, could keep up with the processor's requirements, and the controller which deals with this is called as the cache controller. Cache memory varies from 128 KB to 512 KB or more.

However, this is expensive, hence it is only found in the more expensive chips such as the Pentium IV. This is known as L1 or Level 1 cache. Another solution was to have this cache memory in the same package, but not on the chip itself. This resulted in a saving in cost, and is the main difference between the Pentium IV and the Celeron, both having the same speed. The Celeron was priced as a cheaper alternative to the Celeron A and the Pentium IV, as it did not have any internal cache.

DRAM Controller

The microprocessor needs to attach to the system's main memory to access data and instructions. The DRAM is installed in SIMM sockets on the motherboard. To meet the refresh requirements of the DRAMs, a DRAM Controller is required. The DRAM can be varied by filling up slots on the motherboard.

Input/ Output

I/O is how the computer takes in and sends out data. This includes the keyboard, the monitor, the printer, the scanner, the microphone, the mouse, the modem and floppy discs. The way in which data is given to the processor, and received from the processor is dependant on the I/O circuitary.

Disk storage

This is the storage area, and acts as the reference library, the filing cabinet and toolbox - all in one. There are several types of dics.

The Floppy Disk drive or **FDD** came in two sizes - the 3 1/2 inch in 1.44 MB capacity, and the (obsolete) 5 1/4 inch which came in 1.2 MB capacity.

Floppy disks have been replaced by Flash memory, using USB interface

The Hard Disk Drive or **HDD** is the back-bone of the computer's storage, and is the drive often used to start or 'Boot Up' the computer. It comes in various sizes of capacity (ranging from a few hundred MB to 160 or more GB) and having a much higher data transfer rate (of the order of 30 MB / sec).

The Hard Disk consists of a sealed unit containing several platters, rotating at 5000 - 7000 rpm, with several read / write heads, and having extremely small working clearances. Hence, it is very fragile, and prone to serious damage if subjected to impacts, especially when running at speed. A single Desktop computer may have one or more Hard disks, and even one hard disk may have one or more Partitions.

If a single hard disk is containing more than one Partition, each of these Partitions may contain a different Operating system, and then it is said to have more than one Logical drive, although it consists of only one physical drive.

The Compact Disk - Read Only Memory or **CD-ROM** is one of the most popular modes of disk storage today, when large amounts of data (of the order of 600 - 700 MB) need to be stored. The operating systems of most software makers are now being sold on **CD-ROMs.**

The Compact Disk - Recordable or **CD-R** is a 'one-time recording' CD, and cannot be re-recorded (like a floppy). Thus, once it is recorded on, by a device called a CD- Writer, it is much the same as any other CD-ROM. The advantage of the CD, as compared to other

media, is that it usually does not contain any 'computer virus', which can very easily attack other recordable media, provided it is from an authorised source. With the proliferation of 'Pirated CDs', i.e. CD-Rs which have been recorded by CD-Writers unauthorised to do so, even CDs may contain virus.

Another storage media gaining popularity is the Digital Versatile Disk or DVD. This is having a much greater storage capacity than the conventional CD-ROM (of the order of 10 - 19 GB or even more), and is gaining popularity in applications requiring extensive data storage, such as Video applications.

The **Program** is probably one of the most important parts of the computer. Programs are the instructions which turn a computer from a heap of electronic components into a powerful and interactive tool, that is being used by millions of people today, who probably do not have the slightest idea about how it works!

Programs are of different types, such as Assembly level programs which communicate with the components, and graphical user interface programs like Windows, which allow even non-technical people to carry out various jobs like accounting, designing, drawing and so on.

Power supply : This converts the normal supply of 220 V 50 Hz. A.C. to D.C. voltages of + 5, + 12, - 5 and - 12 V. The normal output of the power supply is around 200 to 220 W for a typical desktop machine. Since a lot of heat is being generated by the power supply, it has its own cooling fan, while another fan is used to cool the chip itself.

Bus

The speed of a microprocessor cannot actually determine the efficiency of the system, since performance depends on the configuration, the speed of the display adapter, the amount of RAM or Random Access Memory, the presence of Graphic accelerator cards and the bus architecture. The bus is how the various components of a computer are connected together, i.e. the access of the chip to the memory, how the peripherals are connected, such as the display adapter, the various drives, LAN adapters, sound card and so on. It can be thought of as the highway which interconnects the components of the computer.

The original PC-XT had an 8 bit wide bus and could transfer 1

million bytes of information per second. The Pentium bus is of 64 bits, and transfers data at speeds of 500 million bytes per second. The bus determines the most performance possible from the processor, it should be contemporary to have adequate support for various devices, and should be able to support upgradation of the chip to higher speeds.

History of the PC

The original PC was developed in a small laboratory in Atlanta, Georgia. In 1980 IBM began the PC project in Boca Raton. It started with a floppy disk based system having 32 kB of RAM. This used the Intel 8088 microprocessor using a clock speed of 4.77 MHz. This speed was selected because it was needed to match 3.56 MHz, which was the colour burst frequency used in TVs.

In 1984, the PC AT was announced, which used Intel's **286** microprocessor, which supported 16 bit bus transfers. The 286 was introduced at a speed of 6 MHz, which was upgraded to 8 MHz. This offered a five times improvement over the PC XT, and supported **DMA** or Direct Memory Access, which is special circuitry used to transfer data at high speed between the I/O or Input / Output devices and the memory of the PC.

With the introduction of the **386** 32 bit processors, memory was separated and attached directly to the local bus, which reduced the constraints on memory size (which was earlier on the chip).

IBM never fully specified the PC AT bus, with the result that various manufacturers worked on varying specifications, that resulted in non-standard products, which didn't match. To remedy this situation, Intel started specifying timing and loading information to force a standard among all the manufacturers, which resulted in the **ISA** or Industry Standard Architecture bus.

A consortium of suppliers lead by Compaq and Intel began the development of the Extended Industry Standard Architecture or **EISA** standard.

With the introduction by Microsoft, of the Windows Graphical User Interface or GUI, display adapter performance became the bottle-neck. This led to the Video Electronics Standards Association or **VESA** bus, which greatly improved the performance by moving the display adapter and its memory to the local bus. The VESA was not intended to displace the ISA bus, but to improve the performance by providing a high speed bus for certain peripherals.

In 1992, Intel developed the Peripheral Component Interconnect or **PCI** bus. The original intention was to provide a standard to support high speed chips interconnecting to the motherboard. The PCI bus would attach to the microprocessor's local bus, through a special chip. Thus, every time the chip was upgraded, only that special bridge chip needed to be changed, while the other devices continued to work automatically, with every new chip that Intel produced at higher speed.

The PCI bus was either 32 bits or 64 bits wide and ran at speeds of upto 66 MHz, with a throughput of 528 MB/sec, which was roughly 42 times faster than the ISA bus.

The **IDE** or Intelligent Drive Electronics is the port which supports the attachment of two hard disks with capacities of 528 MB each.

The **EIDE** or Enhanced Intelligent Drive Electronics removes the limitation of hard disk size, and can support drives of upto 80 GB & more Also it is possible to attach CD-ROM drives on this port.

Serial Ports

The multi- I/O adapter contains Serial communications ports called COM ports. These are typically used to connect the mouse and modem. There were usually two ports called COM 1 and COM 2 , one having a 9 pin D Type connector and the other with a 25 pin D type connector. There can be two more, COM 3 and COM 4, but due to constraints on system resources (IRQ), only two can work at a time. Thus, if there are four COM ports, COM 1 (and COM 3) may be allocated IRQ 3, and COM 2 (and COM 4) may be allocated IRQ 4. Thus only two can be used at any time.

Parallel Port

This is usually used for connecting the Printer, hence it is labeled as LPT 1 or Line Printer Terminal 1. This port may also be used for attaching certain types of Scanners, LAN adapters and so on. The original port was very slow, with transfer rates of 150 KB/ second, but the **EPP** or Enhanced Parallel Port and the **ECP** or Extended Capability Port developed by HP and Microsoft has improved the capability for multiple logical device addressing. USB are presently replacing both the serial & parallel ports.

Display Adapter

The original PCs came with the Colour Graphics Adapter or **CGA** and the Monochrome Display Adapter or **MDA**, which were very soon superceded by the **SVGA**.

Display Adapters are separate cards that plug into one of the bus expansion slots, however the modern systems now have this directly integrated into the motherboard. The display adapter uses memory to display different types of images, e.g. a 256 colour may require 480 KB, while a 16.8 million colour screen requires more than 1 MB, and larger colour schemes require upto 4 MB or more.

The display adapter contains the chip, the memory, the clock generator device and **RAMDAC** or Random Access Memory Digital to Analogue Converter, which changes the digital image data to analogue RGB signals, i.e. Red, Green and Blue which is required by the monitor to display.

SCSI or Small Computer System Interface specifications define a bus that can connect the computer with a variety of peripheral devices.

PCMCIA Cards and slots

As notebook size computers became popular, the devices that connected to them, such as the memory, fax modem, LAN cards and other drives needed standardisation in their method of attachment.

To standardise the method of connections and ports, an organisation called **PCMCIA** or Personal Computer Memory Card Association promoted a single standard for all portable computers and their accessories, which set out sizes for sockets, connectors, bus specification, software drivers (which are programs to operate certain devices) and Plug and Play specifications.

The PCMCIA cards can be installed by simply plugging in to a standard slot on the laptop. They have a compact size, and are now available as SCSI devices, LAN adapters, communications adapter for floppy drives and CD-ROMs, Modems and faxes, sound cards and other uses.

Keyboard and Mouse Controllers and ports

Each motherboard contains keyboard and mouse controller devices and ports. The keyboard attaches to a five pin DIN connector or sometimes through a PS/2 connector, which is usually used on

notebook computers to save space. New standard is the USB port.

ROM BIOS

The Read-Only Memory Basic Input- Output Service device is a chip which contains software that is always required to interact with the hardware. Its primary purpose is to load the basic operating system software from the system's boot or start-up device and perform certain tests during start-up. The operating system's boot device may be either the harddisk or may be through a floppy disk or nowadays even through CD-ROMs.

The contents of the ROM are moved to the faster DRAM memory for execution. Newer BIOS chips can even detect system configuration and resource allocation, to create the **Plug-and-Play** standard.

The Bios contains a set of programs that provide support of the operation of the PC.

There are three main parts :

The first part is when the computer switches on, and it runs tests and initialisation programs to run a self check on the computer. This is called the **POST** or Power On Self Test.

The second part provides the detailed control of all the I/O peripherals such as the drives, which need their own checks.

The third part applies only to the older computers made by IBM, and is the built-in ROM-BASIC, that contains the Basic programs used in early machines.

The Bios chip is usually on the motherboard, and may or may not be upgradable, depending on the make. Certain devices like SCSI controllers have their own ROM-BIOS chips.

Real Time Clock RTC and Calendar with Nonvolatile Memory

Every motherboard has a chip called a real-time clock/ calendar device, which has a small amount of nonvolatile memory, i.e. permanent memory. This device is always running, even when the PC is shut down. Power is supplied by a small battery located on the motherboard. This device supplies the correct time and date information to the operating system software, and stores the basic system configuration.

Integrated Circuit (IC)

A single semi-conductor component can be manufactured from a small piece of silicon which has been suitably treated or doped. This technique can be extended to enable a complete circuit to be produced in a single piece of silicon chip, which is referred to as an Integrated Circuit, or IC, and is of a very small size, weight and cost.

Circuit components are produced by oxidising, etching and diffusing processes. Resistors, capacitors, diodes and transistors can be produced.

One of the common types is the MOSFET or Metal Oxide Semiconductor Field Effect Transistor. The device has three electrodes called Gate, Source and Drain. MOSFETs usually have their gate and source leads shorted (during handling or storage) to prevent any inadvertent damage due to static electricity.

Integrated circuits utilise the resistivity of a silicon layer, which is usually provided as a square. A particular resistance value is obtained by the series connection of several individual resistor squares. Integrated capacitors utilise a layer of silicon dioxide between two conducting areas.

A fabricated integrated circuit will contain various individual components suitably connected by a pattern of aluminium contacts, which is deposited onto the chip.

A suitable package then encapsulates the IC. The connections are usually bent at 90^0 to the pack, to enable insertion into Printed Circuit Boards, or **PCB**s. Inductors cannot be provided in ICs and are thus external components.

Where very large circuits are incorporated in a single chip, the unit is termed as **LSI** (Large Scale Integration) or **VLSI** (Very Large Scale Integration), an example of VLSI being the **Microprocessor**.

Bits, Bytes and Characters

The smallest unit in computers is the binary digit or bit. This can have two values - 0 or no-voltage and 1 or having some voltage. The number of bits that make up a byte are 8, thus the byte is the practical unit of computing, and each of the eight bits that make up the byte are shown in the table below :

Binary pattern	Bit position	Numeric value	Power of 2
00000001	0	1	2^0
00000010	1	2	2^1
00000100	2	4	2^2
00001000	3	8	2^3
00010000	4	16	2^4
00100000	5	32	2^5
01000000	6	64	2^6
10000000	7	128	2^7

There are eight bits in a byte. They are numbered from the right and the numbering starts from 0 and goes to 7. Thus an 8 bit computer deals with data that is handled in terms of eight binary digits at a time. While a 16 bit computer deals with bits numbered from 0 through 15, and a 32 bit computer with 0 through 31.

1 KB	=	2^{10}	one thousand bytes
1 MB	=	2^{20}	one million bytes
1 GB	=	2^{30}	one billion bytes
1 TB	=	2^{40}	one trillion bytes

The Programmable Logic Controller (PLC) is the result of integrating the microprocessor into the field of automation & control. It developed in the automobile industry and is now finding widespread use on modern ships.

The various applications of Computer to ships are :

- Data logging. All parameters can be easily recorded, such as the various pressures, temperatures, levels, flow rates and so on, which confirm the good 'health' of the machinery.

- Combustion monitoring.
  This is for calculating the effect of changes in fuel timing, fuel viscosity, cutting-out a unit and other changes in parameters

- Spare parts inventory control. Programs provide a graphical interface to facilitate easy identification of spare parts, ROB, minimum stock to be always kept, and part numbers for easy identification.

- Purchase of spares, stores and consumables like bunkers, lubes which can be incorporated in a Budget.
  Thus, running costs, annual budget, dry dock costs, and so on can be easily available either as figures, or as a comparison in the form of charts over a period of years, so as to make an analysis of efficiency

- Planned maintenance systems
  These help to carry out maintenance at the required times, and include instructions from manuals, tools to be used, and even an estimate of the man-hours normally required for any job. The program can even give a list of tools required, and some programs can have video clips to explain difficult procedures, such as checking of fuel pump timings, or correct setting of parameters in the control system.

- Communication can very easily be carried out from ship to shore or vice versa, through e-mail, fax, sat-com.

- Control systems like electronic fuel injection, electronic governor and variable exhaust valve timing, are easily controlled by microprocessors, which eliminate the bulky manual controls of the past. These are now quick acting, very accurate, and increasingly getting more reliable.

- Operations like starting and stopping of generators, control of hydraulic systems for pumps and cargo handling systems for bulk cement, chemicals and oil
- Navigation systems like GPS, D-GPS. Satellites are used to provide signals which help to analyse and locate the geographical position of the vessel. This is especially useful in times of bad weather or poor visibility, when traditional visual means of navigation cannot be used.

- Accounting packages for ship's use like Salaries, Provisions and Stores. These programs help to automate long and tiresome calculations, and allow maximum efficiency with the minimum of effort.

- Training packages, which include courses required as per STCW. Databases can be made for all personnel, which allows easy

identification of Certificates, validity, dates for revalidation and so on, which helps keep track of all documents relating to staff.

- Simulators are being increasingly used as tools for developing analytical skills. This is a useful tool, especially in junior / recently promoted engineers, who can develop their skills in trouble-shooting a wide variety of problems, without actually having to experience the same.

- CAD programs for engineering drawings, for surveys and calculating stresses during cargo operations. Such programs can be made taking into account Classification society rules and regulations, to ensure compliance.

  Statutory Certificates. There is a large amount of paper work associated with various organisations like Port State Control, P&I Clubs, Classification societies. For keeping track of various Certificates, Renewal dates, Conditions of Class and so on, Databases prove to be very useful tools.
  Since computers are linked worldwide by the Internet, such information can be accessed very easily from any terminal connected to the Internet from any geographical location.

Basics of Automation

Need for industrial standards :

Standards are generally drawn by associations of manufacturers, distributors, customers and / or a country. It contains detailed information regarding the basic design requirements of a product.

Standards normally establish dimensions, tolerances and other physical characteristics which enables quick replacement of components. There are 5 main standards used in the world for pneumatics and hydraulics.

1. **A N S I** : American National Standards Institute
2. **S A E** : Society of Automotive Engineers.
3. **I S O** : International Standards Organizations
4. **J I C** : Joint Industrial Council
5. **N F P A** : National Fluid Power Association

Standards help interchangeability of components for the end user of pneumatic or hydraulic components.

In today's 21st century shipping scenario, sophisticated computer-controlled systems are completely replacing the older mechanical / pneumatic / hydraulic control systems, of the past century. The sea change taking place is also due to the accelerated phase-out of old ships, which fail to meet the current and future norms.

For today's mariners, 'hands-on' experience, on the latest types of control systems and microprocessor-controlled engines and boilers, is an increasingly significant factor for working on board. To reach the stage of successfully managing these complex systems, the modern seafarer has to be trained systematically. This revised edition is an attempt in that direction.

CHAPTER 2

Sensors

Upon completion of this chapter, you will learn :

- Need for efficient Sensors.
- Pressure measurement – Liquid column manometer, Well type manometer, Bourdon gauge, Diaphragm gauge, Capsule gauge, Differential Pressure cell, Testing procedures.
- Temperature measurement – Liquid in glass thermometer, Bi-metallic strip type, Filled system type, Vapour pressure, Gas filled, Resistance type, Thermocouple, Thermister.
- Level measurement – Gauge glass, Float gauge, Displacement gauge, Hydrostatic type, Pneumatic bubbler, Pneumercator, Igema type, Capacitance type.
- Proximity and Optical sensors.

Need for efficient Sensors

Man is born with five senses : Sight, Hearing, Feel, Smell and Taste. For all practical purposes, these are enough for most human activities. These are the 'Sensors', that we use for Watch-keeping, in the 'manned' Engine room.

However, when we go for Unmanned Machinery Spaces (UMS), we are replacing these natural senses with artificial 'Sensors', which a Controller can 'understand', i.e. it can control the system, on the basis of these values, which are being continuously monitored.

Various parameters need to be measured in the Engine room, such as Pressure, Temperature, Level, Flow, Viscosity and so on. The signal, that is coming back to the system from the Sensor, is the Measured Value (MV). The Controller compares this with a Desired Value (DV), so as to calculate the Error. The Aim, of the Controller, is to reduce this Error or Deviation to the minimum.

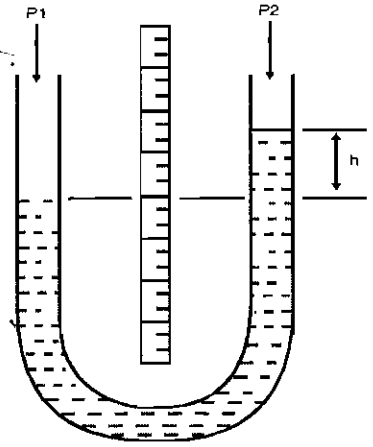
Pressure Measurement

Several types of instruments are used to measure pressure. Pressure measurement includes gauge pressure, vaccum pressure and differential pressure.

The simplest way to measure pressure is the liquid column. Here liquid height is used to measure the pressure, as the density and gravitational accelaration remain constant for that instrument.

Another type of gauge is the elastic type such as the Bourdon tube, diaphragm or bellows. Here, use is made of the elasticity of a specially shaped tube, where the deflection produced is proportional to the pressure which produced it.

Fig. 2



U-TUBE MANOMETER

Liquid Column (U-Tube Manometer)

Liquid, placed in the connecting legs of a U-shaped tube, will indicate the difference in the pressure acting, on top of the two legs. The height of the displaced liquid gives a direct indication of this pressure difference

$$P_2 - P_1 = d \times h$$

where P_1 and P_2 are the pressures on the two sides of the column, d is the density of the manometer liquid and h is the difference in height of the columns. **(Fig. 2)**

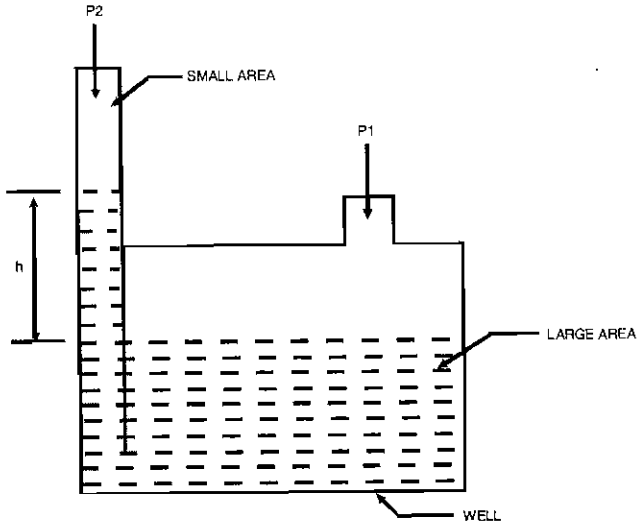
One limb may be open to atmospheric pressure, in which case the liquid level difference is proportional to the gauge pressure, i.e. how much the pressure of the closed end is more than atmospheric

pressure.

A water filled manometer is used for low pressures, such as checking pressure drop across scavenge air coolers, or the exhaust gas economiser. When filled with mercury, higher pressures can be measured. With the two limbs connected across, for example, the inlet and outlet trunkings of a Scavenge air cooler, the difference in height indicates the pressure drop, across the cooler and gives an indication of the degree of fouling of the tubes. The manometer is a simple, reliable and accurate device.

To allow easier readings to be taken, the scale may be enlarged by making one leg into a well, as shown in Fig. 3.

Fig. 3



Cistern or well type Manometer

This type has a large well area, and thus only a single reading is to be noted.

The pressure difference is indicated by the height of the column in the single leg, as a small change in the liquid level in the well gives a large change in the column level.

Provided the ratio of diameters is very large, errors (due to a change, in the liquid level in the well) can be ignored.

$$P_2 - P_1 = d\left(1 + \frac{D_1}{D_2}\right)h = dh + dh \times D_1 / D_2$$

D_2 = diameter of well
D_1 = diameter of leg

As long as the ratio D_1 / D_2 is small, as compared to unity, the term $dh \times D_1 / D_2$ is ignored (the error is negligible) and

$$P_2 - P_1 = d\,h$$

1 m^3 of fresh water has a mass of 1 Mg (Mega-gram) and weighs 9.81×10^3 Newtons. Thus 1 mm of fresh water produces a pressure of 9.81 N/m^2. The relative density of mercury is 13.6 and hence 1 mm of mercury produces a pressure of

$$9.81 \times 13.6\ N/m^2$$
$$= 134\ N/m^2$$
$$= 0.134\ kN/m^2$$

Bourdon Type Gauge

This type of gauge depends upon Hooke's Law for its operation. Within the elastic limit, stress is proportional to strain and deflection is thus proportional to the pressure applied.

The Bourdon tube is the most common type of pressure gauge due to its simplicity and versatility. The gauge consists of an elliptical cross sectioned tube, shaped in a circular are over approximately 270°. One end of the tube is fitted into a block for connection to the vessel whose internal pressure is to be measured, while the free end is sealed and connected to a suitable linkage and moves a pointer on a scale. As the pressure inside the tube is increased, the tube tends to change shape from oval to circular cross section. This causes the tube to straighten, thus moving the free end. The free end movement is only about 5 mm and is proportional to the pressure change. The linkage gives sufficient magnification of the pressure. Over the designed operating range the pointer movement is approximately linear in relation to the applied pressure. The tube may be made from

a variety of materials : Phosphor bronze (copper 95%, tin 5%, with traces of phosphorus) for pressures in the range 1.03 bar to 70 bar;

Beryllium copper (98% copper, 1.8% beryllium, 0.3% cobalt) for pressures in the range 0.3 bar to 350 bar; and

Stainless steel (16% chromium, 10% nickel, 2% molybdenum) for pressures in the range of 2.0 bar to 70 bar, where there is a requirement for corrosion resistance.

The thickness of the tube wall depends upon the range of pressures required. The linkages and quadrants are of brass, phosphor bronze, or bakelite. Vibration is the major cause of damage. Machinery, such as reciprocating pumps, can cause vibrations, creating excessive wear on gauge linkages and gears and work hardening the tube. Small shock absorbing spring loaded dampers may be fitted and filling the case with glycerine damps out vibration.

In bourdon gauges, the normal working pressure is about 60 % of the maximum range, so that inaccuracies (due to changes in the tube material from fatigue) are minimum. Should the gauge be overloaded, such that the pointer goes off the scale, a permanent set may occur in the tube. Thus the link gear could get deformed, and the quadrant and pinion could come out of mesh (subsequently getting re-engaged in a new position), so that the pointer is out of phase or shifted, in relationship to the tube position. A typical bourdon tube arrangement is shown in Fig.4.

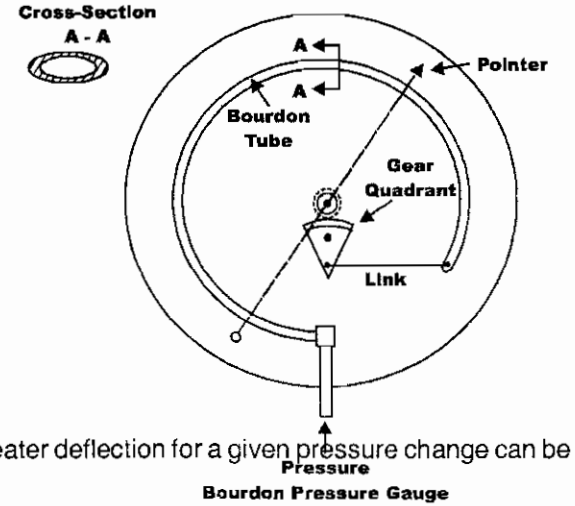


Fig. 4

A greater deflection for a given pressure change can be obtained

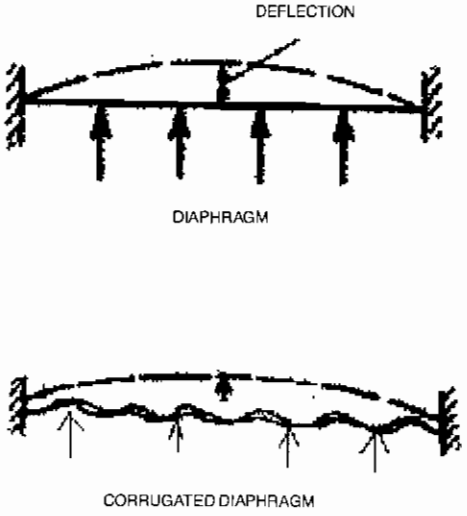
by winding the tube in the form of a helical spring, and eliminating the necessity for any form of amplifying mechanism, such as a segment and pinion, and the friction and lost motion which that entails. This can be used for pressures up to 400 Bar. Gauges are usually calibrated at 20°C and large variations in ambient temperatures can affect accuracy, sensitivity and life. Any gauge used on steam or hot vapours should be fitted with an efficient syphon tube or pipe loop, filled with condensate, to keep the tube from direct contact with the hot medium.

Diaphragm gauge

A modification of the bourdon tube gauge, is the diaphragm sealed gauge, and is used when a bourdon gauge is not suitable, owing to the corrosive nature of the fluid, whose pressure is to be measured.

A diaphragm is fixed between two flanges, to seal off the corrosive fluid from the gauge, and the space is filled with a suitable liquid to transmit the pressure from the diaphragm movement to the bourdon tube. For highly viscous fluids like heavy oil, which solidify in stagnant ends due to viscocity, the use of a diaphragm seal type gauge is recommended. Fig 5.

Fig. 5

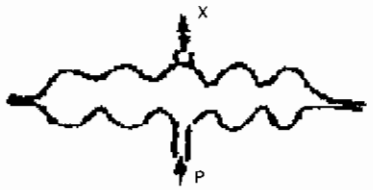


Where fluids to be measured are non-compatible with the gauge or diaphragm materials, the diaphragm and gauge body may be coated with rubber or P.V.C. This reduces accuracy and a hysteresis of 2 per cent of the scale reading may arise. Bellows type gauges are used in control equipment for measuring pressures, due to their compactness and robustness.

Capsule type gauge

Two or more metal diaphragms are joined together at their peripheries to form one or more capsules, as shown in Fig. 6. The diaphragms are pressed or punched out with a central hole and then 'welded together to form a stack. Brass, phosphor bronze, beryllium copper or stainless steel may be used. Pressure measurements are normally in the range 25 mm to 3500 mm water gauge. Diaphragms may be of phosphor bronze or non-metallic material such as Teflon or Neoprene. The movement of the diaphragm is opposed by a spring.

Fig. 6

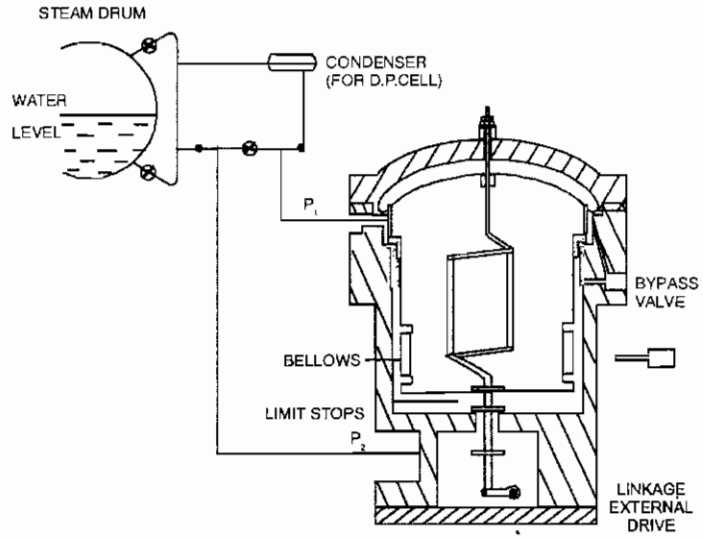


CAPSULE TYPE GAUGE

The Schaffer diaphragm-spring-type pressure gauge can be used for low pressure work. The deflection of the diaphragm is determined by the diameter, metal thickness, corrugation shape, modulus of elasticity, the number of corrugations and the pressure applied.

This gives a linear relationship to the pressure applied, over a wide range with a reasonable sensitivity. By decreasing the corrugation depth and number, sensitivity is increased and linearity reduced.

Fig. 7



Differential Pressure Cell

The measurement of the difference between two pressures, or differential pressure measurement, is done by a D.P. Cell. (Fig. 6)

This type of gauge is frequently used for water level sensing in Boilers. Tappings are made for steam and water sides, and fitted with shut-off cocks.

The steam side is connected to a small condenser, which contains a weir, so that a constant head of water is always acting on the steam side of the D.P. cell.

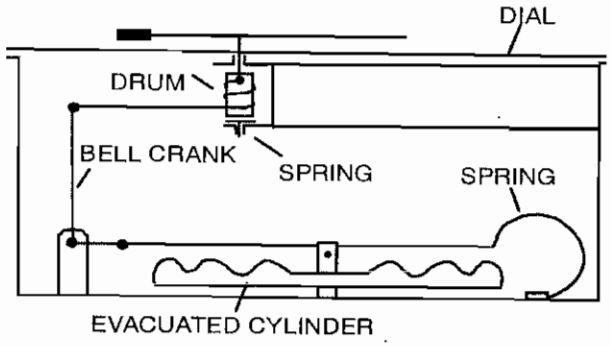
The water side head will vary, depending on the actual water level. Since the steam pressure is acting on both the sides, there is no resultant force due to this.

Thus, the D.P. cell will sense the change in water level, and produce a signal corresponding to the level, which could be used by a controller, or for a remote display unit.

The 'U' tube manometer can also be used for such work.

Testing should be carried out by shutting the downstream connection to the main pipe line and opening the drain.

Fig. 8



Aneroid Barometer

Fig. 8 shows an Aneroid barometer. This consists of a thin cylinder, with the upper surface corrugated, the space inside being evacuated, so that the pressure of the atmosphere tends to collapse it. As atmospheric pressure increases, the centre of the corrugated area moves down, taking the tensioning spring with it and thus moving the bell crank.

This then slackens the chain or wire, operating the pointer spindle, and the return spring on this takes up the slack and in doing so moves the pointer. A drop in atmospheric pressure allows the tensioning spring to lift the diaphragm centre, re-adjusting the bell crank so that it moves the pointer spindle against the return spring.

The diaphragm may be made of phosphor-bronze or cupro-nickel, the remainder of the components being of brass or steel.

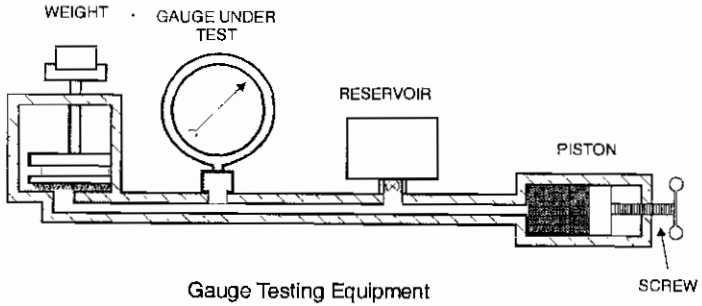
Testing

Ships are now having equipment for testing pressure gauges, in the form of a deadweight tester as shown in Fig. 9.

The system is balanced at atmospheric pressure, by opening the reservoir valve and moving the piston, so that the cylinder is full of oil. The floating piston then takes up its lowest position. The reservoir valve is closed, and the handwheel operated to raise the pressure. The increase in pressure on the underside of the floating piston and the load thus produced is balanced by the load due to piston weight, weight carrier and calibration weight. When equilibrium is reached,

the gauge under test will record the pressure in the chamber. To reduce friction on the piston, it should be spun slowly.

Fig. 9



Gauge Testing Equipment

Another method is to use a very high quality precision gauge in place of the piston and weights, as a comparison.

By varying the pressure over the required range, the accuracy of the gauge under test can be compared, with the precision gauge.

To check a gauge, a definite procedure should be followed, to check the gauge and then to re-adjust it. If the gauge is corroded, or badly damaged, it should be scrapped.

Procedure

To check a gauge, place it in the test rig and test for any leakage.

Raise the pressure to about 25 % above the maximum and hold for 10 seconds. Release the pressure slowly, to minimum, when the pointer should fall to the minimum value.

If it fails to do so, even after tapping lightly, then the tube is in a bad condition and the gauge should be replaced.

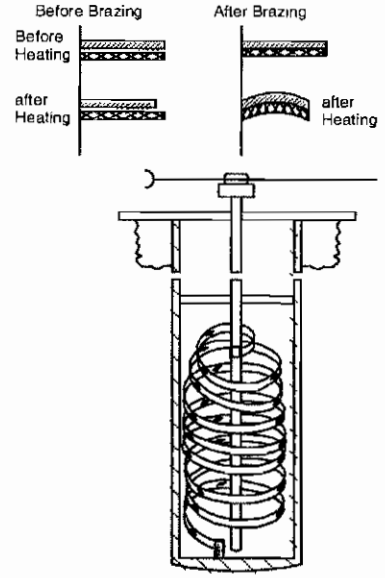
If the pointer does fall to the minimum, raise the pressure again to the gauge maximum and release slowly, watching for jerky movement or a tendency to stick. This indicates wear, leakage or a damaged hair spring.

Following a satisfactory check so far, raise the pressure in a series of steps and then drop it in similar steps, noting any errors that occur. The error should be less than 1.0 % and the hysteresis should not exeed 0.25 %.

Hysteresis is a term used to describe the behavoir of an instrument under varying conditions. Increase the pressure in a series of steps to a maximum, and then repeat along the same points to drop the

pressure from the maximum to minimum. If these points are plotted on a graph, the resultant curve gives the error at each point and the hysteresis of the instrument.

Fig. 10



Bi-metal Strip Thermometer

The accuracy of an instrument may be defined as the closeness the instrument reading approaches to the true value. The sensitivity may be defined as the ratio of output to input, or, the least signal input, that can create an output signal, having the required characteristics.

Temperature Measuring Devices

Liquid-in-glass Thermometer

The volumetric expansion of mercury is about 8 times that of glass. When subjected to heat, mercury will occupy an increased volume. This change of volume forces the mercury up the capilliary tube, attached to the bulb, and the height of this column indicates the temperature of the body touching the bulb. The mercury thermometer can be used from -38° C (freezing point of mercury) to

350° C(boiling point 360°C), and the space above the mercury is evacuated.

To extend the range, up to 600°C, a quartz mercury thermometer is used, the space above the mercury being filled with pressurised nitrogen or carbon dioxide to raise the boiling point of the mercury to above 600°C.

For low temperature measurement, in the range -196°C to aepproximately 100°C, pentane is used. Alcohol is used in the range -80°C to 100°C.

Bi-Metallic Strip Thermometer

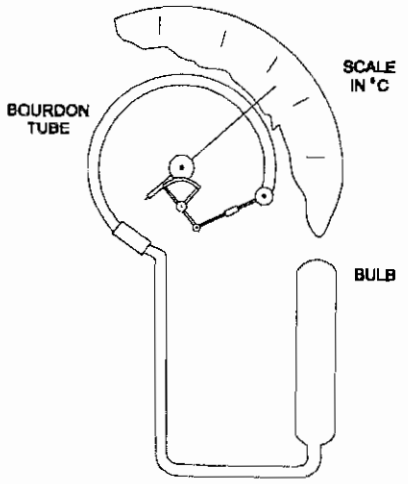
Different metals have different coefficients of thermal expansion. By brazing together two metals, one with a high coefficient (such as copper), and the other a low thermal coefficient (such as Invar which contains 36 % nickel, 64 % iron), a bi-metal strip is formed. fig.10.

When heated, the unequal expansion of the two materials rigidly fixed together causes the strip to deflect. The degree of deflection is a function of the temperature change and by forming the strip into a helical coil and connecting this to a mechanical linkage at one end and fixing the other, the coil can be made to wind and unwind as the temperature varies and move a pointer across a temperature scale. Range is approximately -40° C to 320° C

Filled System Thermometer

This consists of a bulb at the sensing end, a capilliary tube, and a bourdon tube or bellows. The capilliary connects the bulb to the bourdon tube, bellows or similar device.

Fig. 11



Filled System Thermometer

Liquid Filled Thermometer

A typical liquid filled thermometer consists of a bourdon tube, which is connected to a metal bulb, by a small bore capilliary tube.

The internal volume of the bulb and tube is filled with a liquid, having a high coefficient of cubical expansion. The system is completely sealed.

To avoid any problems with vapour pressure effects or a difference in head, between the bulb position and the gauge, the system is pressurized to about 70 bar (in the mercury filled type). The instrument has a linear relationship with temperature change, the range of operation being -39°C to 520°C (for mercury), and the power sufficient to operate a pointer as pen for a recorder or pneumatic transducer.

Vapour Pressure Thermometer

This is similar in construction to the mercury-in- steel type, and has a completely evacuated system, partly filled with a volatile liquid, such as alcohol, and partly with its vapour.

The vapour pressure of the liquid should be suitable for the required

operating temperature range. It has a range of -10°C to 300°C and the scale is non-linear. The system has a considerable time delay in showing a temperature change.

Gas Filled Thermometer

This contains the same basic elements as the two previous types, the system being evacuated and filled with nitrogen or helium under high pressure. If the mass and volume of the gas are considered constant, then the absolute pressure in the system is proportional to the temperature. A change in temperature, at the bulb, is proportional to a change in pressure, at the bourdon tube. The deflection of the tube is proportional to pressure (and hence to the temperature), giving a linear relationship. Ambient temperature changes can produce errors. Range is -193° C to 600°C.

Resistance Thermometer

When a metallic conductor is subjected to a temperature change, the electrical resistance varies according to the law :

$$R_T = R_0 (1 + \alpha T)$$

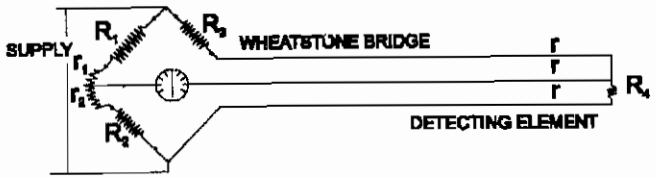
where R_T = Resistance at T° C.

R_0 = resistance at 0°C

T = temperature change in °C

and α = temperature coefficient for the particular metal.

Fig.12.



Resistance Thermometer

The wire used for the resistance may be of copper, platinum or nickel. Platinum is the most suitable, inspite of its high cost, due to its high resistivity. The windings are double wound to eliminate the inductive effect.

Measurement of temperature is carried out by a Wheatstone's Bridge circuit, with either a galvanometer or an electric potentiometer as a detector. Four equal resistances are coupled together in the form of a Wheatstone's Bridge and a current is applied to the bridge by a battery. The galvanometer will not deflect, so long as the resistance of the two parallel paths is equal.

When the temperature of the air or gas, surrounding the resistance wire, changes, then the resistance of the wire will change, according to the previous formula. This will upset the balance of the bridge, causing a current to flow through the galvanometer. The deflection is a function of the temperature change of the external resistance.

In the two wire system, any change in the ambient temperature of the leads, connecting the distance resistance to the bridge, will also affect the resistance on that side of the bridge and produce an error. In most systems therefore a three wire type is used with both sides of the bridge connected to the distant resistance, so that an change in ambient temperature affects both sides equally.

For high accuracy and to allow for automatic monitoring, the balanced bridge form of measurement is used, as shown in fig.5. Here a centre zero galvanometer is used so that by adjusting the slide wire contact, the bridge can be balanced. The slide wire incorporates a scale reading directly in temperature units.

The resistance element would have a resistance of about 100 ohms at 0°C and the range is -200°C to + 600°C. By using tungsten or molybdenum, temperatures up to 1200°C can be measured. Vibration and strain, on the wire, can cause problems. Keep the current flow as low as possible, so that internal heating does not affect the accuracy of the resistance bulb.

Thermocouple

If a circuit is formed, consisting of two dissimilar metallic conductors A and B, and their junctions are kept at different temperatures, a current will flow in the circuit, owing to two different e.m.f.'s being generated at the junctions.

Fig. 13

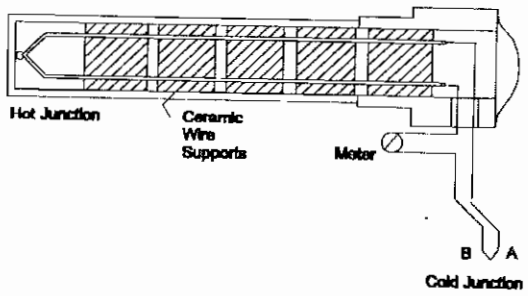


Fig.13 shows a simple thermocouple arrangement consisting of two dissimilar metal wire conductors, with the two ends joined. Good electrical and thermal contact is essential for efficient operation.

If the temperature of one end is varied, while the other is kept at a constant temperature, an e.m.f., proportional to the temperature difference between the two ends, is set up. As the e.m.f. generated is a function of temperature difference, between the hot and cold junctions, allowance has to be made for any temperature variation at the cold junction. The instrument displaying the temperature is a very senisitive millivolt meter, requiring only a small current for full scale deflection.

As the thermoçouple output is only in the region of 100 millivolts, distant reading would need a degree of amplification. Any ambient temperature variation at the head would affect the cold junction, giving rise to error. This problem is overcome by extending thermocouple wires, by cables having matching properties, so that the cold junction can be placed in a position, where the ambient temperature is reasonably stable.

Frequently the cold junction is placed in the instrument case, which may be in an air conditioned control room providing fairly stable conditions. Automatic adjustment of the electrical circuit, to compensate for a temperature variation at the cold junction, can be provided.

Thermocouples are used from -250°C to + 650°C and are commonly found in exhaust gas temperature measurement.

They are cheaper than resistance thermometers, do not require a stable reference voltage, can be made very small, and withstand engine vibration better than most resistance thermometers. However, the latter do not require cold junction compensation and have a high degree of linearity.

Besides the materials quoted in the diagram, other combinations include

Iron (+) and Constantan(-),

Chromel (+) [90 % nickel, 10 % chromium] and Alumel(-) [an alloy with 94 % nickel, 2 % aluminium, plus silicon and maganese.]

These are known as Base Metal Thermocouples and have upper temperature limits of 850°C and 1100°C respectively.

For higher temperatures, up to 1400°C, Rare metal Thermocouples are used, such as an alloy of 90 % platinum, 10 % rhodium (+), and platinum (-).

The leads connecting the thermocouple head to the instrument must have thermo-electric properties to match those of the thermocouple itself and, in the case of the Base metal instruments, they would be of the same material. For the Rare metal type, where long runs and large diameter cables are used to reduce resistance, electrical cost prohibits these metals being used as compensation cables and copper (+) and copper- nickel (-) would be used.

Thermister

A thermistor is an electrical device made of solid semi-conductor materials. The thermistor is used in a similar manner to a resistance thermometer, but its operation differs. With the thermistor, as the temperatures rises, the resistance falls, i.e. it has a negative temperature coefficient of resistance, and the effect is much greater than the increase in resistance, that occurs as temperature rises with the resistance thermometer.

Thus, for a given temperature change a much larger working signal is produced by the thermistor. Being resistance devices they require a reference voltage for evaluation and would be connected into a Wheatstone's bridge circuit.

Thermistors are strong and rugged, and are used for stern tube temperature measurement, and for temperature monitoring in diesel engine bearings.

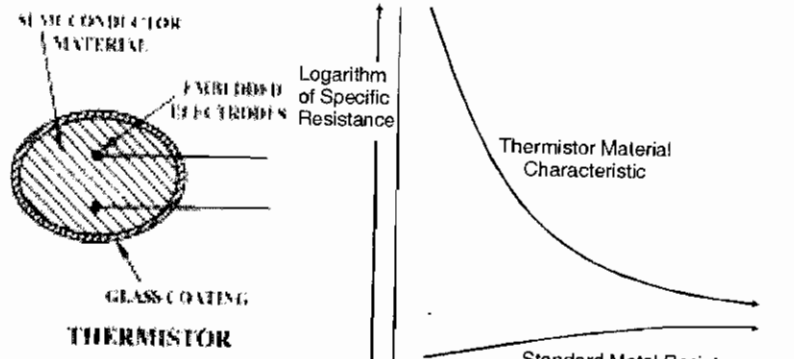


Fig. 14

Fig. 15

Comparison of Thermistor & Resistance Thermometer

All devices measuring temperature, which are in direct contact with the body, fluid or gas are referred to as thermometers, while devices which are not in direct contact are called pyrometers. Engine makers refer to exhaust gas thermocouples as pyrometers, although they being in direct contact with the exhaust gas are strictly speaking thermometers. Sometimes, instruments measuring temperatures below 500°C are called thermometers, in which case, exhaust gas being above 500°C, would probably be called as a Pyrometer.

Level Measuring Devices

There are two basic means of measuring the level of fluids:
Direct and Inferential

Gauge Glass

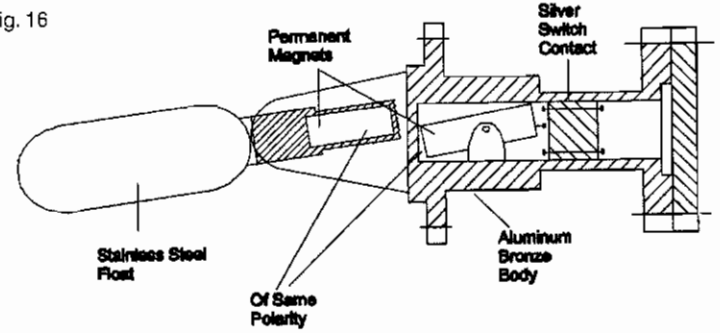
A direct method for indicating tank contents. The ordinary type is susceptible to damage, hence plate type of guage glass is used for high pressure boilers. Gauge glasses are not used for tanks containing inflammable fluids.Heat resistant glasses are available, which must be well protected from damage and have shut off cocks, at top and bottom ends.

Float Gauge

Simple types have a hollow ball or poystyrene disc, which floats

on the surface of the fluid. It has a wire passing out through the top of the tank, over a pulley, with a balance weight and indicator on the free end. It can also operate high and low level alarm contacts.

Fig. 16



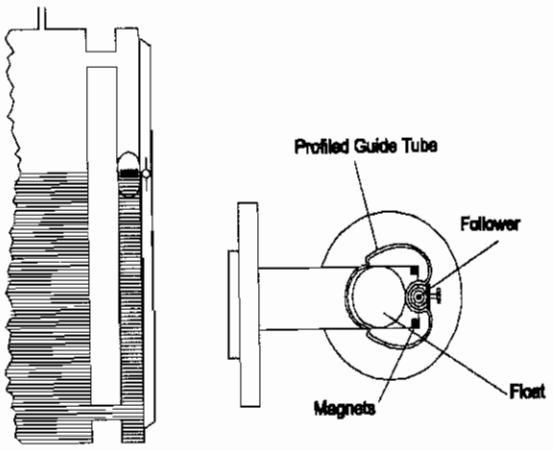
Magnetic Float Gauge

A common application is the magnetic float switch, used for high and low alarm levels in storage tanks and bilge wells. A typical device is shown in Fig.16.

Another float type local reading gauge is shown in Fig. 17.

Here the float, incorporating a magnet in contact with the tank liquid, moves up and down a profiled guide tube. A follower magnet moves in a sealed, evacuated glass tube to indicate contents.

Fig. 17

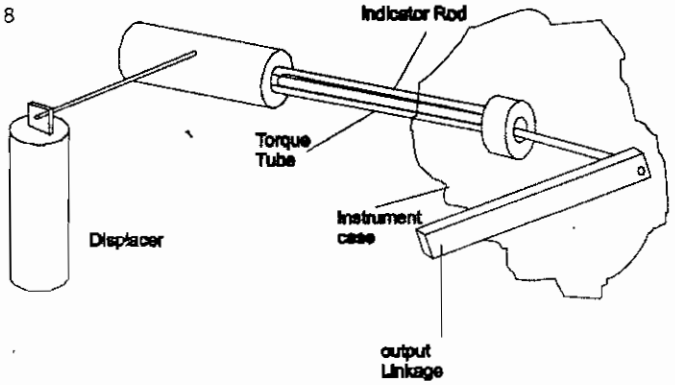


Magnetic Float Gauge

Displacement Gauges

These operate on the principle that a body placed in a liquid is buoyed up by a force equal to the weight of the displaced liquid,The displacer must be part submerged in the fluid and as the fluid rises and falls and moves up and down the displacer, so the apparent change in weight that occurs is used to operate an indicator for the level. One method is to use a torque tube consisting of a tube, sealed at one end and open at the other, and with a flange welded to this end and fixed to the indicating instrument case. A rod is passed down the tube and welded to the sealed end. The free end of the rod extends into the instrument case and operates a pointer or a pneumatic or electrical transducer. The displacer arm is welded to the sealed end of the tube making an angle of 90° with this. As the liquid level varies so the apparent change of weight is balanced by the spring constant of the torque tube as it twists and thus moves the indicator rod. Any change in relative density of the liquid involved has to be compensated for. (Fig.18)

Fig. 18

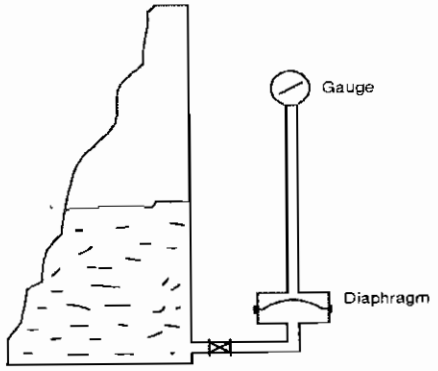


Displacer and Torque Tube Gauge

Hydrostatic Type

It consists of a bourdon type gauge, which is directly connected to the tank, whose level is to be indicated. This is a short distance type, and the gauge tube is usually filled with inert gas. As the liquid level rises or falls, the force due to the varying head of liquid on the diaphragm changes and alters the pressure of the gas in the tube, operating the gauge. Fig.19.

Fig. 19

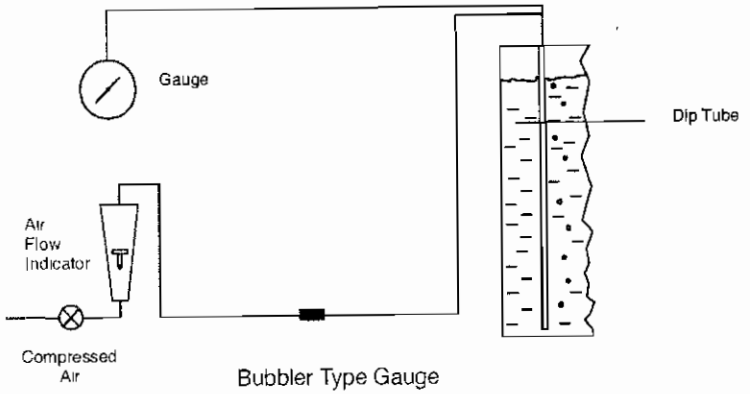


Hydrostatic Gauge

Pneumatic bubbler system

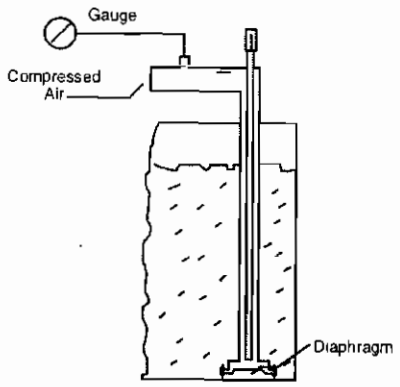
Control air acts via a flow indicator such as a Rotameter and an orifice, on the liquid in the tank, through a dip tube. The purpose of the orifice is to limit the air pressure in the dip tube, to match the maximum height of liquid which the tank holds. The tube extends to about 75 mm above the sludge level line. As the liquid level changes, so the force exerted by the head of liquid (on the air pressure in the tube) changes, i.e. the back pressure changes, which is indicated by the pressure gauge connected. This can also be used to operate indicators, alarms and control circuits. Fig. 20.

Fig. 20



Bubbler Type Gauge

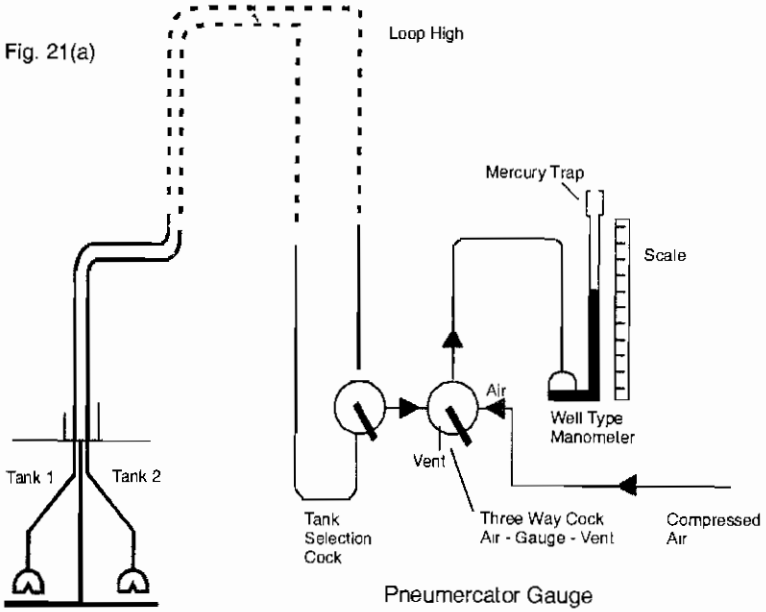
Fig. 21



Diaphragm Bubbler Gauge

Fig. 21 shows a modification of the earlier type. This is to take care of the problem of air bubbling through tank contents, which could be undesirable, especially for applications like boiler feed water, which is deaerated. The varying head of liquid acts on the diaphragm and varies the escape of air from the tube and hence alters the air back pressure in the tube. The back pressure can act as a signal for indication, alarm and control purposes.

Fig. 21(a)



Pneumercator Gauge

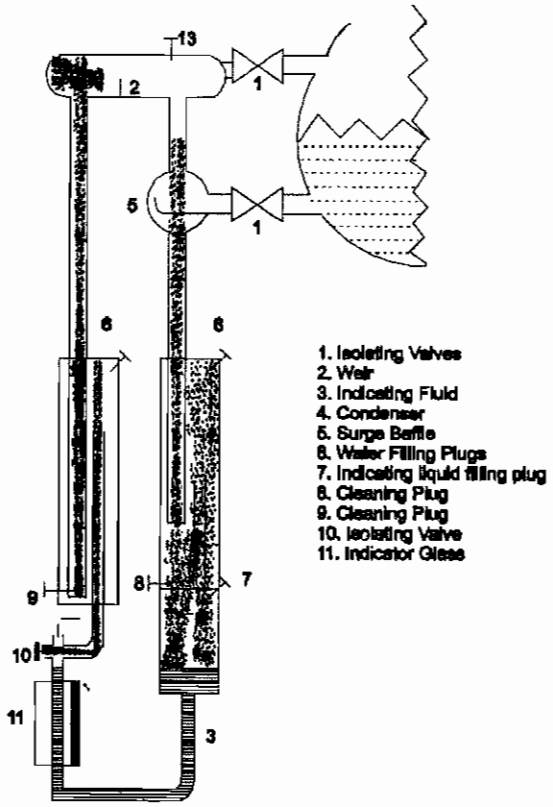
Pneumercator Gauge

Fig. 21(a) shows a typical pneumercator gauge which can be used for measuring the levels of liquids in tanks. It is basically a well type mercury manometer. Control air is used to transmit the force, exerted by the head of liquid in the tank, to the manometer well. A hemispherical bell is placed near the bottom of the tank, just above the sludge line. It connects from the bell, via a well protected pipeline, and is run up to the top of the engine room or into the uptake space and then down to the control room, where the gauge is located. The 'loop high' into the engine room is to prevent any reverse flow, i.e. tank contents passing back into the gauge should tank over-pressure occur. The gauge has a trap, to prevent the mercury from being blown out, in case of over pressure.

One gauge can be used to measure the contents of a number of tanks, with a selector cock. The cock is first turned to Drain, when the line is vented, then Supply, during which the control air fills the system, and the gauge shows full reading. The cock is then turned to Measure, which connects the gauge to the tank. The head of fluid in the tank exerts a force on the air in the line in the Bell, and this pressure is transmitted to the Manometer well.

Level readings will depend upon relative density and ambient temperature. Gauges may be calibrated for a liquid of a particular relative density or have an adjustable scale to accommodate a small range of relative densities.

Fig. 22



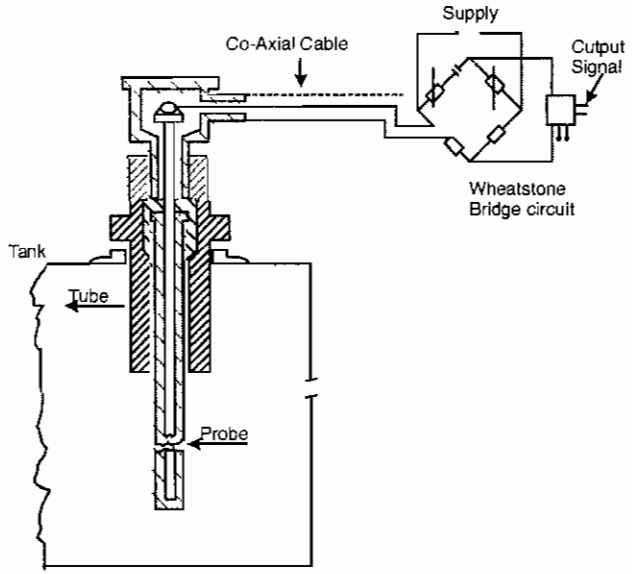
Distant Reading IGEMA Level Gauge

The Igema Gauge

This is a remote reading boiler water level indicator. It consists of a manometer having connections to the steam and water spaces of a boiler, as shown in Fig. 22. The steam connection has a small condenser fitted with a Weir. This provides a constant head of water to one leg. The other leg is subjected to the varying level of water in the boiler. For equilibrium, the forces acting on either side must be equal. Thus in the left hand leg, the constant head of water up to the condenser Weir, plus the head of the coloured Igema indicating fluid must equal the head due to the Igema fluid and the varying water level of the boiler in the right hand leg. As the level of water in the boiler rises, the head of the right hand leg increases and forces some of the Igema fluid, from the reservoir, into the left hand leg, increasing

the height of the fluid in the gauge. This in turn displaces some of the water from the Weir, which over-flows back to the boiler. The Igema fluid is non-misceble with water and has a higher relative density, as well as a distinct colour, so as to make it easily readable. The gauge would have some form of lighting, to make it more easy to read.

Fig. 23



Capacitance Level Gauge

Electrical Capacitance Gauge

Fig.23.shows a Capacitance gauge. A capacitor is a device, which stores electrical charges. It consists of two plates separated by a medium called as the Dielectric, in an electrical circuit. If the medium in the space between the plates (Dielectric) is changed, the value of electrical charges that the plates can store, i.e. the Capacitance will vary.

Thus the relative proportion of liquid to air in the tank, i.e. the level of liquid, is directly proportional to the Capacitance. The dielectric constant is called as the relative permittivity. Air has a relative permittivity of 1.

A probe is inserted into the tank and represents one plate of the capacitor, while the tank side represents the other plate. The

capacitance will be least, when the tank is empty, i.e. air fills the space between the probe and the tank side. It will be maximum, when the air has been displaced by the liquid, when the tank is full, since the dielectric constant of the liquid is higher than that of air. Thus, as the liquid rises and falls, the electrical charge storage capacity will vary.

The probe is connected to a Wheatstone's Bridge network.

This measures the degree of change in capacitance (and hence level) and produces a signal representing this. The instrument can be used not only for measuring changes in level, but also as a high and low level detector and also for detecting an oil/water interface in an oil/water separator.

Proximity Sensors

There are two types of signals : the conventional analog signal and the digital or discrete signal. Digital or Discrete signals provide a finite output, i.e. a 1 or a 0, whereas the analog signal has an amplitude, which varies from minimum to a predefined maximum value.

A Proximity Sensor is a discrete sensor, that can sense when an object has come near it. There are four main types of proximity sensors : the inductive type, the capacitive type, the ultrasonic type and the optical type of proximity sensor. It is important to understand how they work, in order to know which type to use in a particular application.

The Inductive type of proximity sensor works on the principle of the change in inductance (and power loss) of a Coil, as a metallic object passes near it. When a metallic (steel, iron, aluminium, tin, copper etc.) object is near the coil, the alternating magnetic field in the target produces circulating eddy currents inside the material. To the Oscillator, these eddy currents are a power loss. As the target moves nearer, the eddy current loss increases, which causes a loading effect, that decreases the amplitude of the output from the Oscillator.

When it increases beyond a certain value, the drop in amplitude will cause the Oscillator to stall (cease to oscillate), which makes the trigger circuit switch the output to ON. Sensing range depends on the diameter of the coil used : Small diameter coils (¼") have a range of 1 mm, whilst larger coils (3") have a range of about 50 mm. As the metal used (steel, iron, copper etc.) may differ in density,

permeability and values of resistivity, the sensing range may vary, depending on the characteristics of the metal used. Obviously, such sensors cannot be used for non-metallic objects !

Optical Sensors

Optical sensors use light, hence they are capable of sensing objects that are opaque, regardless of the colour or reflectivity of the object. They operate over longer distances, unlike proximity sensors, will even work in a vacuum and can sense any material, whether magnetic / non-magnetic, metallic, porous or conductive. Since they use focussed beams, they can be operated in close proximity of other sensors without fear of 'cross talk' or interference.

There are fundamentally three different types : Thru-beam, Diffuse reflective and Retro-reflective types.

The **Thru-beam** type is the most commonly used, being used in burglar alarms or 'sensitive' areas where any movement is required to be detected. The main disadvantage is that two separate units are required, the Transmitter and the Receiver, with associated wiring. This may be inconvenient or not practical.

The **Diffuse reflective** type has both emitter and receiver located in the same unit, light from the emitter being 'bounced' back from the target, to be picked up by the receiver. When no target is present, the light does not return (OFF). This can create a problem with transparent objects or dull finish / black surfaces.

Retro-reflective (Reflex) The retro-reflective optical sensor is arguably one of the most sophisticated sensors in use today. The sensor works on the principle of blocking of a light beam. The light being blocked, by the target, is light returning from a reflector. Generally, this type of sensor would not work well, if the target has a glossy surface, since this type of surface would also reflect light. This problem could be minimised by using a polarising filter, together with a suitable reflector, as specified by the manufacturer.

CHAPTER 3

Control Fundamentals

Upon completion of this chapter, you will learn :

- Control fundamentals
- Terminology
- Open and Closed Loop control
- Discontinuous, on-off or two step control
- Differential gap and Transfer lag
- Feed-back
- Ratio control
- Cascade control
- Condition monitoring
- Vibration and noise monitoring
- Temperature monitoring
- Ferrography, Spectrometric analysers, Atomic absorption spectrometer
- Particle counter, corrosion monitoring and performance trend monitoring
- Piezo electrical accelerometer
- Different types of control actions : On-Off or two step
- Simple Proportion action, Proportional band adjustment
- Integral and Derivative action – Three term control
- Tuning of P+I+D systems

Control Fundamentals

The Aim, of a Controller, is to reduce the Error or Deviation to the minimum. To achieve this, various means are available. The error is amplified by a factor called the **Proportional Gain.** This factor is sometimes used in another way, where the percentage of the reciprocal of this value is called as the **Proportional Band.**

Thus **P(%) = 100/ kp ,** where kp is the Proportional Gain factor.

Terminology

1. **Controlled condition** :
The variable, that is being controlled, is called the controlled condition. E.g. speed in an engine governor, temperature in the engine cooling system.

2. **Monitoring element** :
The element which measures the controlled condition (variable), and produces a signal corresponding to it, which can be used by the control system. Also called as Sensor, transducer.

3. **Measured value** :
Actual value of the controlled condition, as measured by the sensor.

4. **Desired value** :
the value of the controlled condition, that the operator wishes to maintain.

5. **Deviation** :
the difference between the desired value and the measured value. This signal is sent to the Comparator, in order to initiate some corrective action.

6. **Comparator** :
This is that part of the Controller, which compares the actual measured value with the Desired or Set value, and produces a deviation or error signal, which can be used to operate the Correcting element.

7. **Offset** :
This is sustained deviation, which occurs, when the measured value stabilises (reaches equilibrium) at some point other than the Set value. This could change, with change in load conditions. Offset occurs in simple proportional control.

8. **Control point** :
In a simple proportional control system, the controlled condition will stabilise at some point of equilibrium, other than the Set point, which is termed as the control point.

9. **Error signal** :
The signal produced by the Comparator, after comparing the measured value with the set value.

10. **Motor element** :
The actuator, which is operated by the Controller, to take corrective action, based on the deviation.

11. **Feedback** :
The transmission of the measured value to the Comparator is termed as the feedback.

Loops

Open loop control

This is the simplest type of control, where the input to the process is independant of the output. Since the output is not sensed, the input is usually dependant on some other variable, e.g. time.

A marine example is the conventional oil purifier or centrifuge. The purifier desludges after a fixed time interval, irrespective of the amount of sludge that may or may not have built up. The purifier does not check whether or not the sludge has been removed, or even if there is any sludge at all.

Manual open loop is one where a human operator is essential. If he is replaced by a control system, it becomes an automatic open loop.

Advantages of open loop

Cheaper (than closed loop).
Simpler, thus easier to troubleshoot / repair
No hunting
Suitable for systems, where precise control is not essential.

Disadvantages of open loop

Not suitable for complex systems, having considerable load changes.

Excessive deviation from set point.

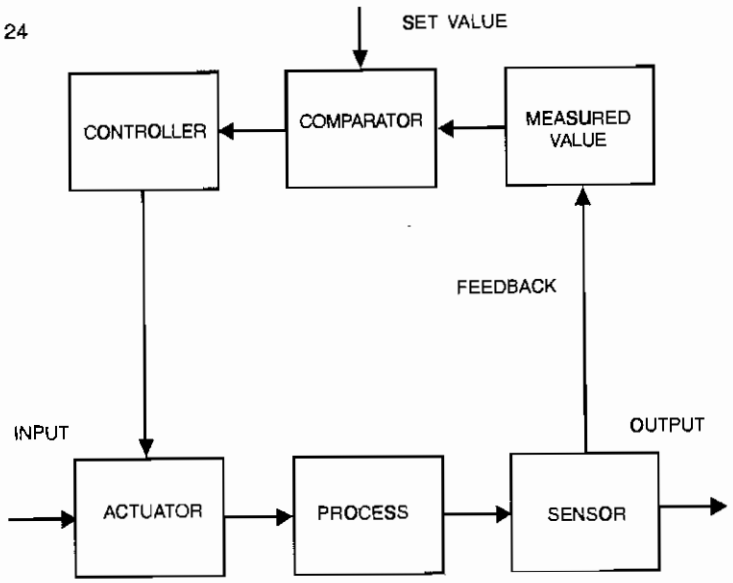
Closed loop control

In a control system, if there is some means to monitor the output, and generate an error signal, which can now correct the input, (i.e. we 'close' the loop), then this form of control is called as closed loop control.

If a human operator does the job of observing the output and taking the necessary corrective action, then it is a manual closed loop.

When the operator is replaced by a controller, that uses a sensor (to provide feedback of the controlled condition), a Comparator to get the Deviation (from the measured value and the set value), and a correcting signal to the Motor element to take appropriate action on the input, then it is called as an Automatic closed loop.

Fig. 24



CLOSED CONTROL LOOP

Advantages of closed loop

Finer control, with less chances of deviation.
Suitable for systems having considerable load changes.

Disadvantages of closed loop

More expensive (than open loop).
Possibility of hunting (loss of stability).

An example of a manual closed loop could be water level control. An operator observes the water level, and accordingly opens or shuts the feedcheck valve, to maintain the desired level.

If the same work was done by a Feedwater Controller, then it would be an Automatic closed loop.

Process Control

This is control over variables like pressure, temperature, viscosity, level, flow and so on.

Kinetic Control

This is control over variables like displacement, velocity and acceleration.

Discontinuous, on - off or two step control :

This is a form of control, where the correcting element has only two predetermined positions, and it can change from one to the other, as the deviation changes.

This form of control is applied, when precise control is not necessary and any fluctuations that occur are tolerable. This control is more stable, unlike in continuous control, where the correcting element is continuously changing.

For discontinuous or two-step control to work effectively, the demand capacity has to be large in comparison with the supply capacity.

Also, the Transfer lag should be high, which prevents oscillation.

Differential gap :

The difference between the deviation required to change the state of the controlling element, and the deviation required to revert to the previous state, is called the differential gap or overlap.

e.g. an air conditioning compressor may cut-in, when the temperature rises to 25°C, but may cut-off only when the temperature falls to 15°C.

If the desired value is 22°C, then when deviation is 25 - 22 i.e. 3°, it will cut-in, and when the deviation is 15 - 22 i.e. -7°, it will cut-off.

The differential gap aids in reducing the frequency of operation, but increases the amplitude of variation of the controlled variable, i.e. the actual value is deviating from the desired value.

If there is no transfer lag, and the difference between the supply capacity and the demand capacity is not much, then a differential gap or overlap is essential.

Transfer lag :

The delay in sensing the effect of a control action; or, the time between change in input and the sensing of the corresponding change in output.

Feedback :

This is the transmission of a signal representing the controlled condition for comparison with a signal pre-set by the operator, and which is intended to determine the value of the controlled condition. Feedback can be considered as the transmission of the measured value to the Comparator.

Ratio Control

Here the ratio between two or more measured variables is maintained constant. In a boiler combustion control system, air flow and fuel pressure are to be constantly matched, such that the air to fuel ratio is always maintained constant, in order to provide efficient combustion.

The air may be kept slightly in excess of that required by the chemical or stoichometric formula, in order to improve the firing, i.e. prevent visible smoke. However, if there is a requirement to use the exhaust gas for Inerting, then a very precise control is required, to keep the oxygen level in control.

Usually the air flow is controlled by dampers to match the fuel pressure, while fuel pressure itself has to match the steam demand.

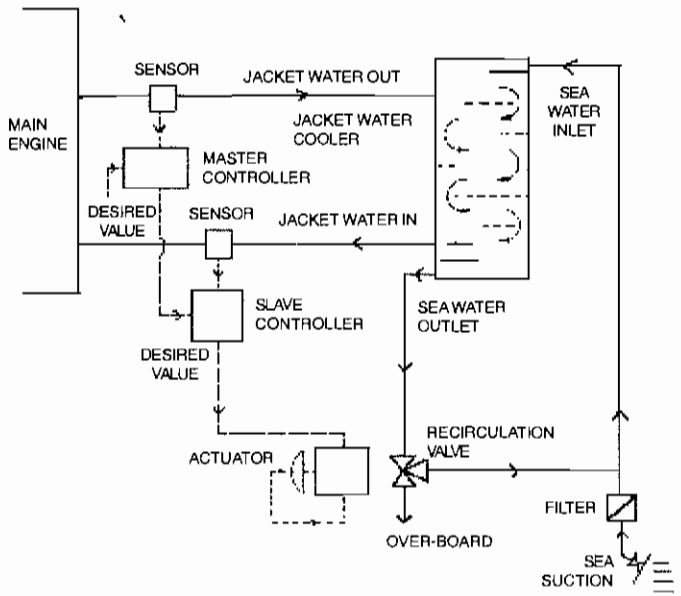
Cascade control

This consists of two controllers in series, one master and the other slave. The same controlled condition is sensed at two different points in the circuit. The slave controller is doing the actual control of the parameter, but it's set point is determined by the Master Controller, which itself has a set point set by the operator.

The purpose of Cascade control is to deal with the large capacities involved, which initiate time lags. Thus, in the example of the cooling system shown, any change in the sea water temperature, would only be sensed by the outlet temperature sensor, after it had over-cooled the engine, in a conventional control. In cascade control, both quantities are controlled, i.e. the change of sea water temperature which affects the inlet to the engine, as well as load changes of the engine, which affect the outlet temperature of the engine.

Derivative action can deal with this to a certain extent, but where more than one large capacity is involved, cascade control is essential.

Fig. 25



Condition Monitoring and Signature Analysis

Condition monitoring implies determination of the condition of a machine and its change (or deterioration) with time, in order to determine its suitability for continued operation.

The condition may be determined by measuring physical parameters, like vibration, noise, temperature, extent of oil contamination, wear debris and so on.

A typical pattern, called as its signature, emerges from these parameters. Any change in this pattern, usually indicates the possibility, of excessive wear or breakdown.

If properly analysed, this becomes a valuable tool, to determine if the machine is in need of any maintenance, and can prevent future unscheduled breakdowns.

This results in less down-time, and can result in considerable savings in terms of damage repairs, as well as from the safety angle.

The monitoring of parameters may be either continuous, or at pre-defined intervals.

In the case of machinery like pumps, turbines, and compressors, the rotating components are subjected to considerable fatigue and wear, and an analysis of the vibration level of bearings often predicts the end of the service life, of a particular bearing, thus planned maintenance can help to prevent breakdowns, long before the effect of worn bearings can be seen on the performance of the machine.

Vibration and noise monitoring :

The vibration signature of a machine is related to the general health of the machine. Measurement of vibration levels in bearing housings, relative movement between the shaft and the bearing and the noise emitted by the machine can provide useful information regarding faults like unbalance, misallignment, excessive wear of bearings causing whipping of the shaft and so on.

Piezo-electric sensors are used for vibration monitoring and microphones for noise monitoring. Frequency analysis instruments are used for obtaining the frequency spectrum of the vibration.

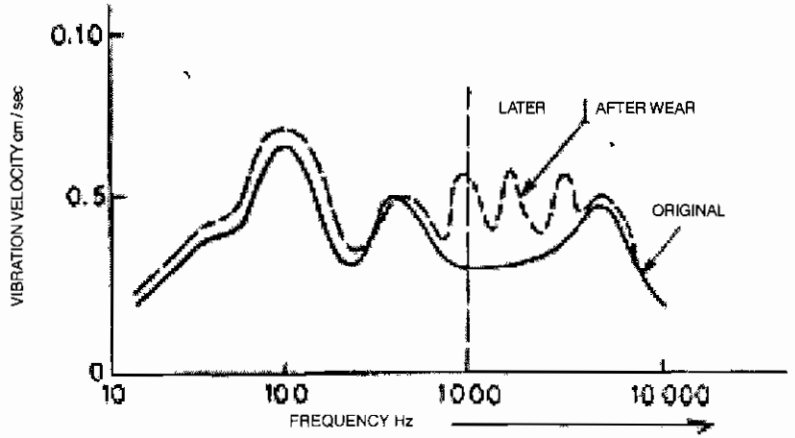
The proper location of the transducer for measuring radial vibration and axial vibration is essential. Improper locations result in wrong readings, due to the flexibility of components, on which these transducers are mounted, which would affect the transmitted vibration levels.

A proximity transducer of capacitive or inductive type is used for recording the motion of the shaft relative to the bearing.

The monitoring systems for noise are similar to those for vibration.

A damaged bearing would emit a sound signal different from a normal bearing, however, the audible noise level is reached after the bearing has been considerably damaged. Prior to reaching this stage, bearings emit sound in the ultra-sonic range, which is about 40 k Hz, thus ultrasonic detectors have been found to be of help in early detection of bearing condition.

Fig. 26



VIBRATION MONITORING

In order to ascertain if the vibration levels are satisfactory or not, some general guide lines are : -

If vibration amplitude is increasing, this could be due to unbalance.

Fairly large axial vibrations, could be the result of mis-alligned couplings, mis-alligned bearings or bent shafts.

Radial and Axial vibration occuring together could be due to roller bearing defect (ultrasonic frequencies of 20 to 60 kHz).

Side bands near tooth meshing frequencies, (tooth meshing frequency = number of teeth x rpm) could be due to worn or damaged gears.

Excessive radial vibration could be due to unbalaced reciprocating forces and couples.

Noise level :

Element	Good	Acceptable	Poor
Roller/Ball bearing	75 dB	85	90
Spur gears	95	105	110
Single helical gears	85	100	105
Epicyclic gears	75	90	95

Temperature monitoring :

Temperature is a useful guide to indicate the health of a machine. Different sensors may be used :-

viz. Thermometer, thermo-couple and infra-red pyrometer.

It is advisable to maintain continuous records of temperature measurements, to enable deterioration of the conditions to be ascertained as early as possible.

Ferrography :

This involves the separation of wear debris of ferrous type magnetically, and arranging in the order of particle size for examination.

A sample of oil, for example, is diluted with a solvent, which breaks down any gel, which may be around the wear particles. The sample is fed to a transparent precipitator tube, on either side of which are the poles of a magnet. The magnetic force attracting the particles is proportional to the size and magnetic susceptibility. The larger particles are deposited at the entry region of the tube, and the smaller particles and the oxides of iron get deposited later on in the tube.

The amount of material deposited is measured by the attenuation of light from a light source placed below the tube. The light passing is detected by photo-electric transducers.

Wear severity is indicated by difference between optical density at two points, distant 5 mm from each other, along the tube length.

In normal rubbing wear, the majority of the particles are small. As the severity of the wear increases, so does the number of bigger particles. In a Ferro-graphic analyser, the oil containing the wear debris is sorted by size along an inclined glass slide. The magnetic field gradient increases as the particles move downwards. The larger particles get deposited first.

The slide is analysed by an optical microscope. Red light is directed through the objective of the microscope.

Opaque objects appear red. Green light is transmitted through transparent objects, which appear green. Other objects appear in various shades of yellow- green. This helps in distinguishing metallic objects from oxides.

Further, the slide is heated to 330° C, when low carbon and alloy steel particles appear to be bluish, Cast Iron appears light brown in colour, bronze appears dark brown, and chromium and aluminium are bright white colours. Thus particle identification is possible.

Spectrometric Analysers

This technique is called SOAP (Spectrometric Oil Analyses Programme).

Oil samples containing wear debris are taken from the sump and analysed, by any of the following methods : -

1. Atomic absorption spectrometer
2. Emission spectrometer

The **Atomic absorption** spectrometer is based on the principle that, every atom absorbs light of its own specific wave length only. From the combustion of the oil sample, metallic elements in the oil sample are atomised.

In a light beam passing through the flame, certain wavelengths get extinguished, due to their absorption by the free atoms of the metals in the sample. Since each element has its own characteristic wave length, and it is possible to measure the amount of light absorbed, we are able to get the concentration of various elements, in the oil sample, in parts per million (p.p.m.).

The monochromator is tuned to accept light of a certain wave length. A meter indicates the difference between the original light received directly by the light source, and that received by the monochromator via the sample.

The difference is the amount of light absorbed, and indicates the concentration of the particular element present. The procedure has to be repeated for each of the elements to be detected.

In an **Emission spectrometer**, it is possible to analyse, simultaneously, a number of elements.

A rotating Graphite disc carries the oil sample, which is subjected to high voltage excitation. A spark excites the various metallic elements, after which they emit their characteristic wave lengths, the intensity of which is measured by various detectors.

Elements which can be analysed by this method include copper, iron, chromium, nickel, lead, sodium, aluminium, magnesium, silicon and silver.

If the composition of various components in the crankcase, viz. Bearing material, is known, then it is possible to identify the source of the wear debris in the oil sample.

Particle counters

The size of various particles of the wear debris can be determined by means of particle counters, which work on the light blockage principle. The oil sample is passed through a sensor, consisting of a light source and a photo diode. A particle passing through the window blocks light according to its size, since light travels only in straight

lines. Accordingly an electrical signal is sent from the photo diode, according to the amount of light received, which is proportional to the size of the particle. The turbulent flow in the tube results in rotation of the particles, thus the probability of getting the largest particle size is very high.

Corrosion Monitoring

The parameters to be measured are : -
The rate of corrosion,
the total depth of corrosion, or
the thickness of good material remaining.

Direct observation is the oldest method, based on visual inspection. It has its limitations, since deep corrosion may not be accurately analysed.

Chemical and electro- chemical methods include analytical, potential or galvanic methods. These methods are used to monitor water quality in boiler feed water. The ratio of metallic ions in a sample, may indicate the extent of corrosion.

Non- destructive Testing methods (NDT) include radiography, ultrasonics, eddy current, thermography and accoustic emission methods.

Ultrasonic methods are being increasingly used for flaw detection and thickness measurements in pressure vessels, piping and well as ship's side plates and other surfaces, where it is not possible to drill holes or cut samples for visual inspection.

The disadvantage of this method is that only a small surface may be tested at a time, thus large area coverage is not possible.

Radiography is useful as permanent records can be kept, however it is slow, and hazardous.

Eddy current technique is primarily a surface technique, penetration below depths of 5 to 6 mm are hard to obain, hence it is not much used for ship's work.

Thermography is used to locate increased surface temperatures due to corrosion damage in furnaces. Electrical resistance technique is based on the principle that, as corrosion decreases the cross-section, so its electrical resistance increases. If it is possible to record the readings on a continuous basis, it is possible to find out the corrosion rates, thus simplifying the maintenance, by taking preventive actions, before corrosion can develop to extreme limits.

Performance Trend monitoring

Monitoring the performance of a system involves the measurement and recording of relevant parameters, and detecting any change in performance.

Parameters like fuel consumption, lub oil consumption, exhaust gas analysis, various temperatures and pressures are continuously monitored.

Deterioration in fuel consumption figures could be due to various parameters changing, which can then be investigated and put right. Similarly, change in the exhaust gas analysis readings could point to faults in the combustion of fuel.

Excessive consumption of lub oil could also be detected, by an increase in the wear rates of piston rings, if an on-line detecting system like SIPWA is fitted.

Selection of condition monitoring techniques

Gear drives	Vibration & noise monitoring, oil analysis.
Roller bearings	Shock pulse monitoring (SPM) and vibration analysis.
Journal bearings	Temperature and oil analysis.
Seals	Particle analysis.
Engines	Vibration, lub oil analysis and performance trend monitoring.
Turbines	Vibration, temperature and wear debris monitoring.
Pressure vessels	Ultrasonics, corrosion monitoring and performance trend monitoring.

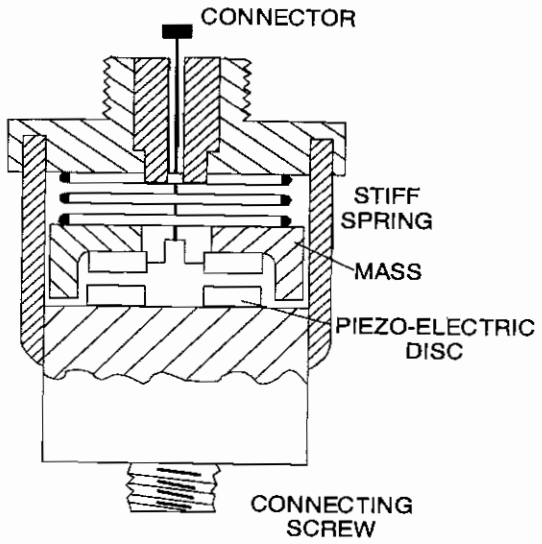
Piezo- electric Accelerometer

This is a seismic transducer, using a piezo- electric crystal. This is commonly used for shock and vibration measurement, especially bearings. The crystal is fairly stiff, and is held in compression by a hemispherical spring. The device is attached to the component, whose vibration is to be measured.

Thus the motion, due to vibration of the component, is transmitted to the crystal, which produces an output voltage, proportional to the amplitude of the vibration.

Such devices are sensitive, light weight and can be used over a fairly wide frequency range.

Fig. 27



Vibration Monitor

Different types of Control actions

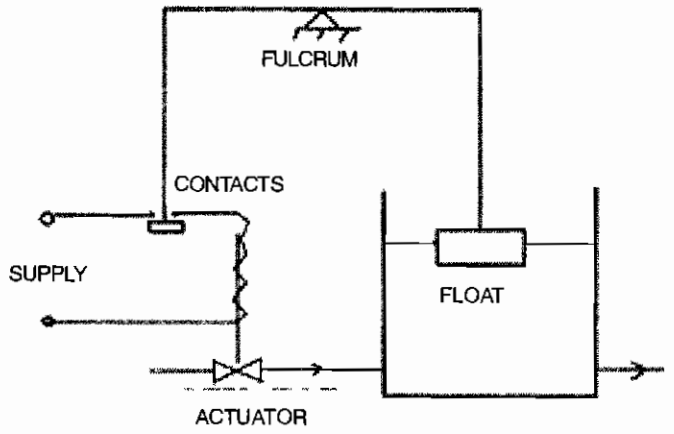
There are several types of Control actions :

1. ON - OFF or Two step control action.
2. Simple Proportional action.
3. Integral action.
4. Derivative action.

Consider the simple system shown in the sketch. The water level in the tank is monitored by the float, which makes or breaks the contacts for the solenoid, which operates the Actuator, for filling the tank.
This is a simple two step controller, and can be used for steady conditions of water in the tank. If, however, the demand varies, then this type of system may be unable to cope with the fluctuating conditions. This is where Proportional action comes into the picture.

Fig. 28

TWO-STEP OR ON-OFF ACTION



Proportional Action

If we replace the simple two step control by a valve, whose opening varies as the deviation, then this type of control will be more effective in coping with varying demand. Since the amount of valve opening is proportional to the change in level, this is called as simple proportional action, as shown in the sketch.

Fig. 29

SIMPLE PROPORTIONAL ACTION

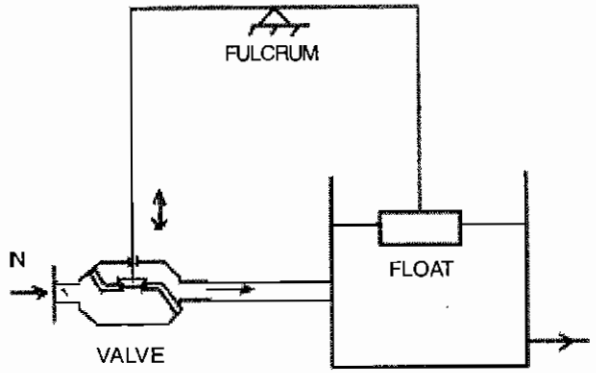
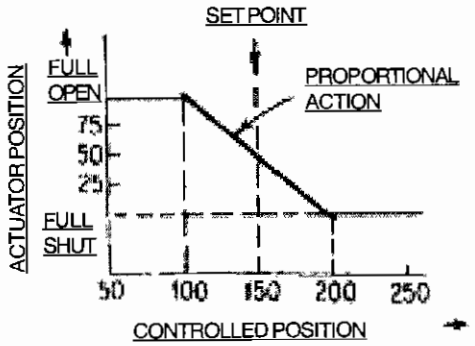


Fig. 30

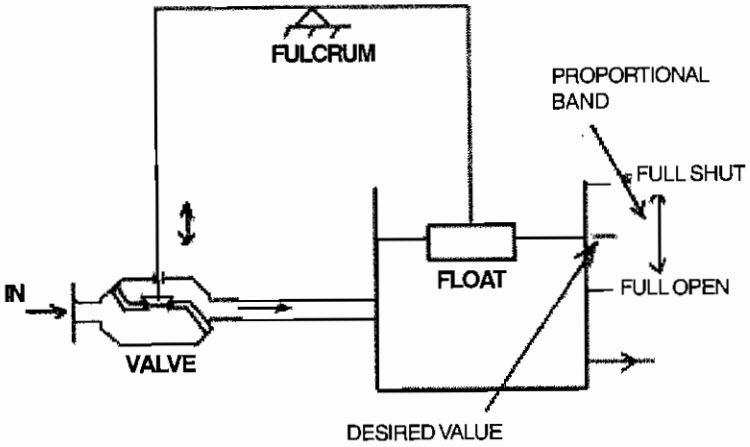


PROPORTIONAL CONTROL

Considering the sketch below, the float can take up a variety of positions. From the lowest point, at which the control valve is full open, through the desired value, till the highest point, at which the control valve is full shut, is called as the Proportional band. Within this band, the opening of the valve is proportional to the deviation from the desired value.

Fig. 31

SIMPLE PROPORTIONAL ACTION

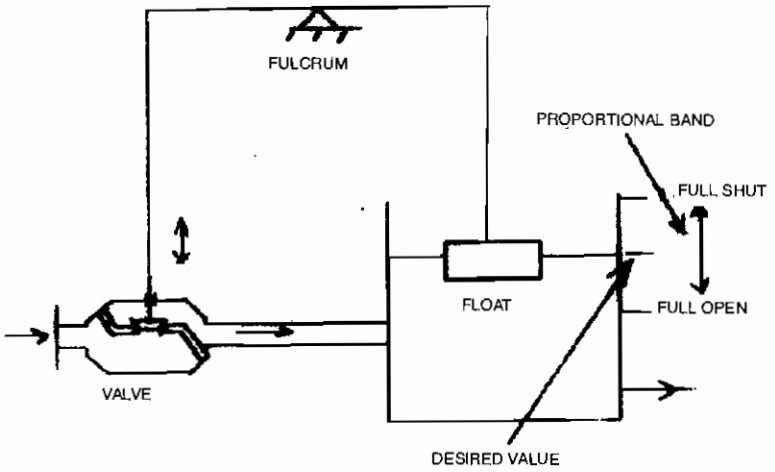


Proportional band

The proportional band can be made wide or narrow simply by shifting the fulcrum away from or towards the float. A wide proportional band means that for wide changes in the level, there is hardly any movement of the valve. This gives a coarse control, i.e. there is stability, but the final value varies from the desired value. This sustained deviation is called as Offset, and is inherent in simple proportional control. The wide proportional band increases the offset, as shown by the sketch.

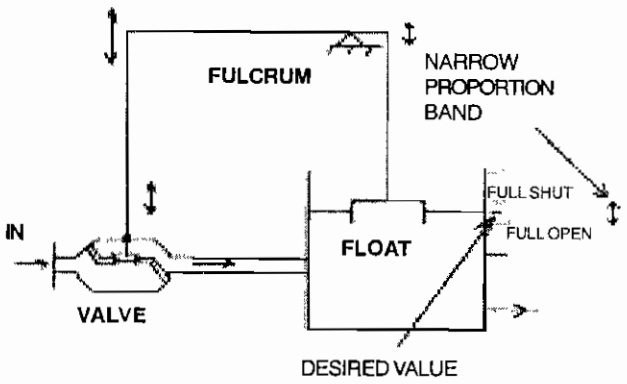
Fig. 32

SIMPLE PROPORTIONAL ACTION



A narrow proportional band will give a finer control, but there will be less stability, i.e. it will keep hunting. If the band is too narrow, the valve ceases to have proportional action, and behaves like an on-off type control.

Fig. 33

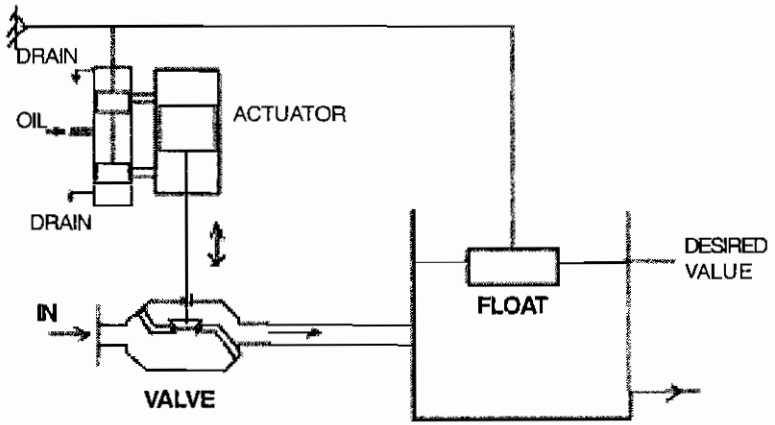


Integral Action

The drawback of simple proportional control is the sustained deviation, called Offset. The wider the proportional band, the more is the Offset. Integral action acts to remove this offset.

INTEGRAL ACTION

Fig. 34



Consider the sketch above. As long as the float is at the desired value the actuator remains stationary. At any other value of the float, the Actuator is either opening or shutting, depending upon which port is uncovered.

Thus, Integral action acts to remove the offset. The system above will only come to equilibrium, when the float is at the Desired value.

Proportional Plus Integral Action

When Proportional action combines with Integral action, the system works more efficiently. Proportional action allows variations in demand to be easily met, since the control action is proportional to the deviation, and thus the valve opening increases with an increase in the deviation, i.e. the difference between the measured value and the desired value.

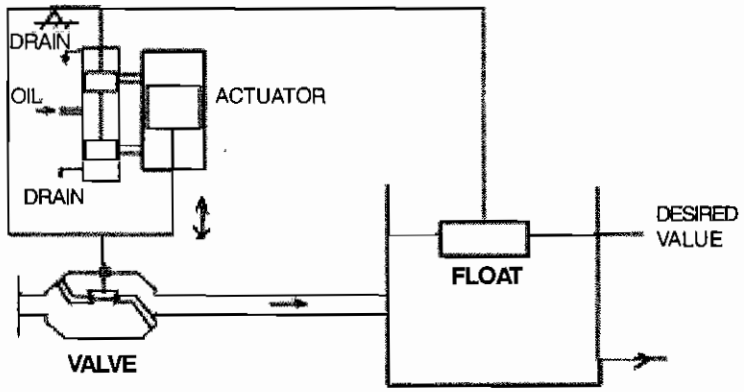
The Integral action acts to remove the Offset, and thus the system will only stabilise when offset has been removed.

The left hand side of the figure shows the proportional action, and the right hand side is the Integral part. Both actions are working on the final control element, i.e. the control valve which regulates the filling of the tank.

The only limitation of this type of control is the time factor. The action continues till the desired value is reached, but it does not do this in the shortest time interval. This brings us to the third term of control, i.e. Derivative action.

PROPORTIONAL + INTEGRAL ACTION

Fig. 35

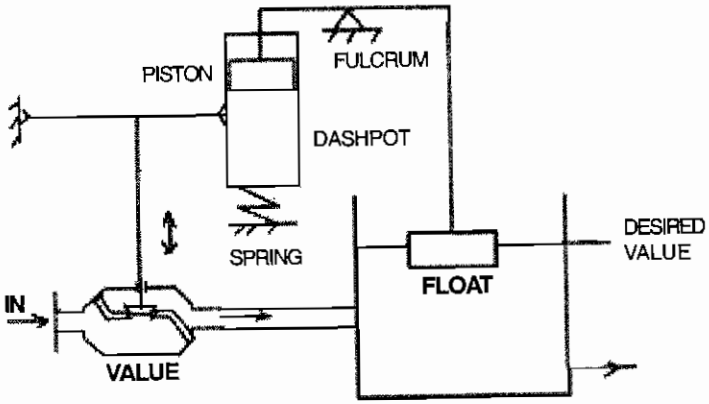


Derivative action

Derivative action is one where the control is proportional to the rate of change of the deviation. This means that derivative action takes place where the rate of change of deviation is higher, thus, if there is a sudden change in the demand, i.e. a sudden drop in level, then Derivative action comes into the picture. Derivative action will effectively deal with sudden variations of the measured value.

Fig. 36

DERIVATIVE ACTION

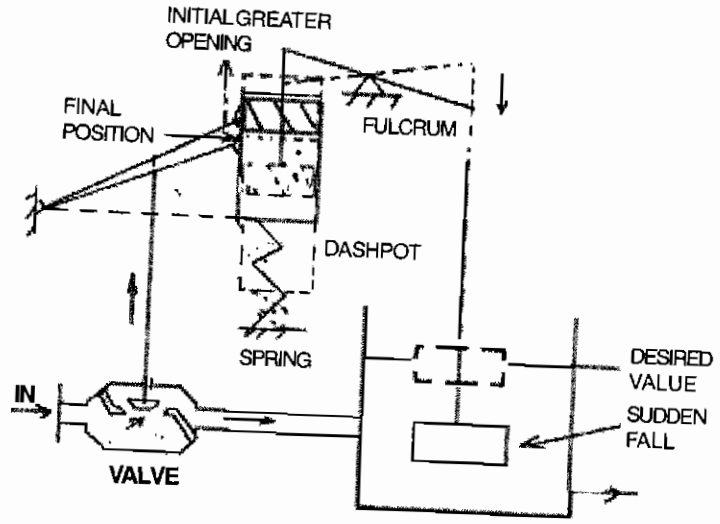


Consider the sketch above. This shows the principle of Derivative action, although it must be borne in mind, that Derivative action does not occur alone, but it is combined with some other action, such as Proportional + Derivative, or Proportional + Integral + Derivative (three term).

The piston works inside an oil filled cylinder called a Dashpot. The spring tries to pull the dashpot downwards. Consider that a fall in level has occurred. This causes the float to fall, and on the other side of the fulcrum, the piston will be pulled upwards. Since the piston is inside highly viscous oil, the cylinder is initially dragged upwards, along with the piston, against the spring force. This causes the control valve to open fully, thus coping up with the sudden increase

in demand. Since the oil can flow slowly past the piston, the cylinder now tends to descend, under the action of the spring, till the piston touches the top of the cylinder. Thus the control valve now takes up a position that is determined by proportional action only. The temporary increase in valve opening was the Derivative action, which was in response to a sudden drop in level. Had the drop in level been gradual, the control valve would have come to the last position directly (i.e. with the top of the piston touching the cylinder), since oil would have had the time to leak past the piston, to the other side.

Fig. 37



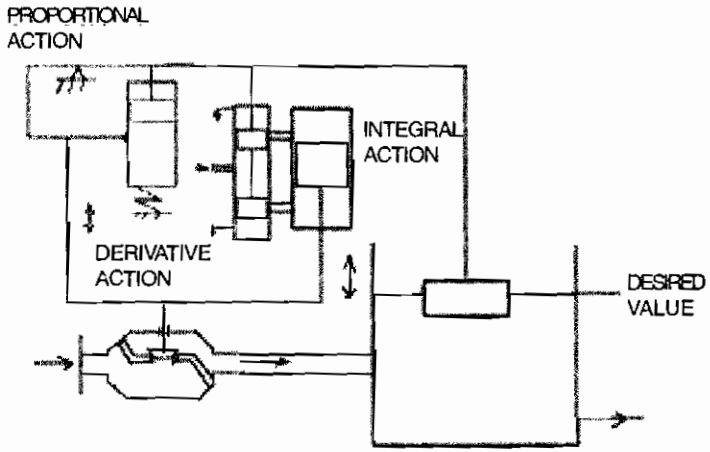
Three term control

All three forms of control can be combined to get a three term controller. This is Proportional plus Integral plus Derivative action. This form of control combines all the advantages of each of the different actions. The proportional feature ensures that load changes can be effectively dealt with, the integral action seeks to eliminate offset, while the derivative action comes into play when there are rapid changes in demand, i.e. sudden changes in the water level.

This form of control is illustrated in the sketch below.

Fig. 38

PROPORTIONAL + INTEGRAL + DERIVATIVE



The tank level can very easily be maintained as shown in the sketch above. The proportional part is the left hand side part. The integral action is on the right hand side, while the central part controls the derivative action.

All three forms of control act on the final element, i.e. the valve which is filling the tank.

Closed Loop systems using P+I+D Control

There are three fundamental problems, with using simple Proportional control :

1. The system may have a large Offset or sustained deviation
2. It may be foo slow in response to changes in Load
3. System may become unstable (hunting) if Gain is changed to reduce Offset

To improve this arrangement, we introduce the integral and derivative functions. Thus, the error signal is amplified by the proportional gain, kp, after which integral and derivative action may be added at the summing junction, to produce the final correcting signal. If a particular function is not needed, e.g. Derivative action may not be needed if the rate of change of deviation is very slow, then its value is set to zero in the equation.

Although the integral and derivative functions imply the use of advanced mathematics for their solution, in-depth knowledge of calculus is not necessary for a marine engineer, in order to understand or apply these in practice. For operating or trouble-shooting a system, there is no need to go into the design aspect, but simply concentrate on the application itself.

Tuning of PID Controllers

Although general concepts and the effects of adjusting the gain, the derivative and the integral functions are the same, there are minor differences in the way the PID Controller is adjusted. In digital controller, for example, PID calculations are not done continuously, but are triggered by a timing function. When the timing trigger occurs, the Measured Value (MV) and Desired Value (DV) are sampled and digitized, after which gain, proportional and / or derivative action functions are calculated and summed, to produce the final correcting signal to the actuators.

The PID process is then 'paused', till the next trigger, after which the calculations are redone and so on. The integral or reset time, the derivative action time and the gain (or the Proportional band setting) can now be re-adjusted (if necessary). The effects of these changed settings will be evident in the next trigger, after fresh readings are taken.

Tuning the PID Controller is often a subjective process, despite the detailed mathematics involved, since the Operator must be familiar with the characteristics of the specific system, as well as the response of the system to load changes, which differ from situation to situation. Additional complications arise, if the system has a non-linear response to inputs, such as using field current control for speed control of a shunt dc motor. In case of non-linear response, one possible solution is the use of a so-called 'fuzzy logic' controller.

Another problem could be in the case of dual mode control, wherein the system uses one particular method to adjust in one direction, whilst using a different method for adjustments in the opposite direction. For example, consider temperature control of the cooling water of an engine. A rise in temperature is relatively simpler, viz. by reducing the coolant flow, but a subsequent fall in the temperature would take a longer time to achieve.

In a microprocessor based controller (PLC), instead of the signals being analog (voltage 0 – 10 V or current 4 – 20 mA), they are digital (bytes). Thus, all numerical processing is done digitally. Calibration is not a problem, since the digital settings will not 'drift' with temperature, over time, nor are there any mechanical components which need to be constantly adjusted, to take care of the 'wear-and-tear', since there are no 'moving' parts, the microprocessor being a solid-state device.

Also, using a microprocessor enables digital filtering (called digital signal processing or DSP), averaging, numerical integration and differentiation…in fact, advanced mathematical functions can vastly improve the system's performance, response, accuracy and stability. The actual calculations are generally performed by a mathematical co-processor, which frees up the main processor to solve the ladder logic at high speed, thus the scan time is not affected, which would otherwise have drastically reduced the speed of the operation.

Although the preceding explanation is expected to give an understanding of the fundamentals of control system, in practice, few systems would perform exactly as anticipated. Thus various methods have evolved, to enable the operator to 'tune' the Controller, so as to get a reasonably good control over the process.

For a person who is unfamiliar with a system, there are fundamental approaches to 'tuning' the Controller. As with most standard tuning methods, these will give approximate ('ball park' figures), which need to be further refined, as per the requirement. It is difficult to get the final result in the very first attempt (it may happen due to luck, rather than any method !), but the first results will generally give some 'steady' figures, that are good enough to start the process, allowing the operator to further 'fine tune' the system, with subsequent attempts.

Modelling the transfer function of the system involves taking into account factors such as inertia, windage, spring constants, masses, friction, damping, inductance, capacitance and so on, many of which are extremely difficult, if not impossible, to accurately measure. Therefore most operators adopt a 'trial and error' method to get the system to start working.

Note that :

Increasing the gain will result in a faster response and better accuracy (less offset), but at the expense of stability, resulting in greater overshoot and consequent hunting, both of which are undesirable.

Similarly, increasing the Derivative action time will enable faster response to sudden changes (for a higher rate of change of the variable), but may be unnecessary when there are very gradual changes in the load.

Decreasing the integral time constant (also called reset rate or reset time constant), will cause the offset to reduce. Thus, smaller values will reduce offset at a faster rate. However, too small a value will increase the oscillation.

There are various tuning methods in use :

'Adjust and Observe' Method : This involves making certain initial adjustments of the parameters and observing the response to these. With the machine operating at a steady state, change the Set Point and observe the response. A good response would be an overshoot of approximately 10 – 30 % of the Set Point change. For different values of the Gain (reciprocal of Proportional Band), observe the response of the system. Now increase the Derivative action time to reduce the overshoot to the desired value. Now adjust the Integral

action time, so as to remove any offset or sustained deviation. Once the initial tuning is done, further adjustments can be done in a like manner, to achieve the desired level of control.

Ziegler-Nichols Method : This was developed by two employees of the Taylor Instrument Companies of Rochester, NY in 1942. This utilises a mathematical process of tuning, resulting in an overshoot of ¼ the previous step, for every subsequent step, thus it is also referred to as the '¼ wave decay' method. The main advantage of the ZN method is, that unlike the 'Adjust and Observe' method, the gain, integral and derivative action constants are pre-calculated and this saves time in application.

Oscillation Method : In this method, two parameters, called ultimate gain and ultimate period, are first determined. Using these values, other parameter are calculated, using certain empirical formulae.

Open Loop Method : Here, three parameters are first determined : the dead time, the process time constant and the process gain. Using these values, other constants are calculated, with the system in an open loop configuration. A chart recorder is used to record a graph of the process variable with respect to time. This is a graphical method, which involves certain empirical formulae.

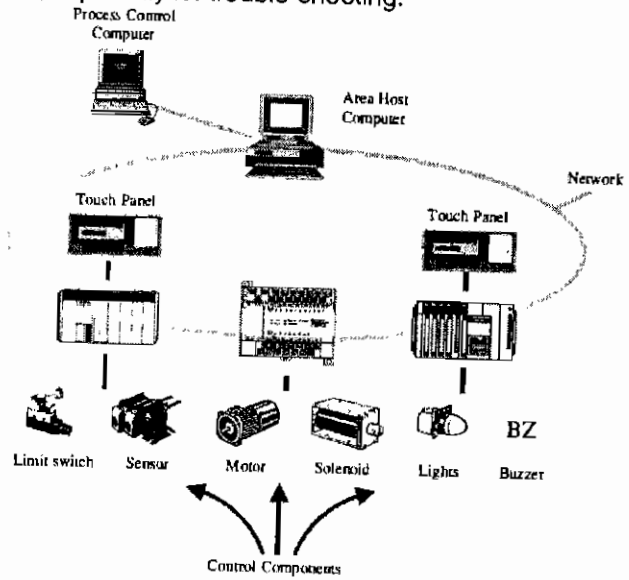
CHAPTER 4

PLC Control Diagram Fundamentals

Upon completion of this chapter, you will learn :

- Control Terminology
- Common Control Components and their Symbols in Control diagrams
- Control Transformers
- Control Wiring
- Reference Designators
- Boolean Logic
- Master Control Relays and Control Zones

Control is a unique area of engineering, that requires the knowledge of specific techniques. Although there are many similarities between control diagrams and electronic circuit diagrams, many of the component symbols and layout formats are different. It is important to understand the fundamentals, before drawing, developing and using control diagrams, especially for trouble-shooting.



CONTROL TERMINOLOGY :

For safety purposes, use of terms generally used in naming control circuits, labelling switch positions or describing modes of operation should be governed by rules, so that the words are defined and explicit, which means that they can have no other meaning.

RUN: A state in which the machine is performing the task, for which it is designed. This state can be reached by pressing the RUN button. It is not to be confused with the ON state, as it is possible for the machine to be ON, but not Running.

STOP : The state in which the machine may be ON, but is not RUNNING. Pressing the STOP button will make a running machine come to the STOP state. In case the machine has already Stopped, pressing the STOP button will have no action.

ON : The state in which Power supply is applied to the machine and its control circuits, which makes it 'ready to run'. This can also be called as the STAND-BY state. This switch is sometimes called as MAINS.

OFF : The state in which power supply is cut-off to the machine and its control circuits. The machine is neither Stand-by, nor should it be able to Start, even if any buttons are pressed on its control panel.

CYCLE : A mode of operation in which the machine can RUN only for one complete sequence of operation. Holding down the Cycle switch should not allow the machine to run for more than one cycle; to do this, the button must be physically released and operated again. This is sometimes called as Single Cycle.

INCH : A mode of operation which allows the operator to very slowly position the machine (in order to improve the accuracy of the operation). INCHing should allow positioning, but not cycling.

LATCH : A feature which allows a machine to continue running, even when the start push-button is released, provided no 'safeties' have activated.

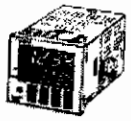
JOG : A feature, which allows the machine to be positioned or 'nudged', to allow for more accurate positioning, without allowing running of the machine.

Components in Control circuits

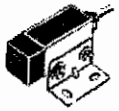
INPUT



Limit Switch



Timer



Photoelectric Switch



Rotary Encoder



Proximity Switch

SWITCHES :

Arguably the most common of control components, switches have many functions. They can be used as an input device, e.g. to start, operate or stop power to circuits. They can be used to select a mode, e.g. Selector switch. They can also be used to provide feed-back to the control system, e.g. limit switches, which let the system know when certain parameters are exceeded. They can even be remotely operated, e.g. solenoid. The most common type of switch is the Pushbutton type, which comes in two varieties : Momentary and Maintained.

The Momentary Switch is ON (when pressed), but is deactivated (e.g. by a spring), when released. Switches for starting / stopping of motors are of this type. The Motor remains 'ON' by means of Latching (a function that will be discussed later) and can be 'Un-latched' by another Momentary Stop switch.

The Maintained type, on the other hand, remains 'activated', when pressed once and is 'deactivated' by pressing a second time. These are sometimes called as **'Push-Pull'** switches. Many PCs have this type of switch, for the Power supply.

Switches are used to 'make' or 'break' Contacts. Contacts can be of two types :

- Normally Open (N/O or NO)
- Normally Closed (N/C or NC)

For a switch in the De-activated position, Normally Open (NO) contacts are non-conducting or 'open', while Normally Closed (NC) contacts are conducting or 'closed'.

In the Activated condition, the reverse is true : for Normally Open (NO) contacts, they will now be conducting or 'closed', while Normally Closed (NC) contacts will be non-conducting or 'open' (in the activated condition).

The same switch could be made Normally Open, Normally Closed or even a combination of the two (viz. one set of contacts are NO, whilst the other set of contacts are NC).

The Actuator, of a pushbutton, is the component that is pressed to activate / deactivate the contacts in the switch. There are of different types :

- Guarded or Shrouded
- Flush type
- Extended pushbutton

Guarded or Shrouded type. Here the pushbutton is recessed ¼ " to ½ " inside the sleeve, so that there is no danger of accidental operation, e.g. by something resting against it. An object smaller than the sleeve (such as a finger) can activate this, without danger of accidental push, such as by a hand resting on it or somebody leaning against it. Drawback could be the difficulty in operating this quickly, e.g. in an Emergency situation.

Flush-type is one where the pushbutton is aligned with the sleeve, but without being recessed. It is much easier to operate, but does not afford the same degree of protection as the Shrouded type. This type is used where it is desired to have the switch back-lighted (for easier viewing in dark / semi-dark conditions).

Extended Pushbutton type is one, where the actuator extends beyond the sleeve, thus making it much easier to operate, however without affording any protection from an accidental push. This is suitable for applications, where there is not much danger from an accidental push, or, where ease of use is more important, e.g. as a direction changing arrangement, wherein a shrouded type may not be ergonomic.

These three types of switches are not used for emergency situations, nor for operations that may occur numerously (e.g. hundred of times

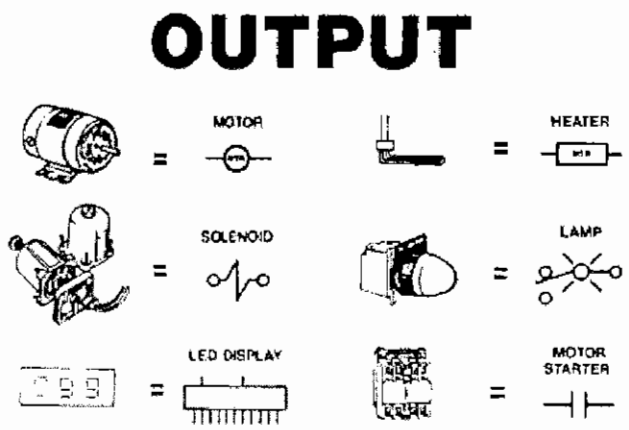
per day). For these applications, the Mushroom head or Palm head type is used. Palm type switches could be Momentary pushbuttons, frequently used in pairs, separated by a distance of 24". This arrangement ensures, that the machine can only be switched ON if the operator uses both hands, thus ensuring that both hands are clear of the machinery (also called as the 'two-handed palm' operation).

Another type of Emergency Stop **(E-STOP)** switch is the **Détente** type. When pushed to Stop, it remains 'in' and needs to be manually Reset, to return to the normal position (e.g. by twisting and releasing it). This ensures that the machine cannot accidentally re-start and create a dangerous situation.

Mushroom head switches have a special schematic symbol. The standard pushbutton symbol has been modified to a curved line on the top.

Selector switches : A Selector switch is also known as a rotary switch. The standard momentary pushbutton symbol has been modified to have a lever added to the top. The direction of the lever shows which side the contacts are 'made' or 'broken'. The selector switch can also be used to make two different sets of contacts.

Limit switches : Normally, these are not operator accessible, meaning that they are usually operated by moving parts, e.g. two different positions of an Actuator, viz. Extended and Retracted positions. They could also be magnetically activated or even by a light beam (photo-cell).



INDICATOR LAMPS :

Control panels are equipped with colour coded Indicator lamps, that tell the operator whether power supply is ON, whether the motor is running and if there is any alarm condition. Red colour is reserved for alarm conditions, while Green is generally used for normal running condition. Amber colour could be used to indicate either a stand-by condition, a paused condition or a situation that is important but not necessarily alarm condition. Usually an Alarm condition is high-lighted by a flashing red lamp and / or siren to attract the operator's attention to an emergency.

RELAYS :

Traditionally, a Relay is an electro-magnetic switch, composed of one or more pairs of Contacts. Fixed Contacts are mounted to the Relay's frame on electrical insulators, while Moving Contacts are operated, to make or break circuits, when the Relay is energised / de-energised, via electro-magnetic coils and springs.

Contacts are named according to their state in the De-energised condition : Normally Closed (NC) are Closed or 'made' (conducting) in the deactivated condition, whilst Normally Open (NC) contacts are Open or 'broken' (non-conducting) in the deactivated condition.

Nowadays, Relays can also be implemented in Software (viz. PLC ladder diagrams), where there are no 'physical' Contacts or Coils, but the switching effect is the same, on the rungs of the circuit. Even though 'software' relays exist, we still need physical relays, at the Output stage, to actually carry out the switching action, when large currents / voltages are involved, since PLCs have a limited voltage / current capacity.

Symbols, in control circuits, are sometimes similar to those of electronic components, which can be misleading. The symbol for a Normally Open Contact, for example, is two parallel bars, which could be interpreted as a Capacitor (in an Electronic circuit diagram.) The Coil symbol has two connection lines, which represent the wire connections to the actual relay coil.

Relays are available in an extremely wide range of sizes, with differing current and voltage values. Reed type 14 pin DIP Integrated circuits (ICs) are capable of switching with very small currents, of a fraction of an Ampere (milli amperes).

Large current handling Relays (e.g. for switching Motors) are called as **Contactors.**

TIME DELAY RELAYS OR TIMERS :

Relays are made with a built-in time delay device. These are designated as TDR or Time Delay Relay and have the same symbol as a conventional relay. The relay would physically have a knob, that allows users to set the time. There are two basic types :

- **Timer ON Delay (TON,** which is pronounced 'Tee ON')
- **Timer OFF Delay (TOF,** which is pronounced 'Tee OFF'.)

The time delay, called Preset, lets the user adjust the switching control action of the Contacts of the relay. So that, for a Timer ON Delay (TON), there is a fixed time delay after the Coil gets energised, for the Contacts to be 'made' (for NO) or 'broken' (for NC).

The amount of time that the Coil is energised does not affect the operation, except in the case of the Coil getting de-energised, before the TON activates the Contacts, in which case the switching action never happens. The moment the Coil is energised, the motor-driven, mechanical or electronic Timer starts 'ticking'. For software timers, the time value resolution, for each 'tick' (measured in milli seconds, seconds or minutes) can also be adjusted, besides the Time Delay value.

Timer ON Delay (TON) is a useful feature, if certain events need to be delayed, as for example, Purging of a Boiler furnace. Switching ON the Forced Draught Fan starts the Purge, after which the Ignition sequence may be started. Here, a Timer ON Delay can be used to set the Purge time. It can also be used for starting Motors, wherein a switching action is required only after the Motors picks up speed (Star - Delta Starters).

Timer OFF Delay (TOF) is the reverse situation, wherein energising the Coil immediately activates the switching (of the Contacts), but on de-energising the Coil, the switching action is **delayed.**

These Timers are useful in case certain actions need to be continued, even after switching off the Coil. E.g. running of the cooling fan for a certain time interval, even after the device is switched off (Projector lamp) for cooling down. Or keeping the light on, for a few seconds after the switch is put off, to enable the operator to leave the space.

CONTROL TRANSFORMERS :

For safety reasons, control circuits are necessarily low voltage circuits, because the components, like switches, lights and relays must be handled by operators / maintenance personnel on a regular basis. Changing of a bulb, for example, could prove to be hazardous, if high AC voltages were to be used, as even accidental contact with the metal base would generate dangerous (even lethal) voltages and currents. Available voltage being from the bus bars, control voltage is thus 'stepped-down', by means of Control Transformers.

Control circuits are also protected by Fuses, which isolate the circuit in case of a 'Short'. Fuses are usually of the 'slo-blow' type, to prevent unnecessary tripping, during transient fluctuations, e.g. when power is switched On. Most control transformers have a fuse block directly mounted on them.

CONTROL WIRING :

Control diagrams are drawn in a standard format, called as the Ladder diagram. All wires are numbered in the diagram, so the corresponding wires will have Wire Markers, to help identify them during trouble-shooting or modification. It is not necessary to use more than one wire colour (except for Earth or Ground), since each wire has a marker, which reduces the costs.

The wire generally used, is of the type MTW (machine tool wire), which has a single PVC insulation jacket, is less expensive, more flexible and thus easier to route, bundle and pull through conduits. This type, however, is not resistant to fuel oil / hydraulic fluid. For wires that may be exposed to oil, THHN (Thermoplastic Heat-resistant Nylon-coated) wire is used, which has a transparent, oil resistant, nylon coating on the outside of the insulation. This, however, not only makes it more expensive, but also increases the thickness and reduces its flexibility, so that it becomes more difficult to use in cramped or restricted spaces, as the larger diameter reduces the number of wires, that can be pulled through a conduit. The wire itself is AWG 14 or AWG 16 stranded copper wire, which is tough, flexible, easy to install and can better withstand the constant vibration experienced in machinery spaces.

REFERENCE DESIGNATORS

For all control diagrams, every component is given a Reference

Designator. This is a label assigned to the component, to assist in identification and location, that is used in the control diagram, the parts list, the schematic diagram and sometimes stamped on the component itself. The Reference Designator consists of an alphabetical prefix followed by the number of the component. The prefix identifies the type (of component), while the number identifies the specific part or component.

Thus, a Relay's Coil and Contacts each have a reference designator label above or below the symbol. This label identifies the component within the diagram, e.g. CR 1 is the Coil of the first relay. CR stands for 'Control Relay' and the number 1 refers to the particular component, each unit having a unique number to identify it.

Standard Prefixes :

T	Transformer	**CR**	Control Relay
LS	Limit Switch	**PB**	Pushbutton
S	Switch	**SS**	Selector Switch
TDR	Time Delay Relay	**M**	Motor relay
L	Indicator Lamp	**F**	Fuse
CB	Circuit Breaker	**OL**	Overload contact or relay

It is generally a common courtesy, to state the last used Reference Designator in a Diagram, e.g. CR 17, F 4, PB 8.

Any further additions / modifications can easily 'pick up' the number to be used from there, so as to avoid having an error, by either skipping a number or having a duplication of the same number.

Similarly, any ***deletions / modifications*** should be clearly noted, so that anyone using the Diagram does not waste time searching for components that no longer exist.

BOOLEAN LOGIC

Relays, in a control circuit, perform control operations, which are logical functions. Such operations can be commonly grouped together as : AND, OR, NOT, NAND, NOR etc.

AND : Where more than one switch is connected in series, the circuit can only be complete if ***all*** the switches, connected in series, are ON.

Thus, we can call this as an AND function, i.e. if Switch 1 is On **AND** Switch 2 is On **AND** Switch 3 is On ...then the lamp will be lit.

By convention, switches (inputs) are shown on the left and the lamp (output) is shown on the right of the rung.

OR : Instead of connecting the switches in series, if we were to connect them in parallel, then the logic would change. Thus, the lamp (output) would be On, if Switch 1 is On **OR** Switch 2 is On **OR** Switch 3 is On...

NOT : This is an invert function. Whatever the signal, a NOT logic would cause the output to be inverted, i.e. if Input was On, then Output would be Off, or if Input was Off, then Output would be On.

We can combine the earlier logical operations as :

AND + NOT = NAND

OR + NOT = NOR

Thus, if the Truth Table for AND was created, we could easily get the Truth Table for NAND by inverting all the Outputs of the AND Truth Table. Similarly, if the Truth Table for OR was created, we could get the Truth Table for NOR by inverting all the Outputs of the OR Truth Table.

Similarly, we can create additional functions AND OR, OR AND

Suppose we take the basic series circuit of AND, to combine with a parallel OR circuit, then we would get the AND OR function.

The opposite of this circuit is the OR AND circuit, where the parallel OR circuit is followed by the series AND circuit, however the result of this is somewhat different.

Boolean Logic is not only used for conventional Relay circuits (using hardwired Relays) but also for PLC / SCADA systems using 'software' relays, timers, markers, counters etc.

MASTER CONTROL RELAYS (MCR) AND CONTROL ZONES

It is often required to enable / disable sections (of the control system), without isolating or cutting-off power to the entire system. Also, if an Engineer requires to test / carry-out calibration on certain sections, without getting annoying alarms, there must be a capability to defeat certain sections of the system, without compromising the safety of the remainder of the system.

A Master Control Relay can be used to enable / disable certain rungs of the circuit, without shutting-down the entire plant. For convenience, the control circuit may be divided into pre-determined Control Zones, such as in a Fire Alarm system, so that entire sections can be isolated and shut-down easily, without affecting the remainder of the system. It goes without saying, that shutting-down of the section should be subject to usual scrutiny / checks by the Duty Engineer, such that safety is not compromised in the manual mode of operation.

CHAPTER 5

PROGRAMMABLE LOGIC CONTROLLERS

Upon completion of this chapter, you will learn :

- The advantages of Programmable Logic Controllers.
- Control diagrams, more commonly called Ladder Logic diagrams, which include Rungs, Rails, Contacts, Coils and other components.
- How to draw simple Ladder Logic diagrams and understand the logic.
- Differences between Ladder Logic diagrams and Electronic circuit diagrams, including symbols commonly used.
- Normally Open (NO) and Normally Closed (NC) contacts.
- Creating a simple Ladder Logic diagram.
- Latch instructions. Math instructions.
- DC Inputs / AC Inputs.
- Relay Outputs.
- Transistor Outputs.
- Communication

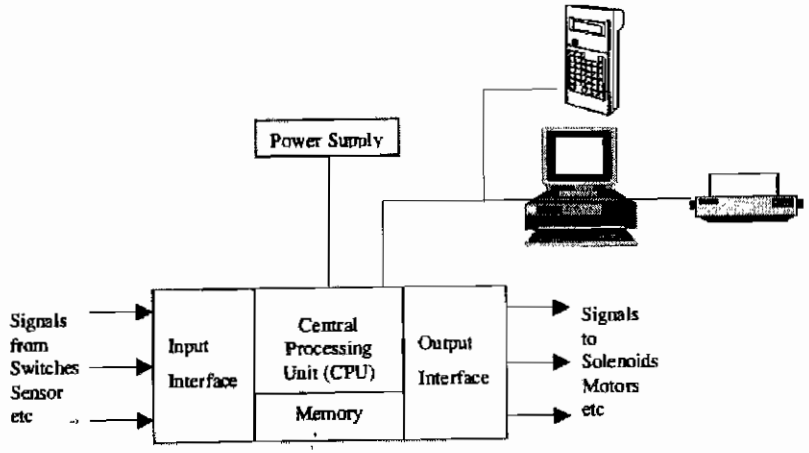
There were numerous limitations in older mechanical control systems, viz.

- 'wear-and-tear' of moving parts
- need to constantly adjust the control system, to take care of the above
- limited range / scope of control actions
- slow speed of action
- limited ability to include complex calculations in the process, to 'fine tune' the control, so as to significantly reduce error
- bulky components
- high costs of manufacture
- difficulty in trouble-shooting

- difficulty in modifying a system for use in differing applications
- no way of 'simulating' the control action, so as to test the system's ability to carryout the intended function satisfactorily
- no way of correcting / modifying control actions, so as to have a more effective control, using above simulation techniques
- no simple way of generating alarms, to alert the operator in case of failure of any component
- lack of redundancy, which reduces the fail-safe factor
- no security feature, to prevent unauthorised changes in control actions
- short TBOs (Time Between Overhauls), which increases maintenance

This forced Manufacturers to look for better and more cost effective ways of control. **Programmable Logic Controllers, or PLCs,** are the result of a logical development of the earlier generation of pneumatic and hydraulic control systems.

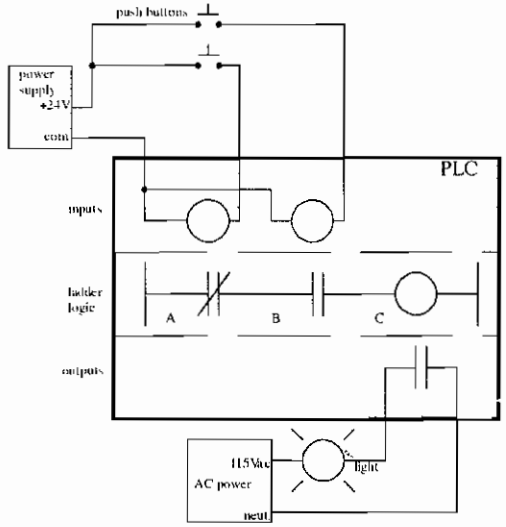
A **PLC** is a Device used for the control of machinery and processes. The heart of the PLC is a microprocessor (CPU). In many respects, PLCs are similar to PCs. They also have Input / Output (I/O) systems, Memory and the Power supply.



The main difference, between a Programmable Logic Controller (or PLC) and a personal computer (or PC), is the Input/Output arrangement (I/ O). These sub-systems connect the PLC to actual Sensors and Transducers. PLCs directly read Limit switches, Temperature indicators and the positions of complex positioning systems. On the Output or Actuator side, PLCs can operate Pneumatic Diaphragm valves through I/P converters, operate Relays to drive electric Motor starters, run Timers, Counters and even Solenoids.

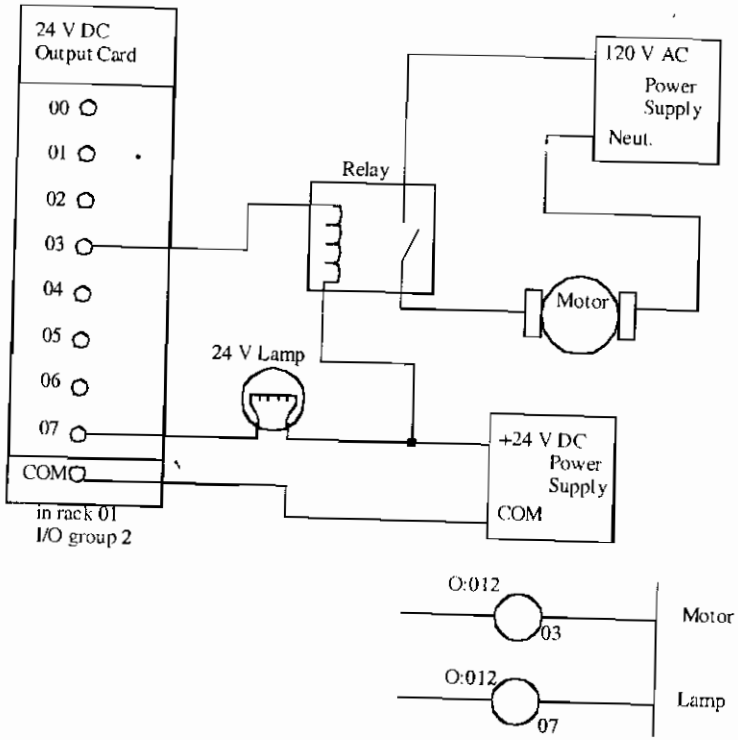
Note that, the last stage of control could still be pneumatic or hydraulic, either for high powered actuators (e.g. hydraulic rams used for steering, hatch cover opening / closing or even cranes / winches) or for safety in hazardous areas (e.g. pneumatic actuators used in pump rooms), but nevertheless the majority of control circuits are microprocessor based.

Input units can be of several different types, depending on the input signals expected from various sensors / transducers, e.g. input modules can accept discrete or analog signals of 4 - 20 mA or 0 - 10 V. Present day Controllers offer discrete signal inputs of both AC and DC voltages, 5 V TTL (Transistor - Transistor Logic) to 250 V (DC) and from 5 - 250 V (AC). Analog signals are converted by ADC (Analog to Digital Converter) circuits, so that the signal is presented as a digital input for the PLC. Push-buttons, limit switches etc. directly provide a digital input.



Outputs are similarly controlled, with Transistors used for TTL and even higher DC voltages, while Thyristors (Thermally-sensitive Resistors) are used for AC voltage outputs. Normally, Output currents are limited to 2 - 3 A, enough for Indicator lamps or similar applications.

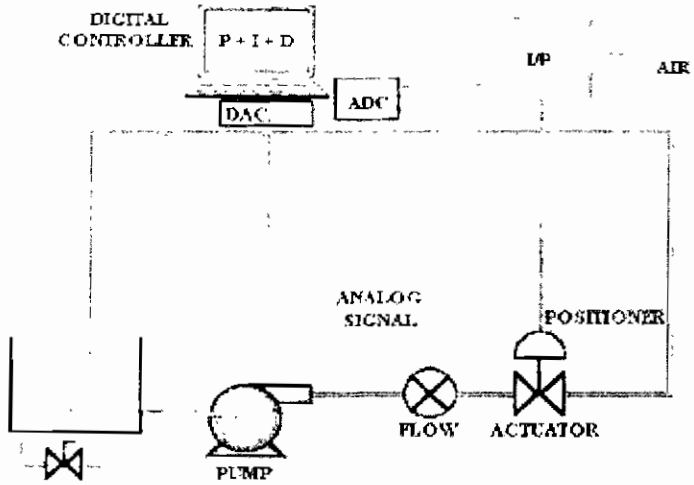
For higher current applications, e.g. Motor Starters, Contactors are triggered by a Coil connected from the Output circuit.



Note that, a DC output is sufficient to operate a Relay's Coil, whose Contacts can switch either AC or DC.

The Central Processing Unit or CPU is the control centre of the PLC, which performs the various operations, viz. scanning and execution of the program, updating Inputs and Outputs etc.

Consider the circuit below :



The parameter measured here is the Flow of liquid.

The Sensor / Transducer is a Flow transmitter, which typically gives an analog electrical output (4 - 20 mA or 0 - 10 V) signal.

This analog signal is first converted to a digital pulse, by the Analog-to-Digital (ADC) card, before it can be used by the Digital Controller.

The measured value (MV) is then compared with the Desired Value (DV), after which a suitable corrective action is generated (three term P + I + D control).

The output signal being digital, this is now reconverted to analog form in a Analog-to-Digital (ADC) card, which can now be used by a field device.

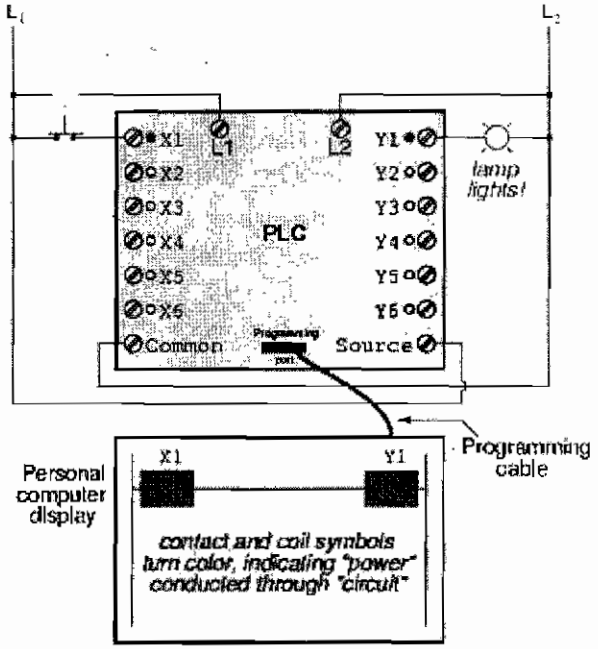
This analog output signal can be directly used, viz. to operate a Solenoid to open or shut a Valve, or start/stop the Pump's Motor (using a Relay), in case of Two Step or On-Off control.

A better method (as shown here), is to reconvert the electrical signal to a conventional pneumatic signal, using an I/P Converter. This can now be used to accurately position the Valve Positioner, so as to have a superior control. An electrical alternative is to use

a Stepper Motor, to position the control valve.

- The pneumatic signal now directly controls the usual Valve Positioner, which is basically a Diaphragm controlled valve, that increases or decreases the flow of the liquid, so as to correct the flow to the desired value (DV).

PLCs are the modern replacements for older 'hard-wired' control systems, such as mechanical, pneumatic, electrical or combination systems, that would need hundreds or even thousands of relays and cam timers. The program controls complex sequencing and is stored in battery-backed memory and/or EEPROMs (Electrically Erasable Programmable Read-Only Memory). The PLC is packaged and designed for extended temperature ranges, dirty or dusty conditions, immunity to electrical 'noise' and is mechanically more rugged and resistant to vibration and impact, than a personal computer.



Often, a single PLC can be programmed to replace thousands of relays. In Programmable controllers, software replaces actual wiring / re-wiring of hard-wired control panels. PLCs express decision making logic in simple 'ladder logic' diagrams, inspired from electrical circuit diagrams.

Drawing the Ladder Diagram

Step 1 :

We have to translate all of the items, that we are using, into symbols, that are used in Ladder logic. Unfortunately, we can't use conventional electrical terms like Switch, Relay, Bell, etc. We use PLC symbols like : Input, Output, Coil, Contact.

First we replace the Battery / Power Source with a symbol called Bus Bars. This symbol connects to all the components and is common to all Ladder Diagrams. The Bus bars are two vertical Bars, one on each side of the Ladder Diagram.



Usually, the left one is the + or Phase voltage and the right one is the Neutral or 'ground', in case of an AC circuit. For DC operation, the left Bar is the Positive, while the right one is the Negative.

The flow of logic is from Left to Right, while moving from Top to Down. Each horizontal line is called a Rung. Thus the PLC 'scans' each Rung from left to right, moving downwards.

Step 2 :

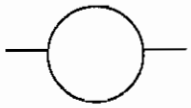
Next we give the Input a symbol. In this example, we have one 'physical' input (i.e. the signal coming from the Switch). We give the Input (the electro-magnetic Coil that the Switch will be connected to, in relay terminology) the symbol shown below.



Input (Contact) symbol

Step 3 :

Next we give the Output a symbol. In this example, we have one Output (the 'contacts' of the electro-magnetic relay, in relay terminology). This 'makes' the Output circuit. This symbol is called as the Output or Coil.



Output (Coil) symbol

The AC / DC supply to the PLC is an external supply (not concerning the logic of the PLC), so we don't put it in our Ladder Diagram. The PLC is only concerned about which Output it turns On / Off (and not what is 'physically' connected to it).

Step 4 :

The PLC must be able to recognise, where everything is located. In other words, we have to give all the connected devices an 'Address'.

Where is the Switch going to be physically connected to the PLC ? (viz. which is the Input, that the Switch is to be connected to ?) We number the inputs I 01, I 02, etc.

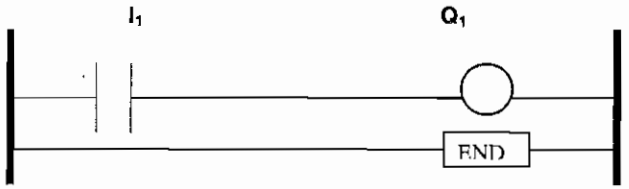
How about the Outputs ? We number the outputs Q 01, Q 02, etc.

The total number of available Inputs & Outputs, how many are Analog and how many are Digital - all this varies with the model / function of the PLC. We must mark, on the Ladder Diagram, the details of each input / output, giving each item a specific Address. Inputs :- I 01, I 02 etc. Outputs :- Q 01, Q 02 etc.

FINAL Step :

We have to make this a logical sequence of events. The program we're going to write. tells the PLC what to do, when certain events take place. This must take care of all eventualities, since the PLC can only work logically, i.e. it cannot work. if there are two different ways to do something. If the path is not properly defined. the output will change in an unpredictable way (this is called as the 'racing' condition.).

The logic should be straightforward. We have to tell the PLC what to do, when an Operator turns on an Input, I 01. E.g. When Input I 01 turns ON, the output Q 01 must also turn ON.

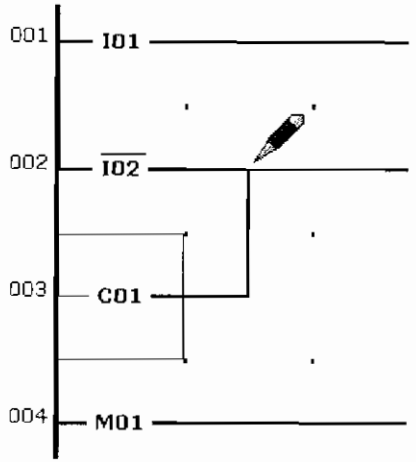


The Ladder Diagram above is the Program, that we have 'written' or rather, drawn. This may be directly done on the PLC if it has a screen / keypad, or it can also be created on Software loaded on a standard PC and transferred to the Device via a cable.

Adding Rungs, Drawing Connections

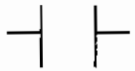
First position the cursor on a rung and add a new rung by **Edit, Insert Rung.**

The counter contact C 01, in rung 03, can be linked to the upper rung 02, by using the Draw tool :



Normally Open (NO) Contact

The Normally Open (NO) instruction is sometimes also called 'examine If On'(XIO). (as in, examine the Input, to see if it's physically On). The symbol for a load instruction is shown below.



Normally Open (NO) symbol

This is used, when an input signal is needed to be present, for it to turn 'On'. When the physical input is On, we can say that the instruction is True.

Normally Closed (NC) Contact

The Normally Closed (NC) contact is sometimes also called 'examine If Closed' (XIC) (as in, examine the Input, to see if it's physically closed). The symbol for a Normally Closed instruction is shown below.



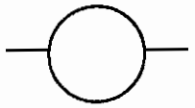
Normally Closed (NO) symbol

This is used when an input signal is not present, for it to turn On. When the physical input is off, we can say that the instruction is True. We examine the input , to see if it's physically closed.

Input	Normally Open	Normally Closed
0	False	True
1	True	False

OUTPUT

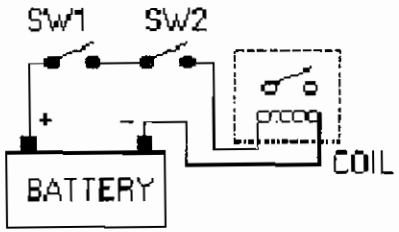
The output instruction is like a relay Coil. Its symbol looks as shown below.



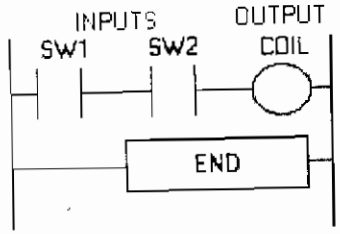
OUT (COIL) symbol

Creating a simple Ladder Logic Diagram :

Let's compare a simple Ladder logic diagram, with its 'physical' externally connected electro-magnetic relay circuit and SEE the differences.



In the above circuit, the coil will be energized when there is a closed loop between the + and - terminals of the Battery. We can simulate this same circuit with a Ladder diagram. A Ladder diagram consists of individual Rungs, just like on a real ladder. Each Rung must contain one or more Inputs and an Output. The first instruction on a rung must always be an Input instruction and the last instruction on a rung should always be an Output (or its equivalent).



Notice, in this simple one rung ladder diagram, we have recreated the external circuit above, with a Ladder diagram. Here we used the simple INPUT and OUTPUT instructions. Some manufacturers require, that every ladder diagram include an END instruction on the last rung.

Latch Instructions

Now that we understand how Inputs and Outputs are processed by the PLC, let's look at a variation. The Latching instruction lets us use a momentary switch, in such a way, that when we push one switch, the Output turns ON and this remains ON (even when we release the first switch), until we operate another (OFF) switch.

Consider the remote control for a TV. It has a button for ON and another button for OFF. When you push the ON button, the TV turns On and remains ON. You don't have to keep pressing the ON button, to keep the TV on. This is called a 'latching' instruction. Only when you push the OFF button, does the TV turn Off (Unlatch).

Math Instructions

Many times in our Planned Maintenance system, we must execute some type of mathematical formula on our data. It's a rare occurrence, when our data is actually exactly what we needed.

As an example, let's say we are carrying out repair jobs. We want to know how many more jobs, that we may have time to attempt, at any given time, so as to meet our daily quota, say 10 items. If say X is the number of jobs completed, at any given moment.

Therefore, we need : Items left for today = 10 - X.

To implement this formula, we obviously need some math capability. In general, **PLCs** include the following math functions :

Addition - The capability to add one piece of data to another. It is commonly called ADD.

Subtraction - The capability to subtract one piece of data from another. It is commonly called SUB.

Multiplication - The capability to multiply one piece of data by another. It is commonly called MUL.

Division - The capability to divide one piece of data from another. It is commonly called DIV.

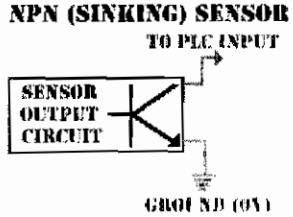
DC Inputs

Let's now take a look at how the input circuits of a PLC work. This will give us a better understanding of how we should wire them up. DC input modules are available that will work with 5, 12, 24, and 48 volts.

We'll first look at how the DC inputs work. DC input modules allow us to connect either PNP (sourcing) or NPN (sinking) transistor type devices to them. If we are using a regular switch (i.e. toggle or pushbutton, etc.), we don't have to worry about whether we wire it as NPN or PNP. We should note that most PLC manufacturers won't let us mix NPN and PNP devices on the same module. When we are using a Sensor (Photo-eye, Proximity, etc.) we do, however, have to worry about its output configuration. Always verify whether it's PNP or NPN. (Check with the manufacturer, when unsure)

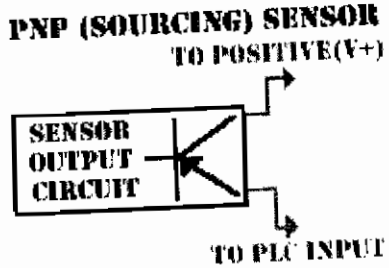
The difference between the two types is whether the load (in our case, the PLC is the load) is switched to ground or positive voltage. An NPN type sensor has the load switched to ground, whereas a PNP device has the load switched to positive voltage.

Below is what the outputs look like for NPN and PNP sensors.

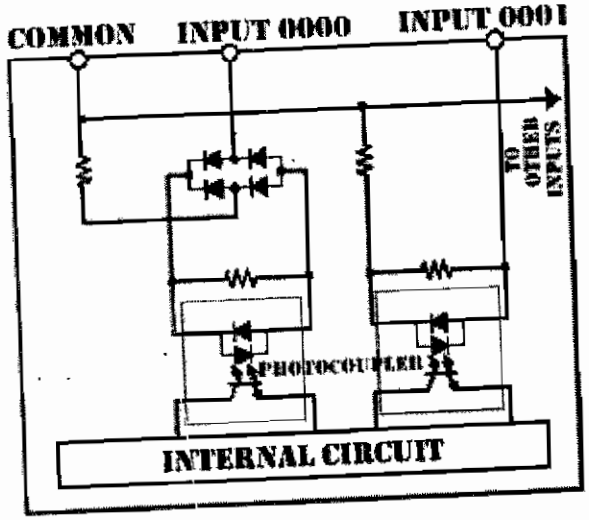


On the NPN sensor we connect one output to the PLCs input and the other output to the power supply ground. If the sensor is not powered from the same supply as the PLC, we should connect both grounds together. NPN sensors are most commonly used. Many engineers will say that PNP is better (i.e. safer) because the load is switched to ground.

On the PNP sensor we connect one output to positive voltage and the other output to the PLCs input. If the sensor is not powered from the same supply as the plc, we should connect both V+'s together. PNP sensors are most commonly used in Europe.



Inside the sensor, the Transistor is just acting as a Switch. The sensors internal circuit tells the output transistor to turn on when a target is present. The Transistor then closes the circuit between the 2 connections shown above. (V+ and PLC's input).



The only things accessible to the user are the terminals labelled COMMON, INPUT 0000, INPUT 0001, INPUT xxxx... The common terminal either gets connected to V+ or ground. Where it's connected depends upon the type of sensor used. When using an NPN sensor, this terminal is connected to V+. When using a PNP sensor, this terminal is connected to 0 V (ground).

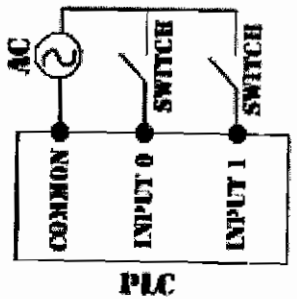
A common switch (i.e. limit switch, pushbutton, toggle, etc.) would be connected to the inputs in a similar fashion. One side of the switch would be connected directly to V+. The other end goes to the PLC's input terminal. This assumes the common terminal is connected to 0V (ground). If the common is connected to V+, then simply connect one end of the switch to 0V (ground) and the other end to the PLC's input terminal.

The photo-couplers are used to isolate the PLCs internal circuit from the inputs. This eliminates the chance of any electrical noise entering the internal circuitry. They work by converting the electrical input signal to light and then by converting the light back to an electrical signal to be processed by the internal circuit.

AC Inputs

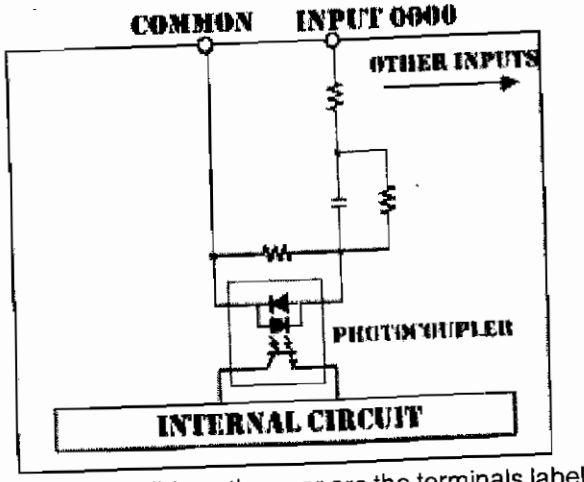
Now that we understand how DC inputs work, let's take a close look at AC inputs. An AC voltage is non-polarized. Put simply, this means that there is no positive or negative. However, AC voltage can be quite dangerous to work with. AC input modules are available that will work with 24, 48, 110, and 220 volts.

AC input modules are less common these days, than DC input modules, the reason being that today's sensors typically have Transistor outputs. A Transistor will not work with an AC voltage. Most commonly, the AC voltage is being switched through a limit switch or other switch type.



We connect an AC device to our input module as shown above. Commonly the AC 'hot' wire is connected to the switch while the "neutral" goes to the plc common. The ac ground (3rd wire where applicable) should be connected to the frame ground terminal of the plc.(not shown) As is true with dc, ac connections are typically colour coded so that the individual wiring the device knows which wire is which. This coding varies from country to country : it's commonly coded as brown (hot), blue (neutral) and green with a yellow stripe (3rd wire ground, where applicable).

The PLC's AC input module circuit looks like this:



The only things accessible to the user are the terminals labelled

COMMON,

INPUT 0000,

INPUT xxxx...

The common terminal gets connected to the neutral wire.

A digital switch (e.g. limit switch, pushbutton, toggle) would be connected to the input terminals directly. One side of the switch would be connected directly to INPUT XXX. The other end goes to the AC hot wire. This assumes the common terminal is connected to neutral. Always check the manufacturer's specifications before wiring, to be sure AND SAFE.

The photo-couplers are used to isolate the PLC's internal circuit from the Inputs. This eliminates the chance of any electrical noise entering the internal circuitry. They work by converting the electrical input signal to light and then by converting the light back to an electrical signal, to be processed by the internal circuit.

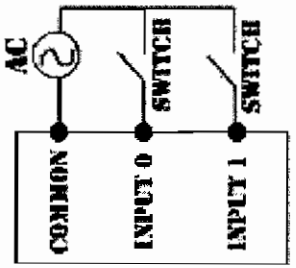
Relay Outputs

We now have a good understanding of how the Inputs are used. Next are the Output circuits.

One of the most common types of outputs available is the Relay output. A Relay can be used with both AC and DC loads. A 'load' is simply whatever is connected to the Outputs. We call it a Load or we say we are 'loading the output' with something. If we connect no load to the output (i.e. just connect it directly to a power supply), we would certainly damage the outputs. This would be similar to replacing the Light bulb, in a lamp socket, with a piece of wire ! If you did this, the Lamp would draw a tremendous amount of current from the Outlet (called 'shorting') and would certainly trip the circuit breaker / blow the fuse. (Please don't try it. Extremely dangerous !)

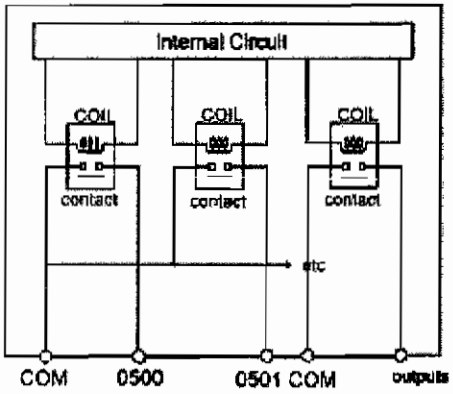
Some common forms of a 'Load' are a Solenoid, a Lamp, a Motor etc. These 'Loads' come in all sizes (electrical sizes, that is !). Always check the specifications of a Load, before connecting it to the PLC's output, to make sure that the maximum current, that it will take, is within the 'specs' of the PLC output. Some types of loads ('inductive loads') are very deceiving. These have a tendency to deliver a 'back current', when they turn 'On'. This 'back current' is in the form of a voltage 'Spike', coming through the system, which can cause damage. Take care ! An example (of an inductive load) is an Air conditioner. Have you noticed, how the lights dim for a second or two, when the A.C. Compressor 'kicks-in'. This is because, when the Air Conditioning Compressor switches On, it draws a very heavy 'starting' current. After this initial 'surge', it subsequently needs much less current, during running, so the lights go back to normal. This could be dangerous, to the PLCs Output relays. As a rough estimate : This surge (or 'kick') is about 30 times the rated current of the load.

A Diode, Varistor or some other 'Snubber' circuit could be used, to help prevent any damage to the Relay.



PLC

Shown above is a typical method of connecting Outputs to the PLC's Relays. Although the diagram shows the output connected to an AC supply, DC can be used as well, since the PLC Output does not directly drive a Motor, but works through a Relay. A Relay is non-polarized and can switch either AC or DC. Here the common is connected to one end of the power supply and the other end of the supply is connected to the load. The other half of the load gets connected to the actual PLC output.



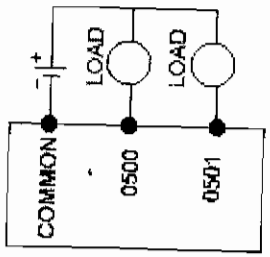
The Relay is internal to the PLC. The Circuit diagram looks like that shown above. When the Ladder diagram tells the Output to turn On, the PLC will internally apply a voltage to the Relay coil. This voltage will allow the designated Contacts to 'close'. Only when the Contacts 'close', does the current flow through the external circuit. When the Ladder diagram tells the PLC to turn 'Off' the Output, it will simply remove the voltage from the internal circuit, thereby enabling the output Contacts to 'release'. The load is than said to have an 'open circuit' (i.e. it will be OFF).

Transistor Outputs

The next type of output is a Transistor-type Output. It is important to note, that a Transistor can only switch a DC current. For this reason, it cannot be used with an AC voltage.

We can think of a Transistor as a 'solid-state' switch. A small (trigger) current, applied to the Transistor's 'Base', switches a much larger current (through its Output). The PLC applies a small current to the Transistor's base and the Transistor's Output 'closes'. When it's closed, the device connected to the PLC's Output will be turned ON. N.B. the above is a very simplified explanation of a Transistor. There are, of course, more details involved, but we don't need to get in 'too deep' here, as we just need a basic understanding, at an 'operator' level.

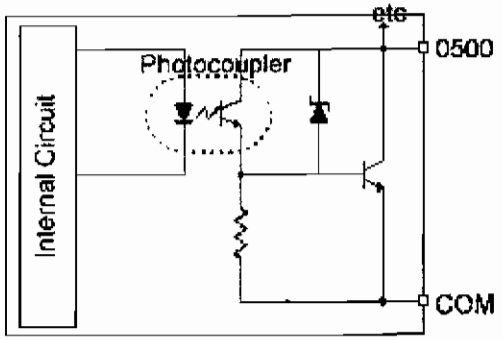
We should also keep in mind, that as we saw before with the input circuits, there are generally more than one type of transistor available. Typically a PLC will have either NPN or PNP type outputs. The 'physical' type of Transistor used also varies from manufacturer to manufacturer. Some of the common types available are BJT and MOSFET. A BJT type (Bipolar Junction Transistor) often has less switching capacity (i.e. it can switch less current) than a MOS-FET (Metal Oxide Semiconductor - 'Field Effect' Transistor) type. The BJT also has a slightly faster switching time. Once again, please check the output specifications of the particular PLC you are going to use, to make sure you never exceed the manufacturer's maximum switching current.



PLC

Shown above is how we connect output devices to the Transistor's Outputs. Note that this is an NPN type transistor. If it were a PNP type, the common terminal would be connected to V + , while V - would connect to one end of the load. Note that, since this is a DC type output, we must always observe proper polarity for the Output. One end of the load is connected directly to V+ , as shown above.

Let's take a moment and see what happens inside the Output circuit. Shown below is a typical output circuit diagram, for an NPN type output.



Notice that, as we saw with the Transistor type inputs, there is a Photocoupler isolating the 'real world' or external devices from the internal circuits. When the Ladder diagram calls for it, the internal circuit turns on the Photocoupler by applying a small voltage to the LED side of the Photocoupler. This makes the LED emit light and the receiving part of the Photocoupler will see it and allow current to flow.

This small current will turn on the base of the output transistor connected to output 0500. Therefore, whatever is connected between COM and 0500 will turn On. When the ladder tells 0500 to turn Off, the LED will stop emitting light and hence the output transistor connected between 0500 and COM will turn Off.

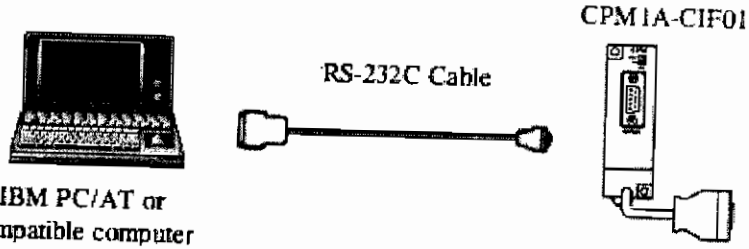
One other important thing to note is, that a Transistor cannot switch as large a load as a Relay. Check the manufacturer's specifications, to find the largest load it can safely switch. If the load current you need to switch exceeds the specification of the output, you can connect the PLC output to an external relay. Then connect the Relay to the large load. You may be thinking, "Why not just use a Relay in the first place"? The answer is, because a Relay is not always the correct choice for every output. A Transistor gives you the opportunity to use external Relays, as and when necessary.

In summary, a Transistor is fast, switches a small current, has a long lifetime and works with DC only.

Whereas a Relay is slow, can switch a large current, has a shorter lifetime and works with AC or DC. Select the appropriate one based upon your actual application needs.

Communication

The most popular methods, of communicating with external field devices, is by using the standard communications methods, viz. RS-232, USB.



All PLC communication systems have their roots in the old Telegraph. Early attempts to communicate electronically over long distances began as early as the late 1700's. In 1810, a German engineer (von Soemmering) was using a device with 26 wires (1 for each letter of the alphabet) attached to the bottom of an aquarium. When current passed through the wires, electrolytic action produced small bubbles. By choosing the appropriate wires to energize, he was able to send encoded messages 'via bubbles'.

In 1839, two Englishmen, Cooke and Wheatstone, had a 13 mile telegraph in use, by a British railroad. Their device had 5 wires, powering small electromagnets, which deflected low-mass needles. The biggest problems with these devices was the fact that they were parallel (required multiple wires). The first practical fully serial binary system gets credited to S.F.B. Morse. In the Morse code, characters are symbolized by dots and dashes (equivalents of binary 1's and 0's). Morse's first system actually had a needle contacting a rotating drum of paper, that made a continuous mark. By energizing an electromagnet, the needle would 'bounce' away from the paper creating a space. Very soon, telegraph operators noticed, that they didn't have to look at the paper, to read the code. They could interpret the code by the sound the needle made, when scratching the paper. So this device was replaced by a sounder that produced click sounds, instead of paper etchings.

Tele-printers came later, and today's serial communications devices are the result.

CHAPTER 6

Marine Equipment

Upon completion of this chapter, you will learn about :

- Electromagnetic Log
- Echo Sounder
- Mechanical Tachometer
- Electrical Tachometer
- Salinometer
- Dionic Water Purity Meter
- pH Meter
- Oxygen Analyser
- Dynamic Viscosity Meter
- Viscotherm

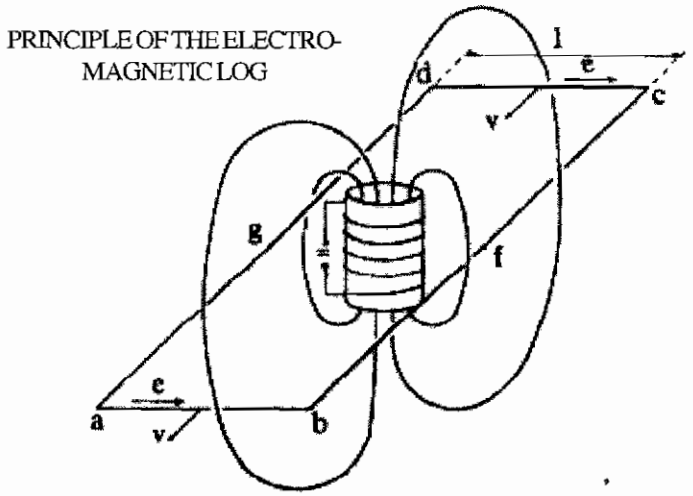
Marine Equipment

The working principle of any device is important, so that the operator can easily understand, not just how the device functions, but :

- how to run it at optimum parameters,
- know its limitations and plan to operate it accordingly
- with the maximum of efficiency
- utilising it to its maximum capability
- understanding the function of each and every component, so that a detailed analysis can be made, during the time of trouble-shooting, which enables the fault to be identified and rectified.

Thus, knowledge of the device can not only help the operator to understand how it works, but also help him to utilise it more effectively, in running the ship safely and efficiently. This reduces the dangers inherent in certain processes to an acceptable risk level and also gives the operator the confidence to run the ship optimally.

Fig. 39



Electromagetic log

The electromagnetic log is based upon the Faraday-Maxwell induction law. The principle of the log is shown in the sketch.

A direct current through the windings of a coil generates a magnetic field. Four conductors (**ab**, **bc**, **cd**, and **da**) are arranged in the form of a loop, around the coil.

If the conductors do not move, relative to the coil, they do not intersect the magnetic lines of force, and no voltage is induced in them.

The direction of the voltage, **e**, depends on the direction of the lines of force, and the direction of velocity of the conductors **ab** and **cd**.

The induced voltage is proportional to the velocity **v**. Should the velocity have an opposite direction, the direction of voltage will also change.

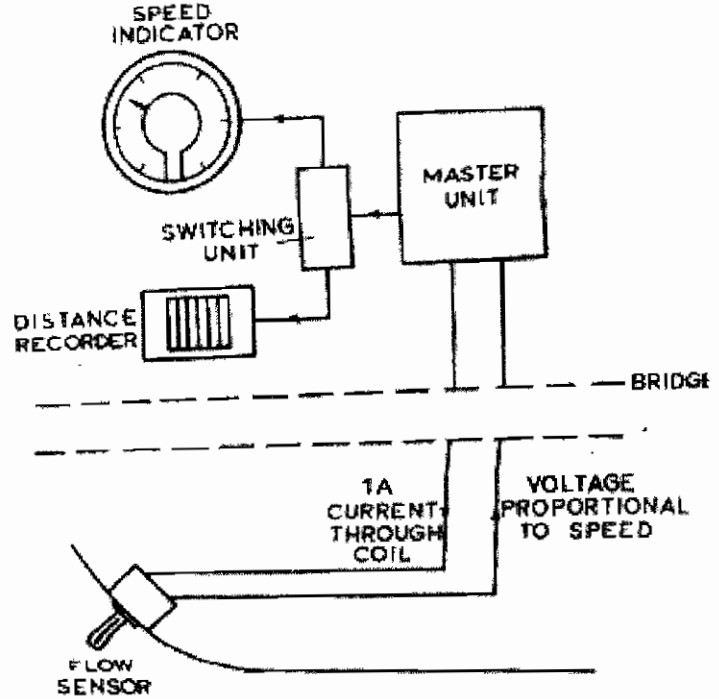
The coil of the electro-magenetic log is inserted in a water-tight flow sensor, which projects through the hull into the water below.

The sensor can be retracted into a tube, either locally or by remote control. The conductors forming the loop are, in reality, the surrounding water, which behaves as a conductor, since it is saline. The water moves relative to the coil, at the ship's velocity **v**.

The signal from the flow sensor gets amplified in the master unit. The velocity can be converted in the master unit, from analogue to digital form, if the readout is digital.

The **speed indicator signal** can be supplied to a special generator which produces a number of pulses per minute, proportional to the magnitude of the signal, and hence proportional to the **distance** travelled.

Fig. 40



ELECTRO-MAGNETIC LOG

Echo Sounder

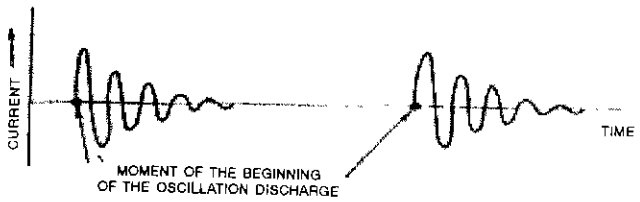
There are two types.

1. Recorder (Echo Graph)
2. Echo meter

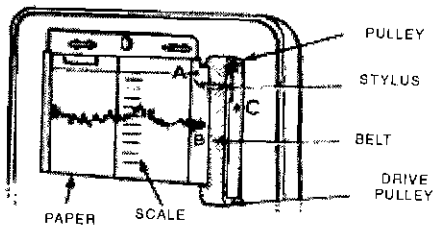
The Echo meter only indicates the instantaneous depth. Echo Graph records the depth on a sheet of paper. If the vessel is moving at a constant speed, the record shows an automatically recorded profile of the sea bed along the route of the ship.

A pulse is sent to the sea bed, and the time taken by the reflected wave is recorded, and this time is proportional to the distance from the sea-bed, i.e. the depth. There is sometimes an alarm provided, which gives audible / visual signal if the measured depth is less than a pre-determined limit.

Fig. 40(a)



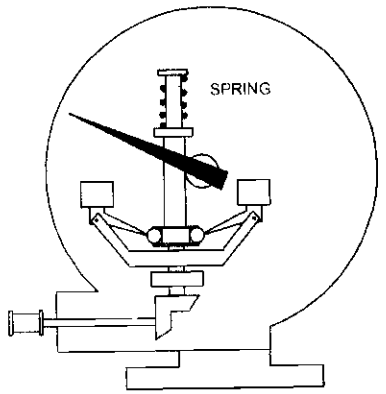
ECHO SOUNDER (RECORDER)



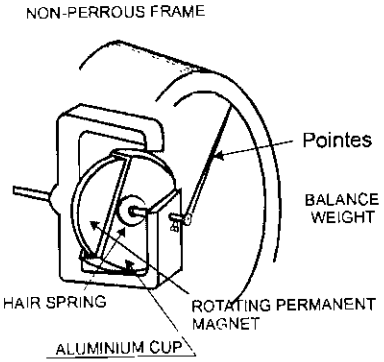
Tachometers

Fig shows a mechanical tachometer. A pair of fly-weights rotate about an axis, and are acted upon by centrifugal force. The movement, of the fly-weights, operates the pointer over the rpm scale. It moves in the same direction for both ahead and astern running.

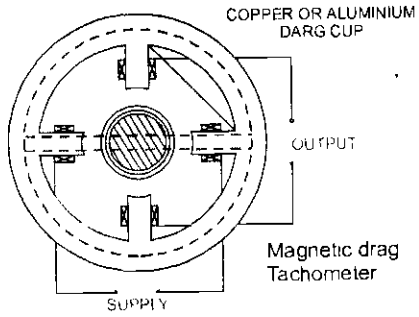
Fig 41(a)(b)(c)



(a) Tachometer (mechanical)



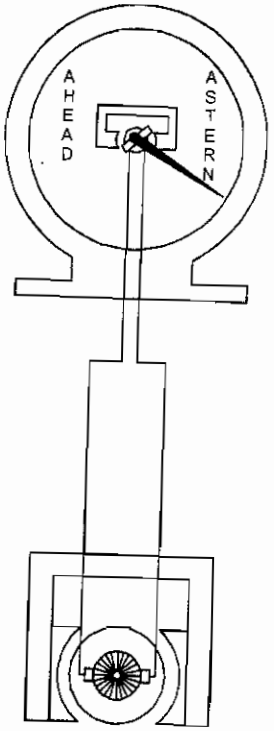
(b)Tachometer (magnetic drag)



(c) Tachometer (a.c)

Fig. 27 shows a magnetic drag tachometer. Here, the effect of the rotating magnetic field is to drag the aluminium cup around with it. The hair spring opposes this. Deflection of the pointer is proportional to the speed of rotation (of the magnet).

Fig. 42



Tachometer (electric d.c.)

Electric tachometer

These are either d.c. or a.c. tachogenerators, driven by the shaft whose speed is being measured.

The D.C. type is a generator with the flux provided by permanent magnets, using a wound armature with a commutator and designed to give an output voltage directly proportional to speed. The output is taken to a moving coil instrument calibrated in rpm. When no current is flowing, the pointer of this meter is positioned by two opposing springs, with zero at the centre of the scale. The direction of the shaft rotation controls the direction of current flow through the coil and hence the pointer shows the direction of shaft rotation as well as rpm (Fig. 28)

The A.C. type produces a voltage, with its magnitude proportional to speed (at constant frequency).

The stator carries two windings with their axes at 90° , one being the supply coil, and the other the pick-up or output coil. The rotor

consists of a thin aluminium or copper cup rotating about a soft iron core. When the cup is stationary, there is no output, as the coils are at 90°. If, when the cup is being rotated at constant speed, a. D.C. supply is provided to the input coil, e.m.f's will be induced by generator action, but the effect is to produce an interacting flux behaviour so that there is no output.

Acceleration of shaft rpm and thus cup rotational speed produces an output voltage in the coils.

Salinometer

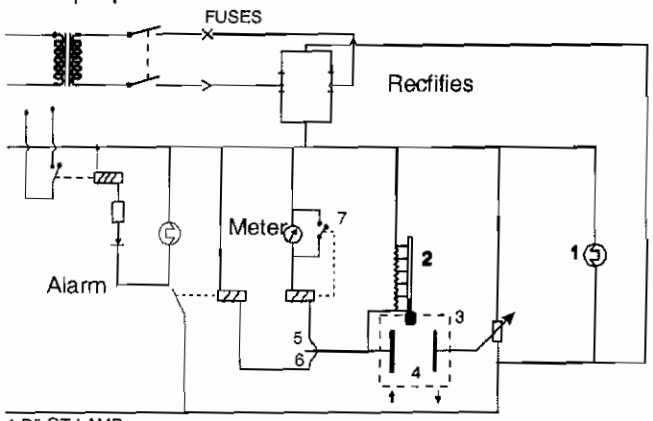
The basic principle is that pure water has a very high electrical resistance and therefore a very low conductivity. Pure water can be considered as non-conductive.

If sea-water is added to pure water, the conductivity increases. This instrument measures the conductivity of water, by passing a sample through a small cell, containing two platinum electrodes.

By measuring the current flow that occurs, due to the conductance effect of any salts present, the salinity can be easily indicated.

Readings are based on the assumption, that the conductivity is due to sodium chloride (NaCl) or Chlorine ions (Cl), but other salts such as magnesium chloride and calcium chloride will almost certainly be present. It is assumed for convenience that all conductivity is due to Sodium chloride. Although this is not very accurate, it is sufficiently for all practical purposes.

Fig. 43



1 PILOT LAMP
2 TEMPERATURE COMPENSATION
3 MEASURING CELL
4 ELECTRODES
5 READ
6 SIGNAL
7 METER PROTECTION

Salino meter

The electrical conductivity of water rises with temperature, about 2.2 percent per °C, and therefore a mercury in glass thermometer is used to adjust the current flow for this. As the mercury rises under an increase in temperature, it shorts out the resistances and leaks off an appropriate value of current commensurate with the temperature variation from the indicator and alarm circuit.

Dionic Water Purity Meter

Specific conductivity is measured in ohm/cm^3 and is the conductance of a column of mercury 1 cm^3 cross sectional area, 1 cm long and is referred to as a *Dionic* Unit.

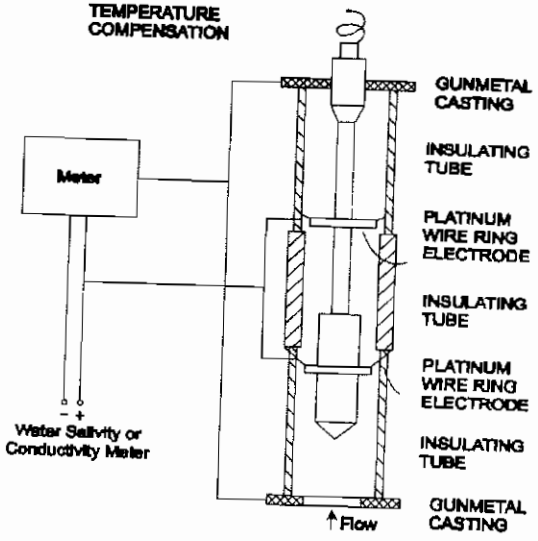
(Fig. 30). The electrical conductivity of water is dependent upon the dissolved impurities. The meter measures the conductivity of two columns of water, in parallel between the platinum and gunmetal collars.

The insulating plunger, operated by the bi-metal strip temperature compensator, automatically varies the water flow, to give compensation to 20°C.

Dissolved Carbon dioxide, can produce errors and should be removed by de-gasifiers.

It is important that the measuring electrodes are kept very clean and electrictrical connections tight.

Fig. 44



pH Meter

Measuring the pH value of any solution, such as the water in an engine cooling water system or a boiler feed water system, is to ascertain whether it is likely to cause corrosion, either by being acidic or even strongly alkaline.

By measuring the concentration of hydrogen ions actually dissociated in the water, we can compute the pH value. A pH value of 0 to 7 indicates an acid, while values of 7 to 14 indicate an alkali, 7 being neutral.

A continuous reading meter operates by measuring the development of an e.m.f., across a thin membrane of special types of glass. This e.m.f. is directly proportional to the difference in the hydrogen concentration, i.e. the pH, between the solutions separated by the membrane.

The Reference electrode has a solution of potassium chloride in a glass tube with a porous plug in the bottom. The reference electrode has a constant e.m.f., regardless of the hydrogen ion concentration around it.

The Measuring electrode produces an e.m.f., proportional to the difference, in the hydrogen ion concentration, between the solution having a constant pH and that of the liquid being checked.

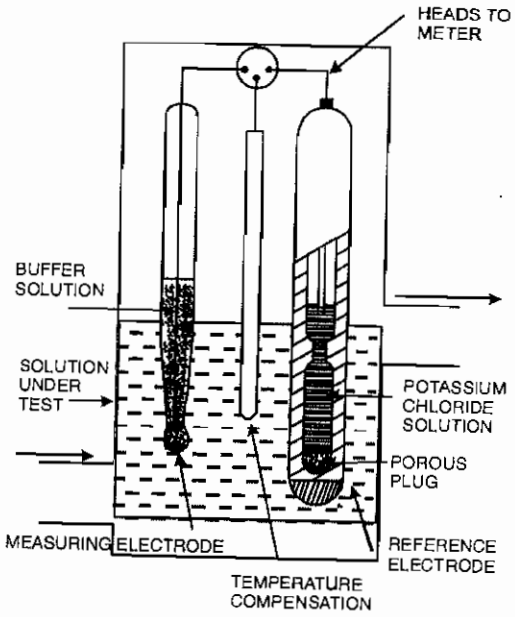
In this instrument, the membrane is a glass bulb. It contains a solution which has a constant pH value (known as a buffer solution - e.g. Potassium Chloride).

Connection is through a silver electrode coated with silver chloride, forming an electrical junction.

As temperature affects the disassociation of hydrogen, temperature compensation has to be provided.

Accuracy can be affected by dampness, strong electric fields, dirty electrodes, distorted electrodes, poor electrical connections and incorrect installation. If the glass electrodes dry out, they may not give a correct reading unless soaked in the buffer solution for about 24 hours.

Fig. 45



pH Meter

Oxygen analyser

Various types of meters may be used for measuring the oxygen content.

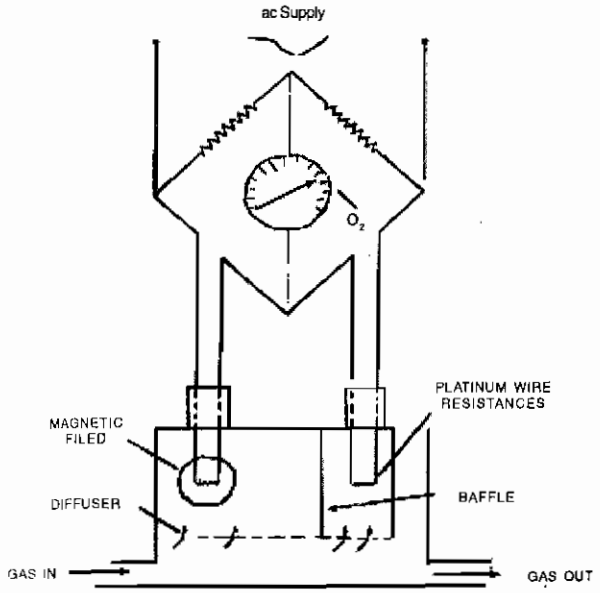
Fig. 31 shows a continuous reading type in which platinum wire elements are mounted in two chambers, one the measuring chamber and the other, the reference chamber.

Oxygen is paramagnetic, i.e. it is attracted to magnetic fields. Thus one filament has a magnetic field, while the reference filament has no field, and attracts only the air. The circuit forms part of a Wheatstone Bridge. The filtered and dried gas is drawn across the elements and the difference in thermal conductivity of the O_2, relative to air, causes a temperature difference in the wires.

This changes the wire resistance, and unbalances the Wheatstone's Bridge. The degree of unbalance is proportional to the O_2 content, and is used to indicate the percentage of O_2 present.

O_2 has a thermal conductivity of 2 .

Fig. 46



OXYGEN ANALYSER

Dynamic Viscosity meter

The fig. 33 shows a means of measuring the dynamic viscosity. The moving part is driven at a constant speed, and the static part tends to be dragged along with a force proportional to the dynamic viscosity of the fluid in the space between. The force on the static part is resisted by the shaft, and the resultant deflection can be read of the scale, which may be directly calibrated in the required units of viscosity.

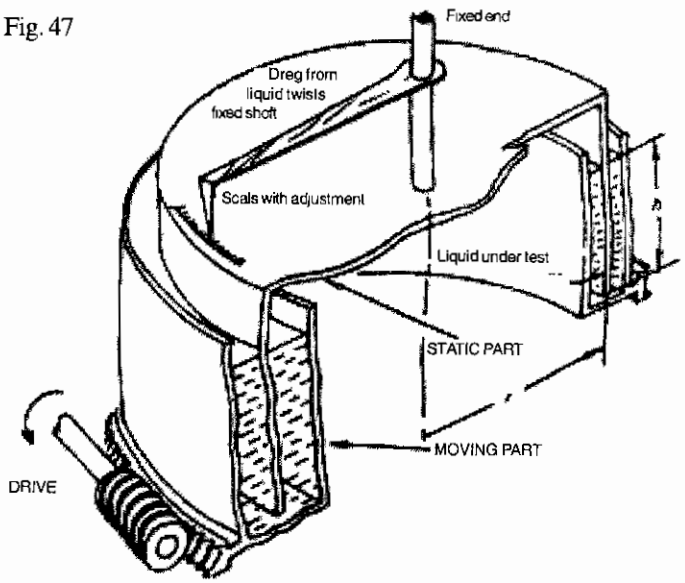
Viscotherm

This is used to measure the viscosity of oil fuel and adjust the heating, so as to maintain the viscosity within a pre-defined range.

As shown in Fig. 34, a constant quantity of oil is taken through the meter by a precision gear pump, driven at 40 rpm by an electric motor, and sent through a capillary tube. It flows under laminar (streamline) conditions, i.e. there are no cross currents or eddies.

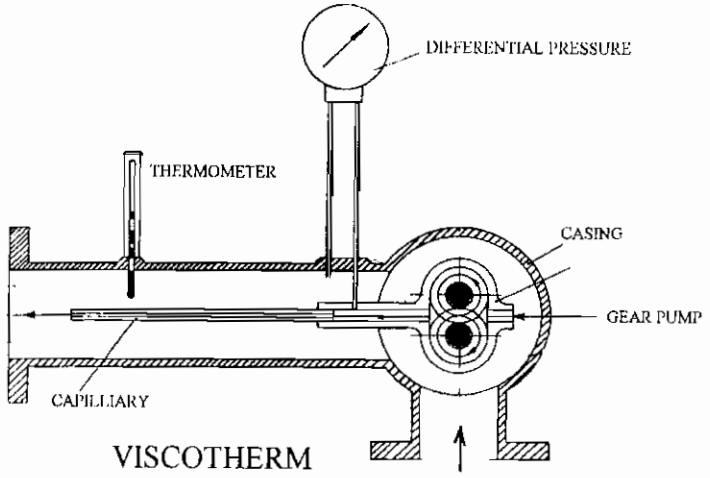
The pressure drop across the tube is measured by a Differential pressure cell and the signal obtained is directly proportional to the viscosity of the oil.

Fig. 47



This signal may then be used to operate a control valve, adjusting the steam supplied to the heater. The unit should be placed as near as possible to the heater discharge. All parts are of stainless steel, so as to provide corrosion resistance.

Fig. 48



CHAPTER 7

Pneumatic Control Systems

Upon completion of this chapter, you will learn about :

- Flapper and Nozzle
- Proportional action
- Split Range Control
- Air treatment
- Air Driers
- Pressure Reducing Valve
- Compressed Air System
- Intercooling
- Lubricator

Pneumatics

The term 'Pneuma' is derived from ancient Greek, meaning 'breath' or 'wind'. Thus, we use the term 'pneumatics' to mean the study or use of air as the control medium. Compressed air is one of the oldest forms of control known to man.

The first man to have engaged himself with pneumatics (i.e. the use of compressed air as a medium) was a Greek named KTESIBIOS. More than 2000 years ago, he built a compressed air impulse catapult. One of the first books, concerning the application of compressed air as energy, originated in the first century and describes devices which were driven by warm air.

Although the fundamentals of pneumatics rank amongst the earliest perceptions of mankind, it was not until the last century, that it's behaviour and fundamentals were researched systematically. Practical industrial application of pneumatics dates back to 1950.

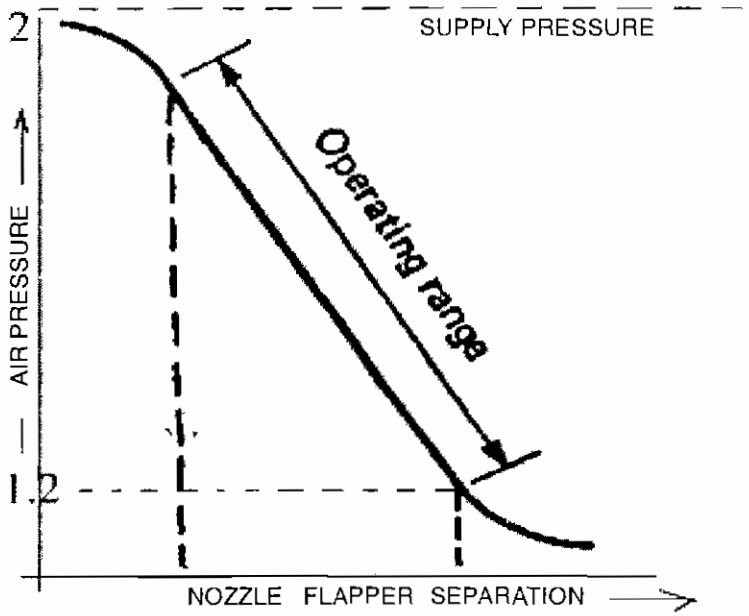
Pneumatics has been widely used on ships and is the preferred medium of control for hazardous areas, such as a Tanker's Pump room or the Cargo area. Air, as a medium, is freely available (unlike oil) and does not pollute the environment.

Flapper and Nozzle

The flapper and nozzle may be considered as one of the building blocks, on which pneumatic systems are based.

Air is fed to the nozzle at a constant pressure via a restrictor. The air leaks to the atmosphere at a rate depending upon the clearance between the flapper and the nozzle. It follows the curve shown in Fig.

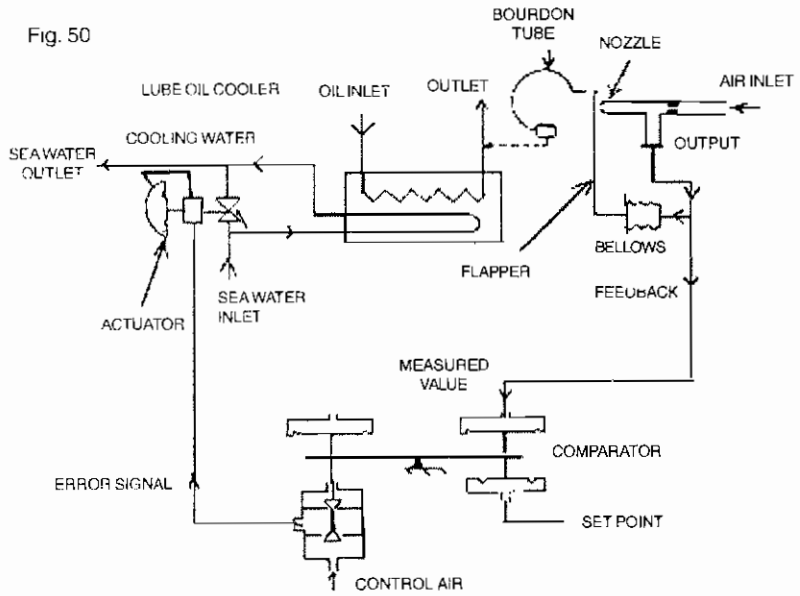
Fig. 49



For example, with the flapper shutting the nozzle P_2 (the back pressure) would be 1.4 bar. When the flapper is moved away 0.03 mm, P_2 is then about 1 bar. As the curve shows, the back pressure P_2 varies in a roughly linear fashion over the signal range, especially between the pressures of 1.2 bar and 1.5 bar. 'Linear' means that P_2 varies an equal amount each time the flapper moves an equal amount.

Feedback means passing back whole or part of the output signal to compare with a set value.

Fig. 50



Here the signal is fed back to the flapper through a bellows. Suppose the signal initially moving the flapper is provided by a bourdon tube at the end of a filled systems thermometer capillary tube.

Assume the temperature at the bulb increases by 2° C and this causes the bourdon tube tip to move the flapper towards the nozzle by 0.015 mm. In Fig. , with no feedback, this movement would occupy the whole linear range so that P_2 changes from 0.2 bar to 0.5 bar and this would be the output signal, the instrument thus having a range of 0.5-0.2 i.e. 0.3 bar for a 2°C temperature range.

If the flapper initially moves the same amount and the back pressure P_2 increases, but in doing so, it causes the bellows to expand, resetting the flapper by a small amount. The nozzle /flapper gap then changes, moving it back towards, but not exactly to, its original position, until a position of equilibrium is reached.

The actual resultant movement of the flapper from its original position may then be very small, say only 0.00025 mm and the output (back pressure) will have increase by a very small amount, say .05

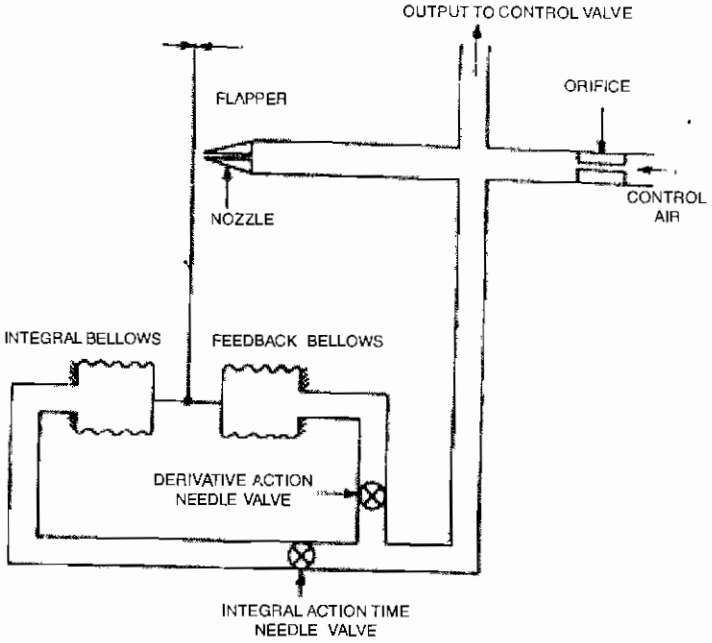
bar so that the output is now 0.25 bar. Thus with 2°C giving a pressure change of .05 bar, the temperature range that can be measured and transmitted has been increased to 12°C.

This has increased the Proportional band, and results in a wider band. The wider the band, the coarser the control. However, a wide band gives stability. The narrower the band, the more the accuracy.

The feedback is the measured value, that is sent to the Comparator, where this is compared to a set value or set point signal.

The Comparator thus gives an error signal to the actuator, which takes corrective action, based on the deviation that occurs. The actuator thus sends more water through the cooler or bypasses it, depending on whether the measured value is above or below the set point.

Fig. 51



THREE TERM CONTROL
PROPORTIONAL, INTEGRAL AND DERIVATIVE ACTION

Proportional action is defined as an action which occurs that is proportionate to the deviation; consequently, it only works if there is deviation, or, to simplify further, simple proportional control has an inherent problem - offset.

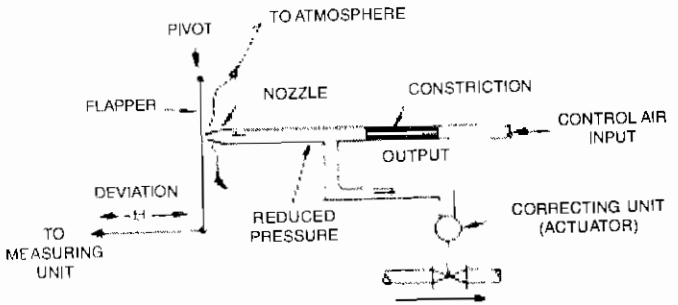
Offset is sustained deviation, which occurs when the measured value stabilises at some point other than the desired value. This could change, with a change in the load conditions, hence offset varies with the load.

This brings in the second term of control, viz. **Integral** action. The Integral action is such that it always opposes the offset. However, in trying to reach the desired value, it takes a lot of time. This is where the third term comes in - Derivative Action.

Derivative action is action, where the control is proportional to the rate of change of deviation, i.e. this is to do with the time factor. Whenever there is a sudden change in deviation, i.e. a quick response is required, Derivative action comes into the picture.

In simple pneumatic controllers, integral action is usually provided by means of the Integral bellows, which cancel out the feedback bellows, at a rate which is controlled by a time setting, called as the Integral action time. The Derivative action is carried out by a needle valve, in the line, which delays the filling of the feedback bellows, thus causing a reduction of the feedback (full opening of the valve) in case of sudden changes in deviation, as shown in the sketch.

Fig. 52



NOZZLE AND FLAPPER

Split Range Control

The output signal from a controller may be split into two or more branches to control two or more correcting units.

In a central air conditioning system, an automatic controller controls two valves to maintain the air temperature to the accomodation constant.

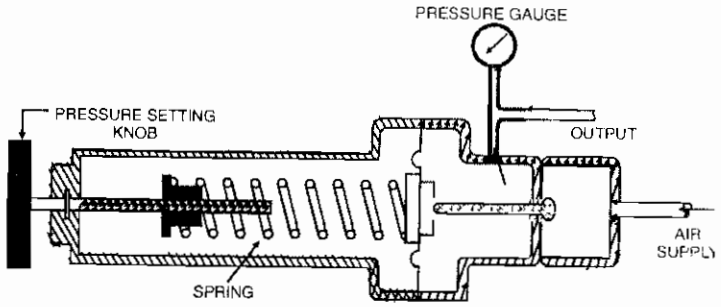
If the air temperature is above a certain value, say 20°C, then the refrigerant circuit will operate to bring down the temperature to the desired value, say 15°C.

If however, the air temperature should fall below 10°C, then the second circuit, i.e. the heating circuit, will now operate to bring up the temperature.

There is a certain range of values, over which neither heating nor cooling is done, here in this example from 10°C to 15°C. This is referred to as the Dead band, and is there to ensure no overlapping of both the systems, i.e. the heating and cooling should not be both working simultaneously, as this would be a waste of energy.

The Fig. 43 shows a means of remote adjustment of set point by generation of a pneumatic signal. This could also be used for remote operation of a valve. The output signal pressure acting on the diaphragm produces a force equal to the spring force set up by the manual control knob. Changing the spring pressure enables the operator to change the output signal.

Fig. 53



Remote Set Point Adjustment

Pneumatic system air treatment

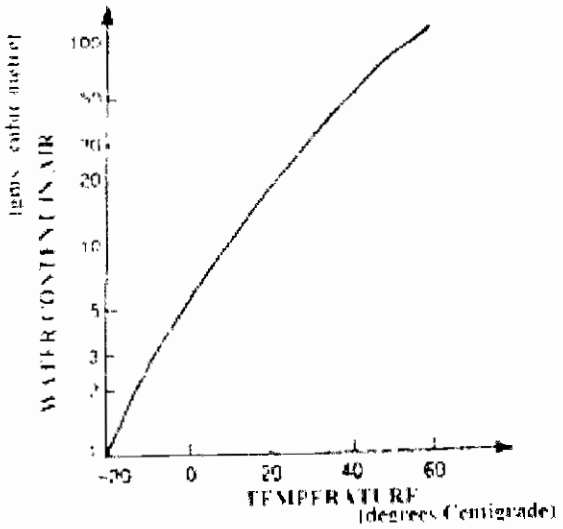
Air supply for the control equipment on a ship requires a high degree of solid particle filtration, oil and water removal. The presence of solid particles greater than 5 microns in size can cause blockage of orifices, nozzles, and damage to diaphragms and 'O' rings.

Oil may be found in droplet, gaseous and vapour forms, but whatever the form it can create problems by blocking orifices and nozzles, clogging up dryers and building up layers in the pipe lines, eventually blocking them. Moisture present in the air can cause blockage of orifices and nozzles if in droplet form, and cause corrosion, the products of which may cause blockages, and under low temperature conditions, such as a sudden pressure drop, or in low ambient temperatures, it can freeze in the pipe lines or actuator mechanisms.

Oil combines with water to form a sticky white emulsion, which can be observed while draining the air receiver. This is undesirable as it tends to attract dirt particles to form an abrasive mixture, that can cause sliding surfaces to seize, or can even cut rubber components like o-rings.

The compressor should preferably be of an oil free design, and may be reciprocating, rotary, screw type or even some variant of the rotary type, such as the liquid ring type. The supply pressure is usually 7 bar.

Fig 54



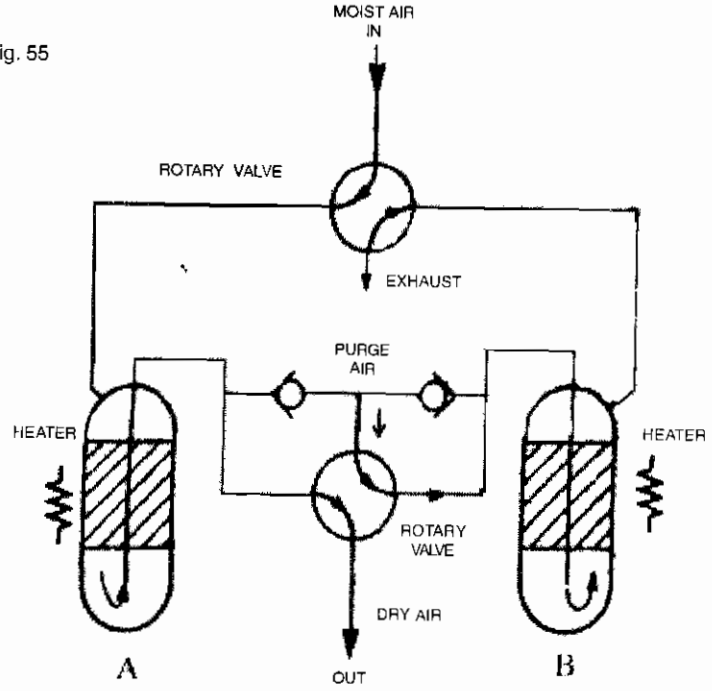
Air Dryers

Control air needs to be absolutely dry. Since atmospheric air always contains moisture, and this moisture can condense whenever the temperature falls below its Dew point, the air needs to be dried beforehand.

The driers used vary, and can be of the refrigerated type, or of the deliquescent type. Some driers use the adsorption principle.

Adsorption dryers work by use of an adsorption material which is granular, such as Silicon dioxide. Moisture collects on the surface of the adsorption material. This can further be removed by heating, so usually two columns are used. One is carrying out the drying, while the other is regenerated, by heating and passage of purge air to remove the moisture from the material. Every few hours, the columns are changed over.

Fig. 55

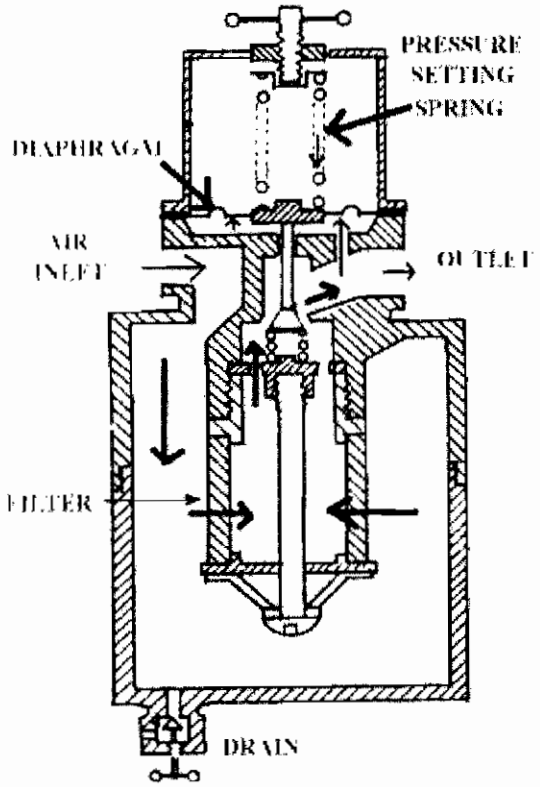


ADSORPTION DRIER

Pressure Reducing valve

The Fig. 46 shows an air line filter and reducing valve, in which the incoming air passes first through a fine mesh filter and then a reducing valve which drops the pressure from inlet P_1, which may be the air receiver pressure to a working control air pressure P_2, which will be lower than P_1 .

Fig. 56



PRESSURE REDUCING VALVE

Compressed air system

Most Pneumatic systems use compressed air as the operating medium, although some use nitrogen.

The slow response of any pneumatic system, as compared to a hydraulic system, means that large quantities of compressed air need to be stored, in an Air Receiver.

The volume of the Air Receiver is chosen, such that there is minimal deviation in pressure, arising from changes in loads. The compressor replaces the air used.

Deviations in compressed air pressure are smaller, and compressor control is easier, if a large Air Receiver is used to feed all the loads. On board ship, compressed air is produced at a central location, usually in the engine room, and the compressed air supply is distributed around the ship in a similar manner to electricity and water.

Since the Main engine is usually started by compressed air, the Starting system may be combined with the Pneumatic control system.

Behaviour of air subjected to changes in pressure, volume and temperature is governed by the general gas equation given below:

$$\frac{P\,V}{T} = \text{constant}$$

Inter-Cooling

The Air Compressor increases air pressure, by reducing its volume, with a resultant rise in temperature.

This rise in temperature causes the work done to be increased.

This is an unnecessary waste of energy, and can easily be reduced, by reducing the temperature of the hot air after the first stage of compression.

The reduction in temperature causes the resultant change in volume, which shifts the curve as shown in the figure.

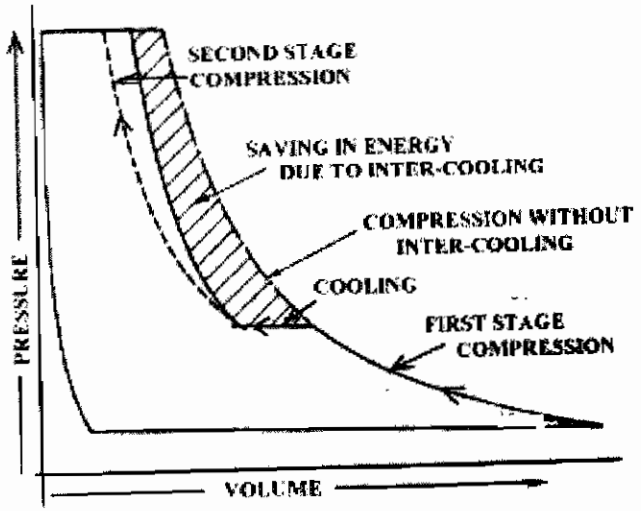
The second stage of compression thus starts on a different curve. In this way, the area under the curve, which is the work done on the air by the compressor, reduces. This improves the efficiency of compression.

A Pneumatic system therefore incorporates some method of removing this excess heat due to compression.

For small compressors, simple fins on the compressor provide the extended heat transfer surface for air cooling.

For larger systems, a separate cooler (usually employing sea water as the cooling medium) is used.

Fig. 57



The Need for Driers

Atmospheric air always contains water in the form of water vapour, the actual amount varying from day to day according to the humidity. The maximum amount of water vapour that can be held in a given volume of air is determined by the temperature. If the temperature falls, any excess water vapour condenses out as a liquid. The point at which this condensation occurs is called as the Dew Point.

This problem also occurs when compressed air is cooled, due to pressure drops in the system. Thus the resultant water droplets would cause fine orifices to ice up and corrosion to take place, which further causes control valves to jam and also corrosion to form in iron pipes and other components which are susceptible to corrosion. Any inter-cooler must, therefore, be followed by a water separator. Often, the Inter-coolers and the After-coolers are combined with water separators.

Dry cool air must be stored in the Air receiver. There is a pressure switch provided, which is used to start and stop the compressor as required, maintaining the required system pressure.

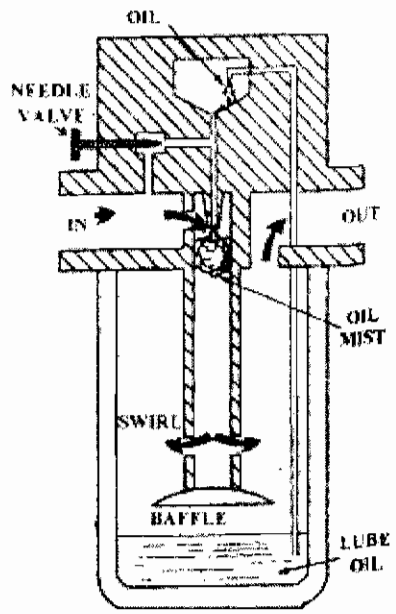
Compressed air is sometimes given a light oil mist, to reduce the chances of corrosion and to lubricate moving parts in actuators, valves, cylinders and so on. This oil mist cannot be added before the Air Receiver, as the mist would form oil droplets in the Receiver's relatively still air. The lubrication is usually provided locally, using a unit, which provides the lubricating mist along with further filtration and water removal. This forms the Secondary air treatment.

Lubricator

Any moving part requires some lubrication, if wear is to be minimised, heat due to friction is to be eliminated and chances of seizure reduced. In the case of pneumatic control valves and components, a carefully controlled amount of lube oil is often added to the compressed air locally, just before use, to lubricate moving parts. This oil is introduced as a fine mist, by a device called Lubricator. Note that the oil should only be added to thoroughly clean and dry air. If oil combines with water, a sticky white emulsion forms, which jams sliding surfaces and clogs orifices and nozzles.

The construction of a typical lubricator is shown here. As compressed air enters the lubricator, the venturi effect causes its velocity to be increased, accompanied by a local reduction in pressure, in the upper chamber. This pressure differential, between lower and upper chambers, causes a suction effect, that draws the lube oil up a riser tube. The lube oil emerges as a fine spray, mixed with the compressed air. The needle valve adjusts the pressure differential across the oil jet and hence controls the oil flow rate.

Fig. 58 **(Diagram)**



LUBRICATOR

CHAPTER 8

Control System Components

Upon completion of this chapter, you will learn about :

- Poppet valves
- Spool valves
- Rotary valves
- Actuators
- Pneumatic Valve Positioners
- Valve characteristics
- Equal percentage
- Quick opening
- Piston and Rotary Actuators
- Materials
- Boiler water level control

Components

Components that make up a control circuit have different functions. Most Control valves are of the Finite position type, viz. they are either open or shut. Trouble-shooting a fault in a control system can be a time-consuming job, if the functioning of components is not properly understood.

Besides malfunction of the component, there can be other causes of failure. Without doubt, dirt causes more components to fail than any other single cause. Dirt also includes foreign substances.

In a pneumatic system, dirt and foreign substances score the honed cylinder tubes, precision-finished valve liners and valve seats, ground and polished piston rods, valve stems, and other precision parts. Heat causes considerable trouble to the components of a hydraulic system. Heat may cause valve spools to stick, packings to deteriorate, oil to break-down, deposits to cling to the finished surfaces, excessive leakage and deformation of components.

Control Valves

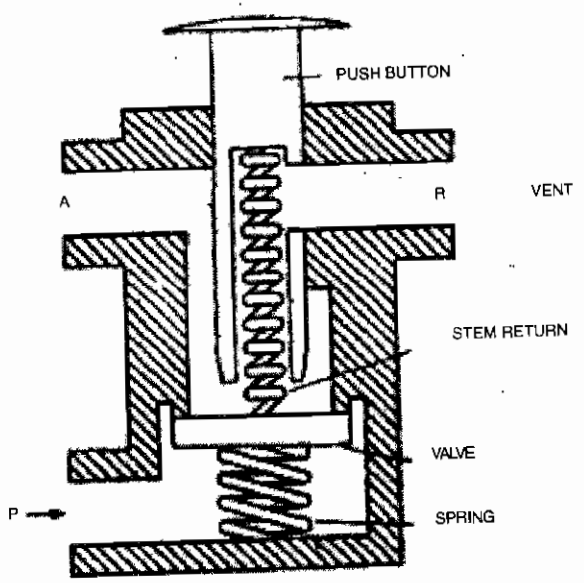
The most common types of valves used in control systems are called as the finite position type, i.e. they are either open or they are shut. (Infinite position valves can take up a variety of positions between open and full shut).

The valves are of several types :

Poppet valves

These are the simplest type. They are usually push button operated, and require a considerable force to operate, if the system pressure is high, or the valve area is considerable. Hence these are usually used in low pressure pneumatic systems (not in hydraulic systems, which have large pressures).

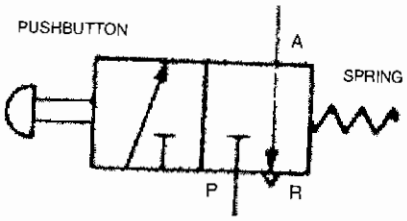
Fig. 59



POPPET VALVE

Poppet valves are simple, cheap and fairly robust. They are particularly suitable as a means of manual control, in the form of push-button operated valves.

Fig.60

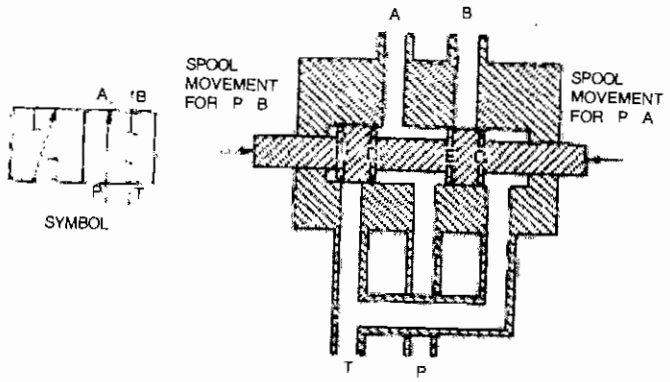


SYMBOL OF A POPPET VALVE

Spool valve

Spool valves consist of spools, which move within a valve body. There are raised areas, called lands, which are used to open or block access to ports. Spool valves can be balanced very easily, since the system pressure can be made to act on opposing faces. Thus there is no requirement of high operating force, as in poppet valves. Consequently, spool valves find use in pneumatic applications, such as Sulzer's air distributer, as well as in hydraulic applications.

Fig.61

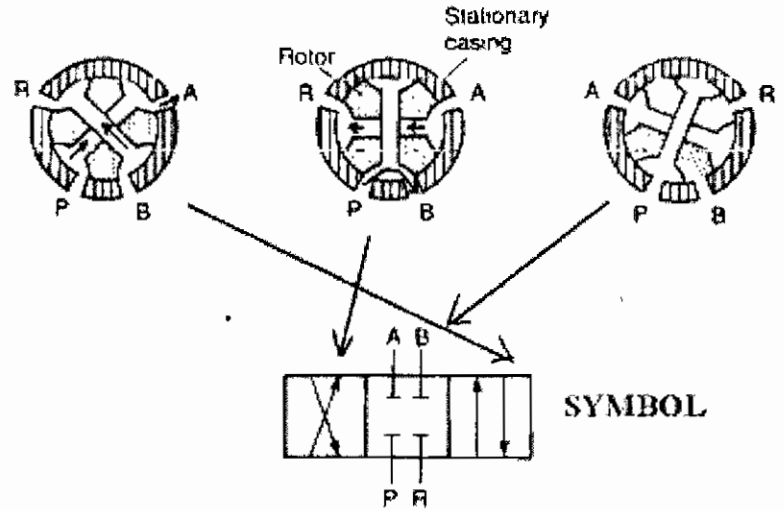


SPOOL VALVE

Rotary valve

The third type of valve is the rotary valve. This consists of a rotating spool, which alligns with ports in the casing, that are connected or blocked by means of passages. These are simple, compact and basically used in pneumatic circuits.

Fig. 62



ROTARY VALVE

Valve operation

Valves can be operated by a variety of ways :
Push button
Solenoid
Lever
Pilot pressure
Roller
Usually, springs are provided to return it to the normal position.

Actuators

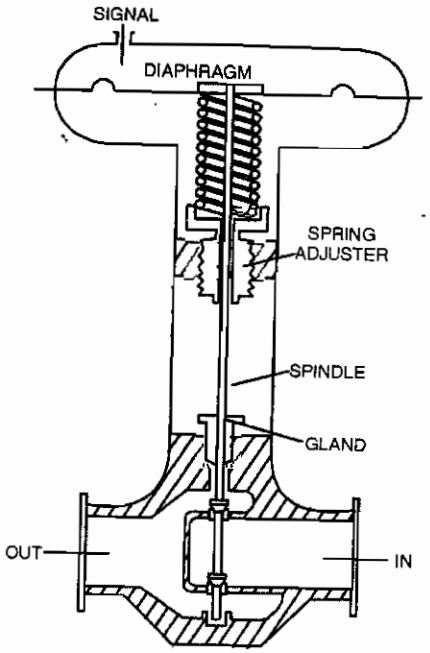
An air pressure signal from the automatic controller is applied to the topside of the diaphragm. An increasing air signal pressure pushes the diaphragm plate and spindle down against the force exerted by the spring to shut the valve.

With the valve shown, the flow passing through will decrease. The diaphragm/spring unit is called the motor element or actuator, while the valve is called the correcting element.

When the air pressure is reduced, the spring forces the valve stem upwards, thus opening the valve. This is called a air-to-close (ATC).

In case of failure of air, the valve will remain open. This could be used in a case like a cooling system. In an emergency, failure of air may lead to over-cooling, which is preferable to shutting the cooling, which can be dangerous. In case of failure of air, the full coolant flow is maintained, and this can be manually throttled by another hand operated valve, which can be in series with this automatic valve.

Fig. 63



Double Ported Diaphragm Operated Valve

The air pressure range on the diaphragm is normally 0.2 to 1.0 bar, although higher pressures can be used and the travel is usually about 90 mm maximum, otherwise non-linearities occur in spring and diaphragm behaviour.

The design can be arranged so that air is supplied on the underside of the diaphragm, using a special seal between the valve stem and diaphragm case. An increasing air pressure on the diaphragm moves the actuator stem upward, while loss of air pressure moves the stem downward. This is then known as reverse acting or air-to-open, since the valve opens as the air pressure increases.

In some designs instead of changing over the air supply to the underside of the diaphragm, the valve body can be disconnected from the yoke and the actuator stem, the bottom flange taken off and then the complete body inverted and reconnected. As air pressure increases on the top side of the diaphragm, the valve now opens.

This is still direct acting, but has the air-to-open (ATO) feature. This is of importance should the air supply to the valve or control system fail, the spring will then cause the valve to shut or open depending upon whether it is ATO or ATC respectively. This feature is also known as fail safe shut, or fail safe open, the spring providing the fail safe force to shut or open the valve. The choice will depend upon the requirements of the system. It is possible by the use of locking valves to lock the valve in a fixed position if air supplies fail. Air is then trapped in the diaphragm giving 'fail safe as is'. There is a requirement that valves must have a means of hand control or a hand operated by-pass. Usually a hand jack is fitted either operating on the diaphragm centre against the spring or on a direct side linkage onto the spindle. This is to provide a means of control should there be a complete failure of air supplies.

The diaphragm case would be of steel, zinc plated, or high tensile iron, the yoke of high tensile iron and the actuator stem of chrome or cadmium plated steel.

The gland seal or stuffing box is can be responsible for a heavy friction load on the actuator, giving poor positioning control and thus poor plant response. The valve may have a packing requiring lubrication, or self-lubricating packing. When graphited asbestos or a metallic packing is used, a lantern ring is fitted in way of the lubricator and it is essential that the correct number of packing rings are placed beneath the end above the lantern ring to bring these two components in line. To improve lubrication where temperature permits, Teflon

impregnated asbestos may be used. The upper temperature limit is 230°C, and with metallic packing the upper limit is 480° C.

Moulded Teflon 'V' ring packing or a similar PTFE material can also be used up to 230°C and does not require lubrication. Here the stem is highly polished, and this type of material has a soapy texture, which minimizes friction. A spring, which must be placed in the bottom of the box before the packing is inserted controls the gland pressure.

A double ported valve is used in which the flow passes across two valves (the lower slightly smaller diameter than the upper to allow fitting) in such a way that the out of balance forces are nearly balanced out.

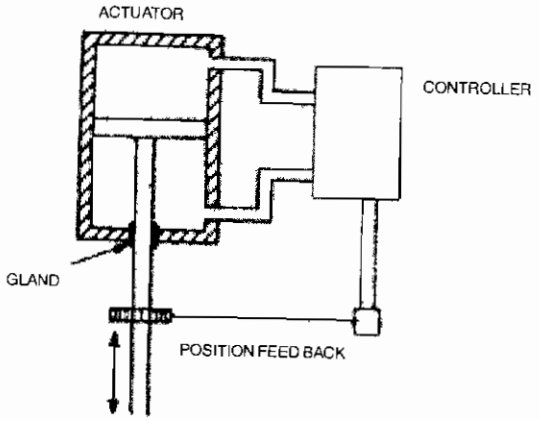
The size of the actuator can therefore be smaller but the more complex path brings increased flow resistance.

It is also more difficult to obtain tight shut off of even if the seats are lapped in, as a change in temperature will cause unequal expansion allowing leakage.

Valve Positioner

A valve postioner is used, when very precise movements of the actuator are required. The actual movement of the valve spindle is fed back to the controller, which uses this to accurately position the Actuator.

Fig. 64



VALVE POSITIONER

Valve characteristics

The relationship between the valve lift and flow through the valve is called the valve characteristic.

The three common flow-lift characteristics are as follows:

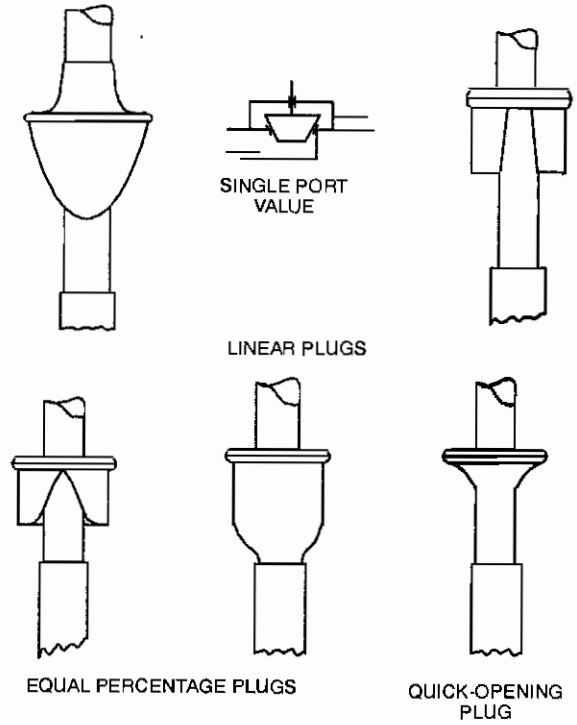
Linear

Here the valve produces a flow that gives a straight line graph when plotted against lift, i.e. at 40 per cent lift, flow is 40 per cent of the maximum flow.

These plugs have limited applications and would be used when the pressure drop across the valve remains stable and only the load varies, i.e. when a large proportion of the system pressure drop is across the valve.

The design may be modified to give a parabolic characteristic in which the valve gives a fine control at low flow rates. The design is produced as a contoured plug or in the form of straight sided V-port plug.

Fig. 65



Characteristics Valve

Equal Percentage

With this valve, equal increments of travel give equal percentage changes in the existing flow.

Thus when the valve plug is near the seat, and the flow is small, then the change in flow is small, while when the flow is large, the flow change is large. This is one of the most common types of valve in use as it gives good control over a wide range of pressure changes and is particularly useful when flow conditions are difficult to measure, as it has a useful flexibility for matching or sizing to systems. It would also be used when most of the pressure drop takes place in the system and when there are high friction losses. It is produced in either the contoured form or Vee port.

Quick Opening

This valve has a small lift, but it offers the minimum obstruction to flow, with the port area increasing rapidly as the plug leaves the seat.

They are used with on-off controllers and may have a slightly shaped plug or a disc valve.

The flow lift relationship tends to be linear and some, by means of shaping, are designed for this to prevent hammering action in the pipe lines.

The contoured shape plugs are used primarily for situations where corrosion and erosion present problems, as the surfaces can easily be surface hardened and allow a free flow for fluids with solid particles. The edges of the Vee port designs could be severely damaged under this condition.

All the above valves may be supplied as single or double ported design, with top and bottom guiding. Valves may suffer from dynamic instability causing the valve to jump vertically or vibrate horizontally, causing wear, noise and fatigue failure. It is basically due to changing flow patterns, over the surface of the plug, turbulence at the inlet due to the flow impinging on the plug, pressure drop and valve configuration.

Some valve bodies can be adapted for low flow rates retaining the standard size valve body but using smaller valves and seats. These are known as restricted trim valves and may be contoured, Vee ported or fluted. The capacity of the valve using this trim style is usually about 40 per cent that of the full body size. Such designs are used when it is not economical to reduce pipe line size to fit a smaller valve and with certain vapour liquid conditions.

Piston Actuators

These are used when high operating forces are required. They provide long strokes, with sufficient power to overcome high resistance due to flow with high system pressures.

Electric Valve Actuators

These are used when long distances make the use of hydraulics expensive, and when fast acting Actuators are required.

Rotary Actuators

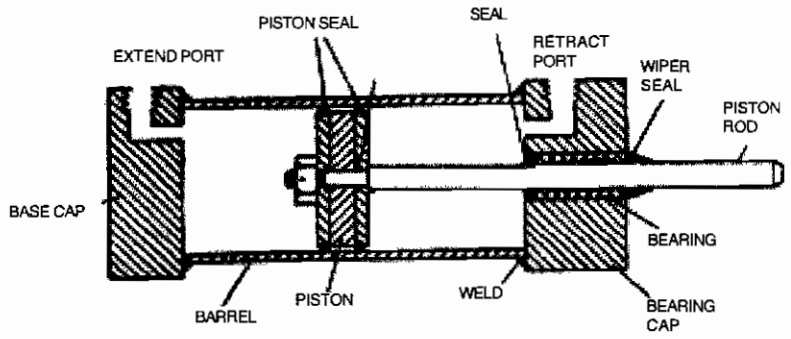
These are the hydraulic / pneumatic equivalent of electric actuators. They produce large torques, at very low speeds, and are also safe to use in a hazardous atmosphere such as on tankers. The torque is specified in different ways :

Starting torque is the torque available to move from rest

Stall torque is that required to stop, and

Running torque is that available while moving.

Fig. 66



HYDRAULIC ACTUATOR

Materials

The valve body may be of cast iron for temperatures in the range of - 29°C to 350°C, carbon steel for temperatures of 450° C, stainless steel when corrosion resistance is required and temperatures vary from - 100°C to 560°C, chrome molybdenum steel for temperatures of 620°C or nickel steel for low temperatures down to - 100°C.

Chromium molybdenum steel has good creep resistance. Extremely erosive conditions may require tungsten carbide, ceramic and urethane elastomers.

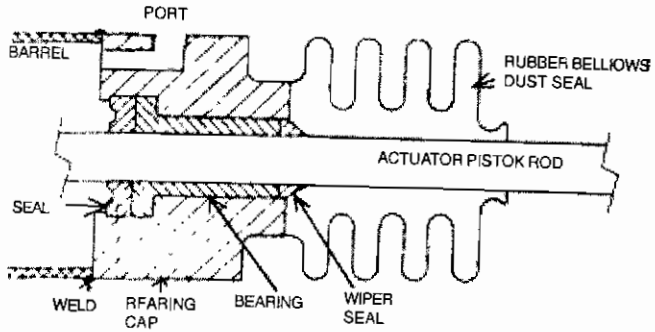
The choice of material will depend upon temperature, the lubricating properties of the fluid and the compatibility of the materials to prevent galling or sticking and tearing of mating surfaces.

For low temperatures such as - 200°C to high temperatures of upto +525°C Hastelloy can be used.

Cleanliness

The maximum problems relating to hydraulic systems are due to dirt, since small particles can abrade sliding surfaces, block orifices and nozzles, jam moving surfaces, and cut rubber rings. Filters should always be cleaned at the recommended intervals, and oil condition should be checked by testing samples. Dirty oil causes frequent filter blockage, while sludge can cause moving surfaces to seize. Oxidisation of oil or water contamination can frequently cause severe problems in control systems.

Fig. 67



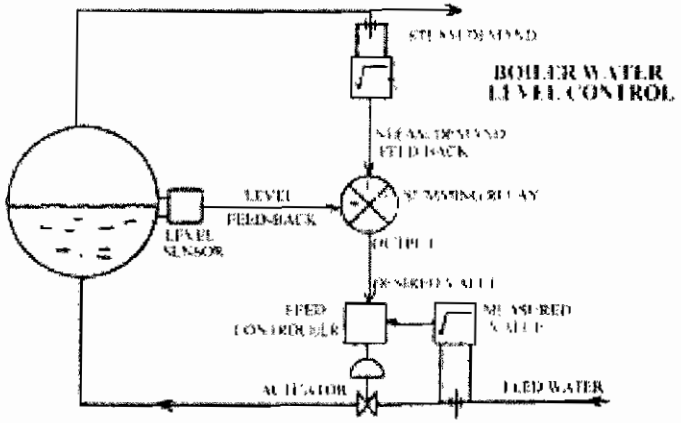
ACTUATOR SEAL

Boiler water level control

The level of water in the boiler drum needs to be accurately controlled. Failure to maintain the level could lead to great danger to the operator. Inadequate water level could lead to a situation called as 'dry boiling', which is extremely dangerous.

Too high a level could result in 'priming', which is undesirable. Level control alone could not be a reliable means, since there is a relationship between the demand of steam and the water level. Also, at the time of raising steam from cold, the level would change, as the water heated up.

Fig. 68



Thus, a summing relay is used, to provide a combined feed-back of level and steam demand, to serve as the desired value for the water level controller.

The flow meter on the feed water line provides the measured value, via a square root extracter, which protects the drum level from changes in feed water pressure.

The effects of change in steam demand are used to anticipate the changes in boiler drum water level, so that necessary corrective action is taken in advance, to prevent large fluctuations in water level.

CHAPTER 9

Starting and Reversing

Upon completion of this chapter, you will learn about :

- Methods of starting a marine engine
- Drawbacks of direct air admission
- Starting air period
- Rules for Starting air systems
- Safety devices
- Main Engine Interlocks
- Typical control system of a marine propulsion engine
- Governors
- Overspeed protection device
- Indicator Diagrams
- Questions

Starting and Reversing

For a slow speed, two stroke, reversible marine engine, starting is usually by direct air admission. For the engine to be reversed, consideration must be given to the adjustment of the cams of the fuel pump, air distributor and exhaust valves.

The various methods of starting an engine are :

Hand cranking :

This is limited to very small engines, e.g. lifeboat engine. A decompression lever is usually provided, to enable the engine to be easily cranked upto a certain speed, thereby gaining some momentum, which is stored in the flywheel.

Injection of fuel is mechanical, and the engine fires, which disconnects the cranking lever automatically.

Electric start :

This is also limited to fairly small engines. An electric motor is used to crank the engine, to achieve a certain momentum, during which fuel is mechanically injected.

Hydraulic start :

This is similar to electric start, except that a hydraulic motor is used for the cranking. An accumulator is used to store hydraulic fluid under pressure, and this is released, to the hydraulic motor, by a lever.

Air motor :

This is similar to the hydraulic cranking, except that air is used as the medium.

Direct admission of Compressed air :

The direct admission of high pressure air, into the cylinders of the engine for starting, is used for larger engines, where the large masses and inertia do not make external cranking practicable.

The direct method of starting, by compressed air admission into the cylinders, is reliable.

Each unit has a separate starting valve, and there is a period of overlap, so that even if any one unit starting valve should not operate, the engine can still be started.

Drawbacks of direct air admission
The drawbacks are that :

a large quantity of compressed air is injected into each unit, and if the fuel injector were to leak, there is a chance of premature explosion, leading to excessive pressure rise and possible engine damage. To take care of this, cylinder relief valve are fitted.

There need to be a minimum number of units, (4), to enable the engine to be started from any position.

Moisture in the starting air will enter the unit, which could create corrosion problems.

In case more than one unit has to be cut- off, there could be starting problems.

Starting air period

Starting air can be supplied from TDC till the opening of the exhaust valve / exhaust ports, in theory. In practice, the valve is timed to open a little before TDC, since there is always a period of 'lag', between the opening of the valve, and the actual action of the air, which must fill the cylinder first, before its turning effect is felt on the piston.

There is also a period of overlap, provided, where more than one cylinder starting air valve is open. This is to give a smoother turning action, and to make sure that the engine will start from any position, and in either direction easily.

Also the torque acting on the engine, is minimum around TDC, therefore if there were no overlap, the torque may not be sufficient to turn the engine, were it to be stopped with the unit at TDC. Since the torque is maximum around half way to BDC, if overlap were provided, and the unit in question were at TDC, the other unit, whose starting period overlaps, would be at a point which would give a higher starting torque, thus making starting easier.

Inverse cams are preferred. It becomes easier to take the rollers off the cam, once the engine has picked up speed and there is no possibility of stuck pilot valves or broken springs causing any cylinder air starting valve to remain in the open position.

Rules for starting air systems

Air Bottles :

Two or more Air bottles are to be provided, which are of approximately equal volume. With both the bottles completely pressed up, they should be able to cater to 12 starts of the Main engine, alternately started in Ahead and Astern directions (since reversing consumes more air), without having to replenish them with Starting air compressors. This is in case of reversible engines.

Where the Main engine is non- reversible, they should be capable of giving 6 or more starts of the Main engine, under the same conditions.

Air Compressors :

Two or more Main Air Compressors should have the required capacity equally divided between them. Total capacity of the Main Air Compressors should be such that they can fill up the Air Bottles starting from atmospheric pressure to the required capacity within one hour.

Safety Devices :

1. Non-return valve in the Starting Air line, closest to the Main engine.

2. Pressure relieving device in the Starting Air line, viz. Bursting Disc.

3. Relief valve in each Cylinder head of the Main Engine.

4. Flame trap or spark arrestor.

5. Means to prevent Starting Air admission to a unit which is already firing.

6. Fusible plug / Relief valve on Air Bottles.

7. Relief valves on each stage of Main Air Compressors.

8. Unloaders on Main Air Compressors.

9. Bursting discs on Inter/ After Coolers.

10.High air temperature alarm for Compressors.

11.Overload Trip for the electric motors driving the Compressors.

Main Engine Interlocks

Interlocks are provided so that the engine can be started or reversed only when certain conditions have been fulfilled. When there is a remote control of engines, it is essential to have interlocks.

This reduces the possibility of engine damage and any hazards to the operating personnel.

a) Turning gear Interlock. This device prevents the engine from being started if the Turning gear is engaged. The Turning gear is provided for turning the engine.

b) Running Direction Interlock. This prevents the fuel from being supplied if the running direction of the engine does not match the Telegraph.

c) Starting Air Distributor in end position. This prevents starting from taking place if the shifting of the Distributor has not been completed.

d) Main Lub. oil pressure, Piston cooling pressure, Jacket water pressure, and important parameters must be above the required minimum.

e) Auxiliary Blower Interlock. The Auxiliary Blower is provided in case of Constant pressure turbocharging. This ensures sufficient air during low speed running, which is essential for proper combustion.

f) Air Spring pressure Interlock. In case of the present generation of engines using Exhaust valves shut by Air Springs, the Air Spring pressure must always be maintained, else the exhaust valve may not close while running.

Typical control system of a modern marine two stroke engine

A modern marine engine is capable of running on heavy fuel oil from pier-to-pier.

The engine also has the capability to stop the vessel's momentum, by means of an air brake.

This allows the engine to be stopped, even while the ship is still moving, and engine to be started in reverse, to break the momentum of the vessel, which is sometimes referred to as Emergency Full Astern.

During this movement, the engine load can be very high, hence the rpm is restricted to a level, which limits the pressures in the cylinders from exceeding the safe limit.

Fuel lever - on 'Stop'

Stop valve is depressed , thus control air at 7 bar flows through Quick venting valve and then onto valve , which is linked to the Telegraph.

Also air depresses valve 4, which sends a pre-set air signal (1.6 to 2 bar) to the Governor, to release minimum fuel for firing.

Air depresses valve 21, which further allows control air to depress valve to operate the Fuel pump Puncture valves.

Air acts to depress valve to allow starting air to engage the pilot valves of the Distributor , as well as supply boost air to Governor during start.

Telegraph - Ahead

Air operates valve which releases control air to shift the fuel pump rollers to Ahead .

Simultaneously , air flows through valve to operate the Reversing cylinder for the Distributor, which axially shifts to Ahead position.

After shifting is complete, air passes through valves 13 and 15, upto valve 17.

At this point , Reversing is complete, and the Engine is ready for the Air start.

Fuel lever - put to 'Start' position

Start valve 3 is depressed , which allows control air to flow and depress valve 17, thereby releasing air to depress valve 20.

This allows control air (through the Turning gear Interlock) to flow through , and carry out the following functions:-

a] Control air depresses valves 10 &11, which being plugged, pneumatically locks the Distributor shaft from shifting .

b] Control air acts to depress valve 23 , which releases control air to open the Automatic air starting valve.

c] Control air acts to depress valve 25, which allows starting air [30bar] to enter the pilot valves of the Distributor.

Additionally, air passes through the already depressed valve to engage the pilot valves on the cams , and to supply boost air to the Governor.

Fuel lever- Put on 'fuel'

As soon as the fuel lever is moved beyond Air Start, stop valve is released, which vents the air from the line, thus shutting valve which stops air to the Puncture valves of the Fuel Pumps, allowing them to function.

This venting is assisted by the Quick release vent valve.

The venting of the line through the Telegraph valve is delayed by 6 seconds, to allow complete reversal of all units.

Additionally , valve 4 is released, which cuts-off the pre-set air signal to the Governor, and instead allows the actual speed setting air signal from valve 2 to pass.

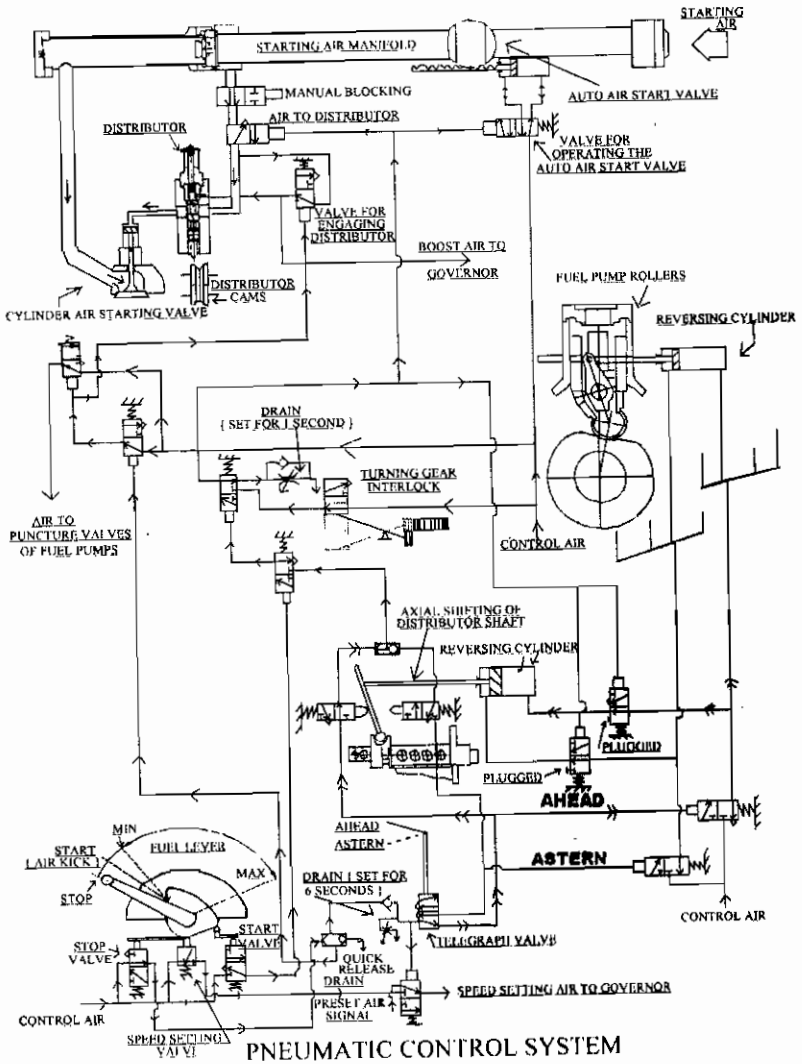
Valve 2 is controlled by the position of the Fuel lever.

Note:

* The Telegraph position and the Running direction of the engine, are sensed elecrically, and in case they do not correspond, fuel is cut off via the Puncture valves, and Wrong way Alarm is triggered off.

* At Finished with Engines, the control air line has to be vented.

Fig. 69



PNEUMATIC CONTROL SYSTEM

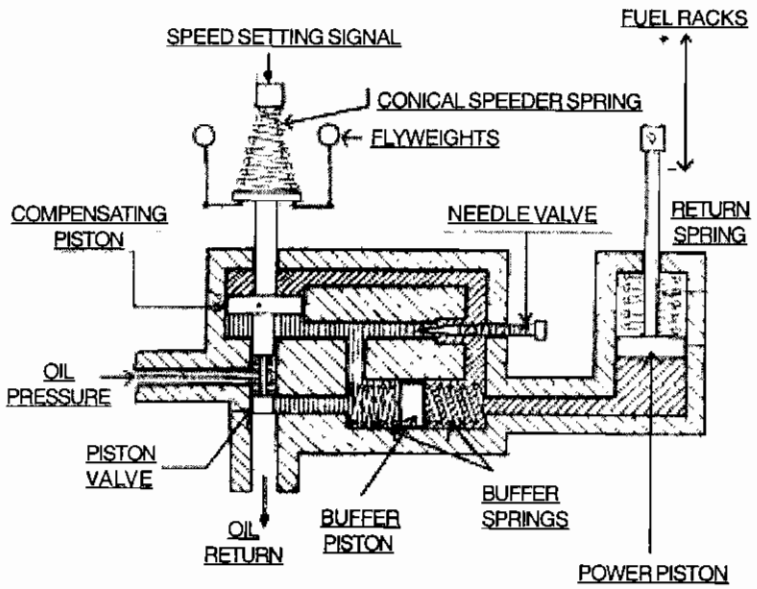
Governors

A Governor can be described as a device for automatically controlling the speed of an engine, by regulating the quantity of fuel.

The mode of control may be mechanical, hydraulic, electrical/ electronic or a combination of these.

The most common type is the mechanical-hydraulic type, as shown in the sketch.

Fig. 70



The sensing of the speed is by means of a pair of flyweights.

The change in speed causes the piston valve to move, allowing oil pressure to drain off / act on the buffer piston, depending on whether it is an upwards / downwards movement.

The movement of the Buffer piston causes a pressure change at the Power piston, which correspondingly moves the fuel rack in the decrease / increase direction.

There is a Compensating piston provided, which tends to reset the movement of the piston valve.

Although the engine may appear to operate satisfactorily because the speed is constant at no-load, changes in load may cause a slow return to normal speed, or too much hunting of the fuel lever, both of which are signs of incorrect compensation.

The compensation in this case may be adjusted by the needle valve shown.

Sensitivity :

The sensitivity of a Governor is a measure of its ability to control the speed of the engine within narrow limits. The higher the sensitivity, more is the hunting, or loss of stability.

Stability :

Stability is the opposite of Sensitivity, i.e. it is a measure of the reluctance to excessive speed variations. Compensation improves the stability, but at the cost of sensitivity.

Relay type :

In the relay type, the sensing done by the flyweights is used to operate a hydraulic power piston, which does the actual work of operating the fuel shaft.

Electronic Governor

A typical electronic governor has a three component governing system for 24 V dc operation. The system includes an electronic speed control, a magnetic pickup, and an all-electric actuator.

The actuator is designed to be installed on the prime mover and mechanically linked to the fuel control. The actuator increases the fuel rack setting, while a return spring moves in the decrease direction.

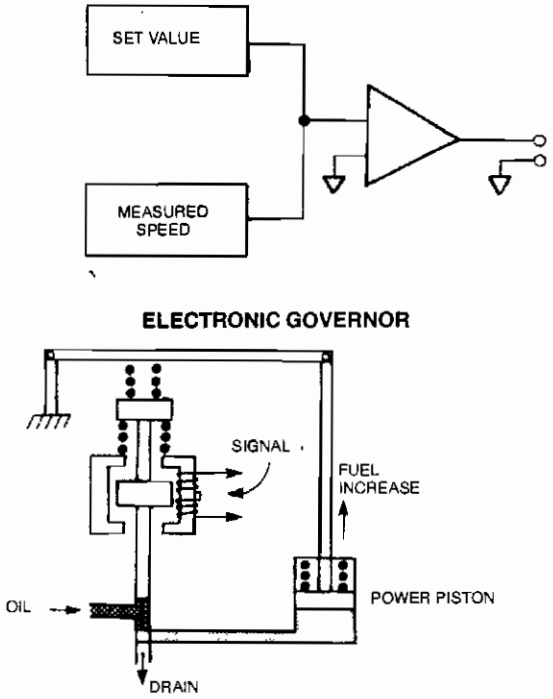
Pulse-width modulating the output to the actuator (instead of having a constant-voltage drive) keeps the driver from becoming too hot.

The speed control is housed in a rugged aluminum casting, which permits installation on the engine skid. In operation, the control unit compares the actual speed of the engine with the desired speed, then calculates an error signal and causes the actuator to correct the condition.

The actuator portion of the system may be installed in a vertical or horizontal position. The control box may be mounted on the skid, but is not designed for installation on the engine itself.

The system is designed to provide electronic control to diesel engines which require torque to operate the fuel rack. The system is particularly suited to engines or turbines which do not provide a mechanical drive for the governor or hydraulic oil supply for the governor.

Fig. 71



Overspeed Protection device

A typical electronic Overspeed Protection System monitors three individual magnetic pickups to provide overspeed protection for steam turbines and diesel engines.

The system replaces mechanical overspeed devices, especially in high-speed applications (above 10,000 rpm) when a mechanical overspeed device becomes unreliable.

Ease of calibration and testing, accuracy, and repeatability are the benefits of this system.

It uses digital speed-sensing technology to change the set point by programming the value from the touch panel on the front.

To insure its accuracy and functionality, each of the three speed-sensing units can be individually tested while the prime mover is running, using the integral frequency generator.

A key lock prevents programming and testing by unauthorized personnel.

Accuracy is achieved through the three digital units which sense and indicate speed to within 0.1%. No mechanical overspeed device can match this accuracy level.

The system's two-out-of-three voting scheme assures that the device will not shut down the prime mover because of transient conditions or conditions affecting only one of the three isolated units.

Each unit is fully isolated and only connected to the other two units via voting relay outputs.

To further decrease the chance of a false trip, each speed-sensing unit is operated from a separate power source.

The device does not have problems of mechanical wear, susceptibility to temperature or other atmospheric conditions, or susceptibility to conditions related to a turbine's rotating element.

It will hold its trip set point and will repeat consistently for long periods with no adjustments, since it does not depend on mechanical moving parts, with their unavoidable friction, wear, and other problems.

When a shutdown has occurred, the system stores and can display the highest overspeed rpm the prime mover reached.

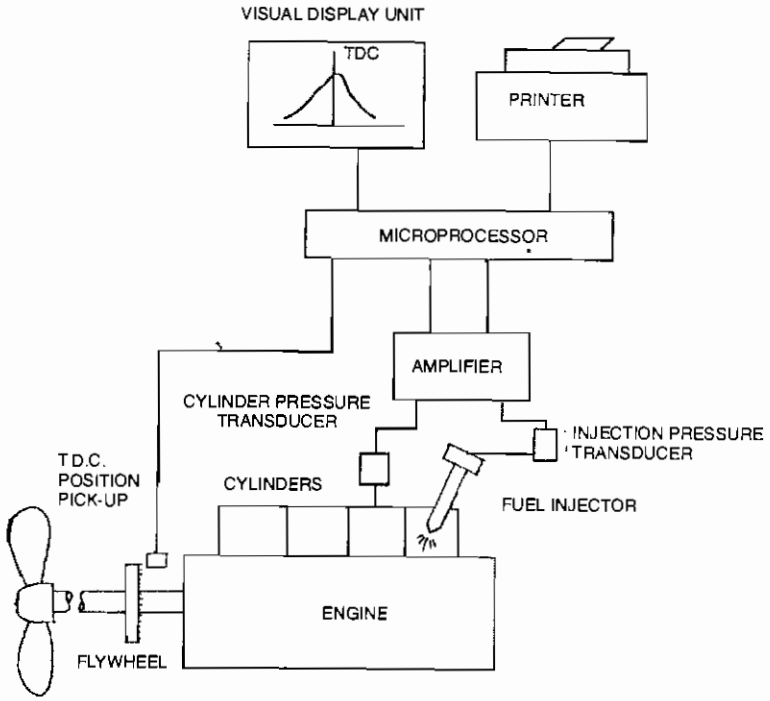
- Sample Time is 5 ms
- Total Response Time is 40 ms (Maximum)

Indicator Diagrams

An indicator diagram is a Pressure Vs. Volume graph, taken from the Indicator cock on the cylinder of a working engine. It is plotted either as a continuous line showing all pressure changes during one complete engine cycle which is called as a power card, or as a Pressure Vs. Crank angle or Draw card.

ELECTRONIC INDICATOR

Fig. 72



Indicator diagrams can be of great assistance in understanding the combustion characteristics of an engine. Examination candidates are required to be familiar with the information which may be obtained from the shape and dimensions of the diagram, as also the typical faults that can be identified from them.

In many modern engines sophisticated measuring devices may be fitted to record cylinder pressures, or there are portable indicators which can store data that can be down-loaded to PCs for further analysis.

There are several advantages to electronic devices, such as elimination of errors due to mechanical springs (used in conventional indicators) which limit the use to slow speed engines.

Modern indicators using pressure transducers can now be used on engines ranging from slow (50 RPM) to high (2000 RPM), without any significant errors.

Another advantage is the ability to superimpose curves taken at long intervals so as to study any changes (e.g. card taken during one loaded passage can be compared with another taken much later during a subsequent loaded passage), and without any possibility of human error, which can be there in mechanical indicators.

Calculation of indicated power, specific fuel consumption, peak pressure, compression pressure and so on is done by the microprocessor, and is thus easy to obtain and extremely accurate.

The T.D.C. postion of No.1 unit is sensed by some type of sensor, usually located close to the flywheel, and which passes on the data to the unit.

Typical faults can be easily traced, such as faults in the injectors, loss of compression pressure due to broken piston rings or ovality of liner, or choked scavenge inlet ports. The limitations of the mechanical indicator was its inability to compare the output, without which any trouble-shooting of faults is not possible. The electronic indicator stores the readings, and these can be easily analysed by the microprocessor, if the software already has the typical faults programmed in, and thus it can easily recognise them.

CHAPTER 10

Motor Controls

Upon completion of this chapter, you will learn :

- Need for a Starter for marine motors
- Components in Starters
- Motor Overload protection for AC Motors
- Speed Control of Motors
- Pulse Width Modulated (PWM) Motor speed control
- Variable Frequency Drive (VFD) for AC Motors

Motor Controls cover a broad range, from Direct On-Line (DOL) Starters for smaller motors to Star-Delta Starters for conventional marine motors. New developments include Pulse Width Modulated (PWM) Motor speed control for DC Motors, which utilise phase angle control, and Variable Frequency Drive (VFD) for AC Motors. Most marine AC motors are of the three phase, Induction type.

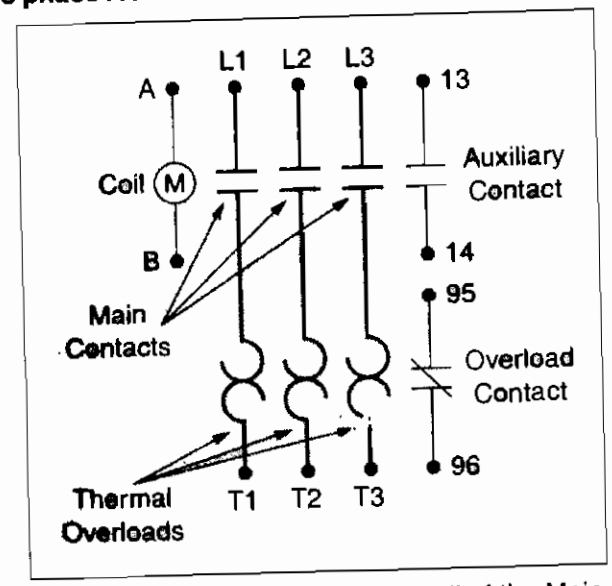
Motor Starter

In its simplest form, the Starter performs two basic functions :

- Allows the low voltage control circuit (AC or DC) to control the high current, high voltage, 3 phase Motor.
- Incorporates safeties and interlocks, such as prevention of immediate re-starting (in case of trip), Overload trip, Single phasing protection etc.

At a minimum, there must be a Contactor with three heavy-duty Normally Open (NO) contacts for power supply to the Motor's Stator, a light-duty Normally Open (NO) contact for auxiliary duty (such as Latching) and a light-duty Normally Closed (NC) contact for other duties (such as Overload / Stop). Thus, terminals are labelled with letters or numbers etched or moulded into the body, next to the relevant screw terminals. Motor controls cover a broad range from simple on-off motor starters to sophisticated phase-angle controlled dc motor controls and variable frequency ac motor drive systems.

3 phase AC Motor Starter



Simple Three-Phase Motor Starter with Overloads

The Coil of the Contactor actuates all of the Main Contacts and the auxiliary contact at the same time. The Overload Contact is independent of the Contactor coil and only operates under an overload condition. More complex starters may have four or more Main contacts (instead of three) to accommodate other motor wiring configurations, and extra auxiliary and overload contacts of either the N/O or N/C type. In most cases, extra auxiliary and overload contacts may be added later, as needed, by 'piggy-backing' them onto existing contacts.

As can be seen in the figure above, standardised system of numbering exists, wherein overload contacts are numbered 95 - 96 and auxiliary contacts 13 - 14.

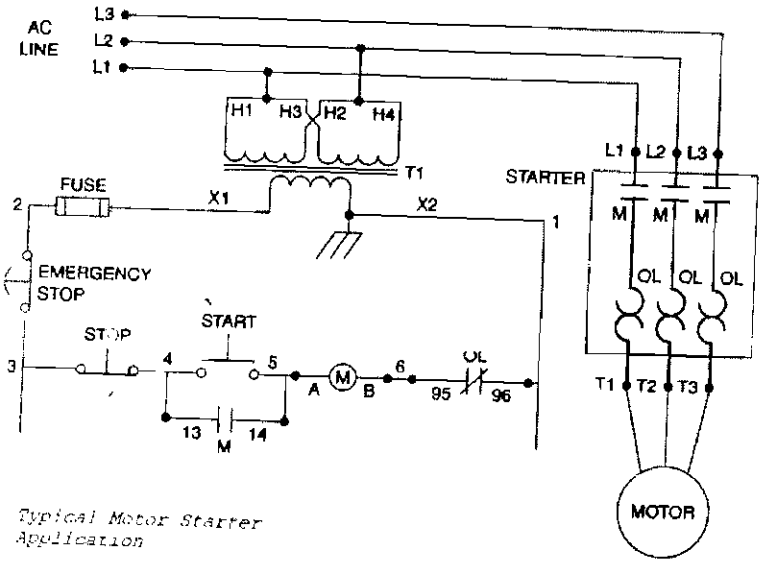
The following figure illustrates one method of connecting the Starter to the Motor's control circuitry. Power from the three-phase line (Mains) is applied to terminals L_1, L_2, and L_3 of the starter, whilst the Motor is connected to terminals T_1, T_2,and T_3.

Pressing the START switch supplies power to the Motor Starter's coil M. As long as the overload contacts OL are normally closed (no overload condition exists), the Starter will actuate and three-phase power will be provided to the Motor.

When the Starter actuates, auxiliary contact M closes, which latches the START switch, such that the START switch no longer needs to be pressed (in order to keep the Motor running). The Motor thus continues to run until either :

a) the Power (Mains supply) fails,
b) the EMERGENCY STOP button is pressed,
c) the STOP switch is pressed OR
d) the overload contact OL opens.

When any one of these four events occurs, the Motor Starter coil (M) de-energises, the three-phase supply to the Motor is interrupted and the Auxiliary Contact opens.



Typical Motor Starter Application

As can be seen in the figure above, overload contacts are numbered 95 - 96, auxiliary contacts 13 - 14 provide the 'latching' for the Start button, while Stop button is a Normally Closed contact.

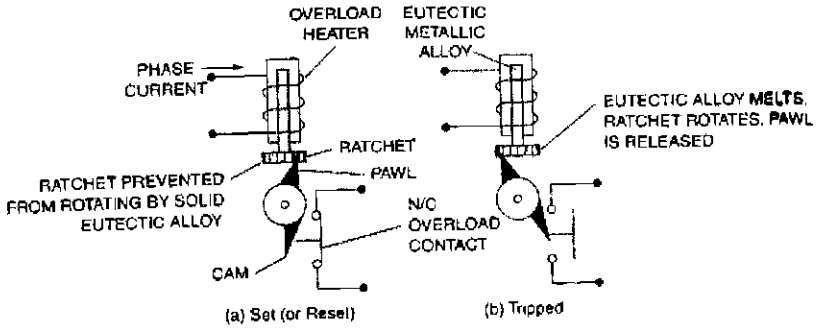
Overload Protection for AC Motors

Rated current is the current, in each phase of the supply line, when operating at the rated load and is always listed on the Motor's Name plate. Motors need to be protected from Overload, to prevent damage to the conductors of the Motor and the feed circuits, in the event a fault

condition occurs, such as a blocked rotor, rotor stall and internal electrical faults.

Conventional Fuses cannot be used as overload protection devices. This is because the Starting currents can (momentary) range from five to fifteen times the rated full-load current, when a motor is started. Therefore, a fuse (sized for rated current) would 'blow', even when the Motor is normally started.

Even worse, if only one of the fuses were to blow, the Motor would go into a 'single-phase' condition. In this case, the motor shaft may continue to rotate slowly (depending on the mechanical load), with very high currents in the remaining (live) phase lines, thus the motor will operate at a drastically reduced efficiency, causing it to overheat and eventually fail. Thus, we would need to simultaneously blow all the three fuses, which supply the three phases powering a motor, which is impractical.



Therefore, the Motor's overload protection must ignore short term excessive currents (that occur during motor starting) and simultaneously interrupt all three phases, if an overload condition occurs.

The solution to the potential single-phasing problem is to connect a current sensing device (called an overload) in series with each of the three phases and to mechanically link them so that when any one of the three overloads senses an over-current condition, it opens a contact called an overload contact). The over load contact is connected into the motor starter circuit so that when the overload contact opens, the entire Starter circuit is disabled, which in turn opens the three-phase motor Contactor, interrupting all three phases supplying the motor.

The most popular type of overload is the thermal overload. Although some thermal overloads use a bi-metallic temperature switch, the more popular type of thermal overload is the eutectic metallic alloy overload. This device consists of a eutectic alloy, which is heated by an electrical coil (called a heater). If the phase current (through the over-load heater) is excessive, the eutectic metal will eventually melt. This releases a ratchet, which 'cams' the normally closed (NC) overload contact, into the 'open circuit' condition. In order to make the trip reusable, the eutectic alloy is sealed in a tube, so that it will not leak out, when it melts. As the temperature falls, the alloy goes back into the solid state. A eutectic alloy is a combination of metals, that has a very low melting temperature and directly changes from the solid to liquid state (like solder). The nuisance tripping problem is overcome by designing the overloads to react slowly. It therefore does not trip, when a short-term overload occurs, such as during the normal start of the Motor.

One major advantage in using the thermal overload is, that as long as the over-load heater is properly 'sized' for the motor, it heats at the same rate as the Motor's windings. Therefore, the temperature of the overload is a good indicator of the temperature of the Motor windings. However, the effect of any ambient temperature change must be considered. If the motor and overload cannot be located in the same area, the overload size must be 'readjusted', using a temperature correction factor.

Another advantage in using this type of overload is that the starter can be resized to a different trip current, by simply changing the overload heater and this can be easily changed in a few minutes using a screwdriver.

Specifying a Motor Starter

Motor starters must be selected according to the :

1. Number of phases to be controlled (viz. Single phase, 3 phase),
2. Motor size (horsepower),
3. Coil voltage, and
4. Overload heater size.

Additionally, optional equipment, such as the number of auxiliary contacts, overload contacts and their types and whether the Starter is to be reversible, i.e. the Motor is to be unidirectional or to have a reverse running option. Most motor starter manufacturers specify the Contact's

rating by the Motor's Horsepower and Line voltage. For most squirrel cage induction motors, knowing these values will give the Voltage and Current, that the Contacts must switch.

By knowing the full-load motor current (from the motor's nameplate), the overload heater can be selected. Because the hearer is a 'resistive' type heater, the heat produced is a function of only the heater resistance and the phase current. Line voltage has no bearing on heater selection. Since the overload spindle heats slowly, normal starting current or short duration over-current conditions will not cause the overload to trip, so it is not necessary to oversize the heater.

Since the Contactor's actuating coil is operated from the control circuitry, the Coil's Voltage must be checked to be the same as Voltage of the control circuit. Also, AC or DC operation must be specified, which may or may not be the same voltage as the Contact rating. Note that Contactors with AC and DC coils are not interchangeable, even if their rated voltages are the same.

Before the advent of modern solid-state motor controllers, most applications, that required variable speed, required the use of DC motors. The reason is that the speed of a DC motor can be easily controlled, by simply varying either the Voltage applied to the Armature or the Current in the Field winding. In contrast, the speed of an ac motor is determined, for the most part, by the frequency of the applied AC voltage.

Since electricity is delivered at a fixed frequency, AC motors were relegated to constant speed applications, while dc motors had to be used whenever variable speed was required (e.g., in Cranes and Lifts). At most, another set of windings could be used to get more than one speed, but most times, flow from a pump could only be controlled by throttling the valves, which is not very desirable, as this leads to inefficient running or may even lead to cavitation.

This situation changed radically, when Variable Frequency solid-state Motor (VFM) Controllers were introduced, which now allows us to easily and inexpensively operate AC motors at variable speeds. Although solid-state controllers have allowed AC motors into applications traditionally done by DC motors, there are still many applications where a DC motor is the best choice, mostly in Emergency Battery powered applications. The universal ac motor, which is used in ac-operated power hand tools and appliances, is actually a spin-off of the series DC motor design

(i.e. it is not an Induction motor) and thus can be controlled in the same manner as a DC motor (i.e. by varying the applied voltage).

DC Motor Controller

The voltage applied to the Armature and the current through the Field, of a DC motor, can be reduced by simply adding a Resistance in series with the Armature or Field. This is an inefficient method, as the increased resistance generates tremendous heat losses (I2R). Also, under heavy mechanical loads, the high Armature currents cause an exponentially high I2R power loss, which reduces the efficiency.

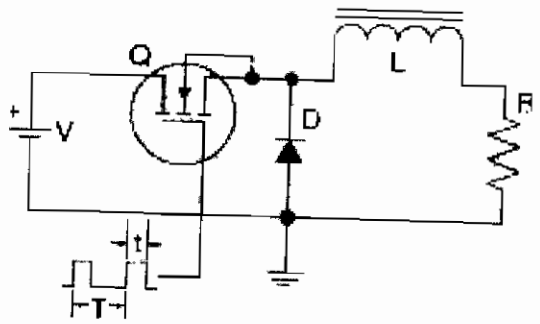
Solid-state electronics has made improvements in the control of DC motors in much the same way that it has with AC motors. There are two solid-state control techniques.

1. Controlling the speed of a DC motor, powered from a DC source.
2. Controlling the speed of a DC motor, using an AC power source.

In both cases, we will control the Motor's speed by varying the average Armature voltage. We will assume the Field is held constant, either by the use of permanent magnets or by a constant Field current.

DC Motor Control with DC Power Source

Consider the circuit below. We have a DC voltage source V, a resistor R, inductor L, diode D and a semi-conductor switch Q (an N-channel insulated gate MOSFET).



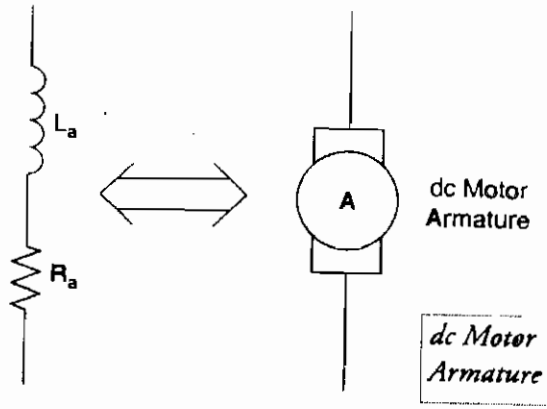
The signal applied, to the gate of the switch Q, is a pulse train, with a constant frequency (f) and constant period (T), but with Varying pulse width (t). The amplitude of the signal, applied to the gate, will cause the

Switch to transition, between Cut-off and Saturation, with very short 'rise and fall' times. The relative values of R and L are selected, such that the time constant (T = L/R) is at least 10 times the period, T of the pulse train applied to the gate of Q. The long L/R time constant has a low-pass filtering effect on the chopped output of the switch Q and will effectively 'smoothen' the current into DC (with a very little AC component).

For the switch Q, the ratio of the On-time (t) to the period (T) is defined as the Duty cycle and is represented as a percentage between 0 percent and 100 percent. If we apply a pulse train with a 0 percent duty cycle to the gate of the switch, the switch will remain Off all the time and the voltage on the resistor R will obviously be zero. In a similar manner, if we apply a pulse train with a duty cycle of 100 percent, the switch will remain On all the time, the diode D will be reverse biased, and, after five time constants, the voltage on the resistor R will be V. For any duty cycle between 0 percent and 100 percent, the average resistor voltage will be a corresponding percentage of the voltage V. For example, if we adjust the applied gate pulses, so that the duty cycle is 35 percent (i.e., ON for 35 % of the time, OFF for 65 % of the time), then the voltage on the resistor R will be 35 % of the input voltage V.

Although this seems to be a rather involved method of controlling the voltage on a load, there is one major advantage in using this method. Consider the power dissipation of the switch during the times when it is either OFF or ON.

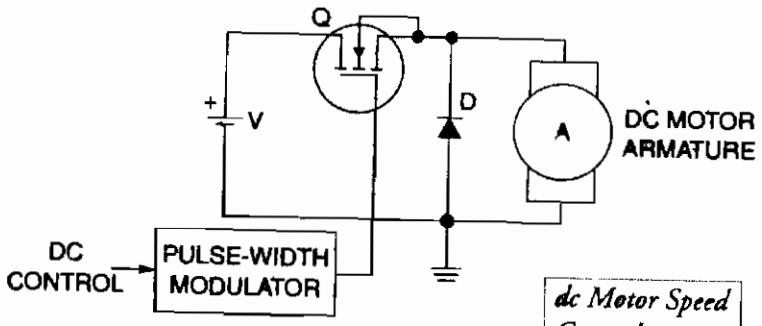
The power dissipated by any component in a DC circuit is simply the voltage drop on the component times the current through the component, or V x I. Assuming an ideal switch, when the switch is OFF, :the current will be zero, resulting in zero power dissipation. Similarly, when the, switch is ON, the voltage drop will be near zero, which also results in near-zero power dissipation, However, when the switch changes state, there is a very short period of time, when the voltage and current are simultaneously non-zero and the switch will dissipate power. For this reason, when constructing circuits of this type, the most important design considerations are the On-resistance of the switch, the Off- state leakage current of the switch, the 'rise and fall' times of the pulse train and th+er the electrical model of the armature of a DC Motor, shown in the Figure below :



dc Motor Armature Model

Since the armature windings arc made from copper wire, there will be distributed resistance in the coils R, and the coils themselves will give the armature distributed inductance L_a. Therefore, we generally model the armature as a lumped resistance and a lumped inductance connected in series. Some armature models include a brush and commutator voltage drop component (called brush drop), but for the purpose of this analysis, since the brush drop is relatively constant and small, adding this component will not affect the outcome.

Since we can model the dc motor armature as a series resistance and inductance, we can substitute the armature, in place of the resistor and inductor, in our dc switch circuit, as shown in the Figure below :



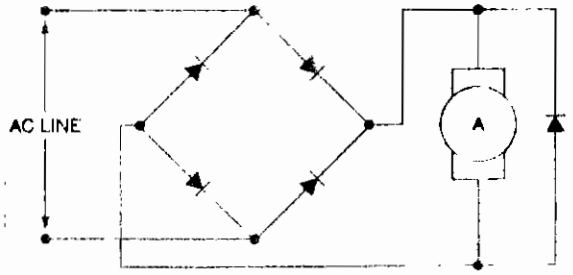
dc Motor Speed Control

A pulse-width modulator has been added. This is a subsystem, that converts a DC control input voltage to a constant-frequency, variable duty cycle pulse train. The complexity of this function block depends on the sophistication of the circuit and can be as simple as a 555 integrated circuit timer or as complex as a single board Microcontroller Since the heart of this entire control circuit is the pulse-width modulator, the entire .system is generally called a Pulse-Width Modulator motor speed controller.

DC Motor Control with an AC Power Source

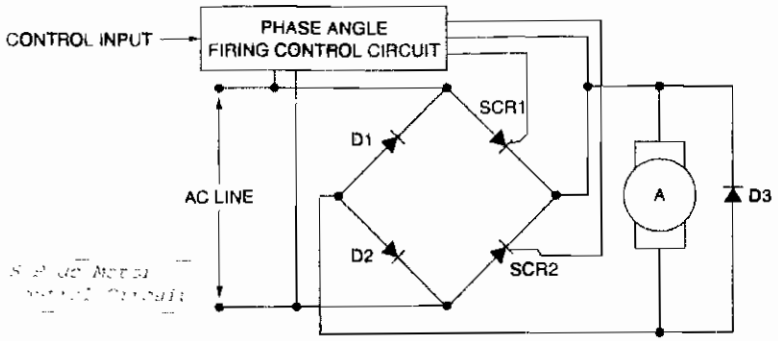
In order to operate a DC motor from an AC source, the AC power must first be converted to DC. It should be noted, that the DC supply, used to operate a DC motor, ***need not*** be a continuous, non-varying voltage. It is permissible to provide DC power in the form of a half-wave rectified or full-wave rectified power or even pulses of power (as in the Pulse-Width Modulation scheme discussed earlier).

As long as the polarity of the Voltage, applied to the Armature, does not reverse, the wave shape is not critical. In its simplest form, a full-wave rectifier circuit will power a DC Motor's Armature. However, in this circuit, the average DC voltage, applied to the Armature, will be dependent on the AC line voltage. If we desire to control the speed of the motor, we would need to have the ability to vary the AC line voltage.



Full-Wave Rectifier dc Motor Supply

To provide control, over the average dc voltage applied to the Motor, without the need to vary the line voltage, we will replace two of the rectifier diodes, in our bridge, with Silicon Controlled Rectifiers (SCRs) as shown :



Note that in the circuit (shown in the Figure above), it is not necessary for all four rectifier diodes to be SCRs. The reason is that the remaining two diodes, D_1 and D_2, are simply 'steering' diodes, that route the return current from the Armature back to the opposite side of the AC line. Therefore, we can completely control the current flow in the circuit, with the two SCRs shown. The Phase angle firing control circuit monitors the sine wave of the AC line and produces a precisely timed pulse to the gate of each of the SCRs; As we will see, this will ultimately control the magnitude of the average DC voltage applied to the Motor armature.

We will begin analyzing our circuit using the two extreme modes of operation, which are when the SCRs are fully OFF and fully ON.

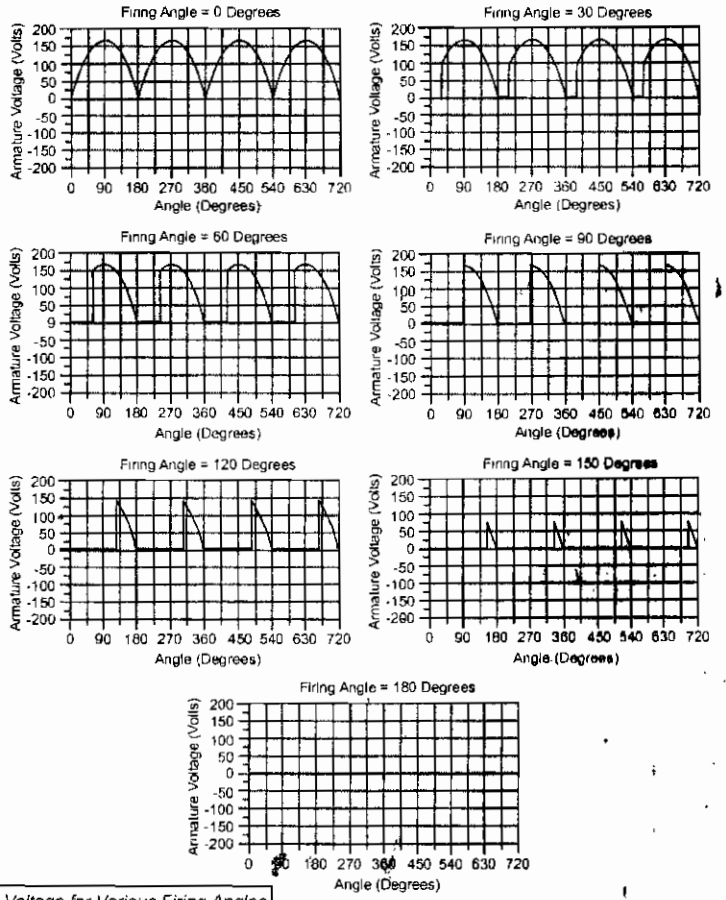
First, assume that the firing control circuit produces no signal to the gates of the SCRs. In this case, the SCRs will never fire and there will be zero current and zero voltage applied to the Motor armature.

In the other extreme, assume that the firing control circuit provides a short pulse to the gate of each SCR at the instant that the anode-to-cathode voltage (VA-K) on the SCR becomes positive (which occurs at 0 in the applied sine wave for SCR_1 and 180 for SCR_2). When this occurs, the SCRs will alternately 'fire' and remain 'fired' for their entire respective half cycle of the sine wave. In this case, the two SCRs will act, as if they are Rectifier diodes and the circuit will perform the same as with the motor operating at full speed.

Now, consider the condition in which the SCRs are fired at some delayed time, after their respective anode-to-cathode-voltages (VA-K) become positive. In this case, there will be a portion of the half-wave rectified sine wave missing, from the output waveform, which is the portion from

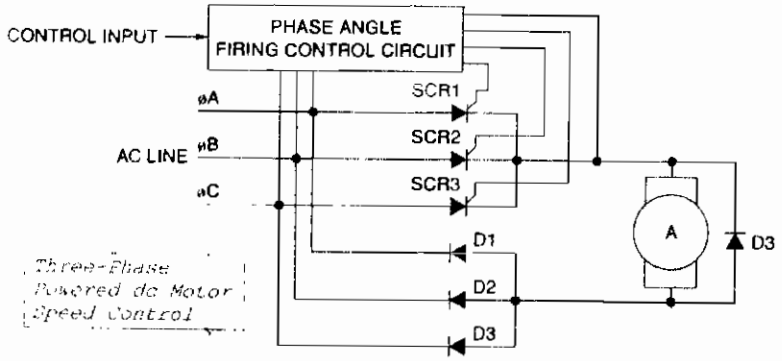
the time the waveform starts at zero, until the time the SCR is fired. When we fire the SCRs at some delayed time, we generally measure this 'time delay' as a trigonometric angle (called the firing angle) with respect to the positive-slope zero crossing of the sine wave of the line voltage.

The Figure below shows the armature voltage for various firing angles between 0 and 180. Notice that, as the SCR firing angle increases from 00 to 1800, the motor armature voltage decreases, from full voltage to zero.



Armature Voltage for Various Firing Angles

The circuit in the Figure is designed to operate from single-phase power. For most small horse power applications, single-phase power is sufficient to operate the motor. However, for large DC motors, three-phase power is more suitable because, for the same size motor, it will reduce the AC line current, which consequentially reduces the wire size and power losses in the feeder circuits. The circuit shown in the Figure is a simplified three-phase SCR control and rectifier for a DC motor, that uses the same SCR firing angle control technique to control the Motor's armature voltage.



There are other techniques for controlling the speed of a DC motor, but most are variations on these two techniques.

DC motor speed control systems are usually controlled by a DC voltage input, typically 0 - 10 volts DC, which can be easily provided by a PLC's analog Output.

Variable Speed (Variable Frequency) AC Motor Drive

If we wish to control the speed of an AC induction motor and produce rated torque throughout the speed range, we must vary the frequency of the applied voltage. This sounds relatively simple; however, there is one underlying problem that affects this approach. For inductors (such as induction motors), as the frequency of an applied voltage is decreased, the magnetic flux increases. Therefore, as the frequency is decreased, if the voltage is maintained constant, the core of the inductor (the Motor Stator) will magnetically saturate, the line current will increase

drastically and it will overheat and fail.

In order to maintain a constant flux, as we reduce the frequency, we must reduce the applied voltage, by the same proportion. This technique is commonly applied, when operating a 60 Hz induction motor on a 50 Hz line. The motor will operate safely and efficiently, if we reduce the 50 Hz line voltage to 50/60 (or 83.3 %) of the motor's rated nameplate voltage.

This principle applies to the operation of an induction motor at any frequency; that is, the ratio of the frequency to line voltage f / V must be maintained constant.

Therefore, if we wish to construct an electronic system, to produce a varying frequency, to control the speed of an Induction motor, it must also be capable of varying its output Voltage (proportional to the output frequency).

Note that : When operating the Motor at a reduced speed, although the Motor can deliver rated Torque, it cannot deliver rated Horsepower.

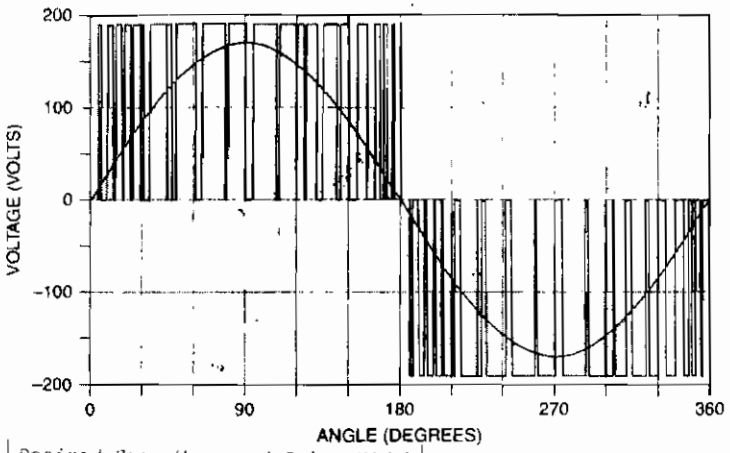
The reason for this is, that the Horsepower output, of any rotating machine, is proportional to the product of the speed and torque. Therefore, if we reduce the speed and operate the motor at rated torque, the horsepower output will be reduced by the speed reduction ratio. Operating an Induction motor at reduced speed and rated horsepower will cause excessive line current, overheating, and eventual failure of the motor.

Earlier, we had seen the technique of using a pulse-width modulation (PWM) technique to vary the DC voltage applied to the armature of a DC motor. Assume, for this discussion, we add a second pulse-width modulator but design it to produce negative pulses, instead of positive pulses. We could connect the outputs of the two circuits in parallel to our 'Load' (i.e. the Motor). By carefully controlling which of the two pulse-width modulators is operating, at any given time and the duty cycle of each, we can produce any time-varying voltage, with a peak voltage amplitude between +V and - V.

Such a device is capable of producing any wave shape, of any frequency and of any peak-to-peak voltage amplitude (within the maximum voltage capability of the PWM. For motor control applications, the desired wave shape will be a sine wave. Practically speaking, in order to do this, we

would most likely need a Microprocessor for controlling the system, that processes the desired frequency, to obtain the correct line voltage required and the corresponding PWM duty cycles, to produce a sine wave of the correct voltage amplitude and of the desired frequency.

The output wave form of such a device, called a Variable Frequency Drive (VFD), is shown in the Figure below. Superimposed on the pulse waveform is the desired sine wave. Notice that during the 0° – 180° portion of the Waveform, the positive voltage PWM is operating and during the 180° – 360° portion, the negative PWM is operating. Also notice, that during each half cycle, the duty cycle starts at 0 p%, increases to nearly 100 % and then decreases to 0 %. The duty cycle of each pulse is calculated and precisely timed by the controller (the Microprocessor), so that the average of the pulses approximates the desired sine wave.



Desired Sine Wave and Pulse-Width Modulated Waveform (One Phase)

The amplitude of the voltage output, of the VFD, is controlled by proportionally adjusting the width of all the pulses. For example, if we were to reduce all of the pulses, to 50 % of their normal width, the average output waveform would still be a sine wave, but would be 50 % of the amplitude of the original waveform.

Since the VFD will be connected to an inductive device, (an Induction motor), the pulse waveform shown, in the Figure above, is generally not viewable, using an oscilloscope. The inductance of the motor will smooth

the pulses, in much the same manner as the DC motor smoothes the PWM output, which, for the VFD will result in voltage and current waveforms that are nearly sinusoidal.

It is important to recognize that the previous discussion is highly theoretical and oversimplified, to make the concept more understandable. When a VFD is connected to an Induction motor, there are many other non-ideal characteristics that are caused by the inductance of the Motor and its magnetic characteristics, such as counter EMF, harmonics, energy storage and power factor, which make the design of VFDs much more complex than can be comprehensively covered in this text. However, present day VFD technology utilizes the versatility of the internal microprocessor to overcome most of these adverse characteristics, making the selection and application of a VFD relatively easy for the end-user.

In-addition to performing simple variable frequency speed-control of an Induction motor, most VFDs also provide a features that make the system more versatile and provide protection for the motor being controlled. These include over-current monitoring, automatic adjustable overload trip, speed ramping, ramp shaping, rotation direction control and dynamic braking. Some VFDs have the capability to operate from a single-phase line, while providing three-phase power to the Motor. Selection of a VFD usually requires simply knowing the line voltage, the Motor's rated input voltage and the horsepower. Most VFDs can be controlled from a PLC by providing an analog (usually 0 - 10 volts) DC signal from the PLC to the VFD, which controls the VFD between zero and the rated frequency.

Direction control is usually done, using a discrete output from the PLC to the VFD. Although VFDs are very reliable, there should be a Contactor, to control the main power to the VFD. Because an electronic failure in the VFD or a line voltage disturbance can cause the VFD to operate unpredictably, it is unwise to allow the VFD to maintain the machine at rest, while an operator accesses the moving parts of the machine.

Various methods of controlling ac and dc motors range from the simple motor starter to the sophisticated pulse-width modulated (PWM) DC motor controls and the variable frequency (VFD) AC motor controllers. New advances in solid state power electronics have made the speed control of both AC and DC motors simple, efficient, and relatively inexpensive.

CHAPTER 11

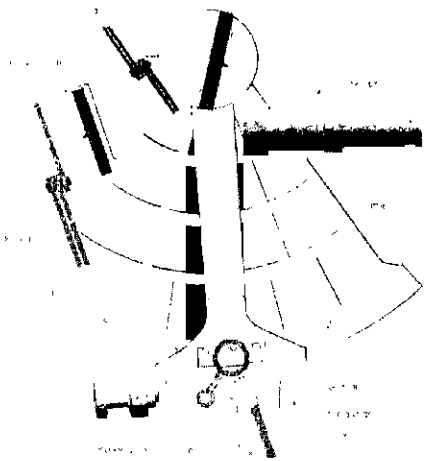
Electronic Navigational Aids

Upon completion of this chapter, you will learn :

- Need for having effective aids, to ensure safe navigation
- Advanced Radar Plotting Aids (ARPA)
- Global Positioning System (GPS)
- Differential Global Positioning System (DGPS)
- Electronic Charts Display and Information System (ECDIS)
- Global Maritime Distress and Safety System (GMDSS)

Navigation is the process of planning, recording and controlling the movement of a vessel, from one place to another. The word 'Navigate' is derived from Latin roots 'navis' (meaning ship) and 'agere' (meaning to move or to direct). Aids to Navigation are devices, which act as guides to assist mariners in determining their position, in relation to land and hidden dangers, each being used to provide specific information.

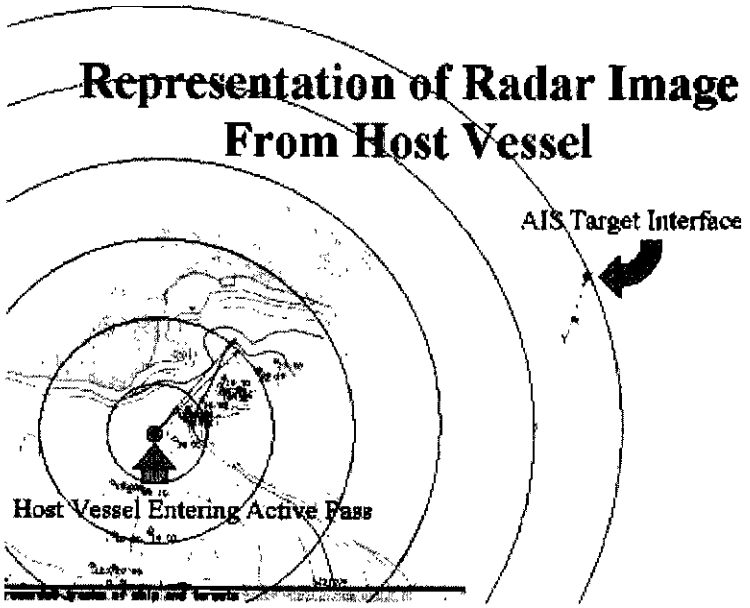
Different navigational techniques have evolved over the ages, but all involve locating one's position compared to known locations or patterns. The earliest form of navigation was by Dead Reckoning (DR) and Celestial navigation, using an instrument called the Sextant.



This was followed by Radio navigation. The term RADAR was coined, in 1941, as an acronym for Radio Detection and Ranging. This acronym (of American origin) replaced the previously used British abbreviation RDF (Radio Direction Finding).

Advanced Radar Plotting Aids (ARPA)

Development of ARPA started after the collision of the passenger liner **Andrea Doria** with the **Stockholm**. A Radar set, with Automatic Radar Plotting Aid (ARPA) capability, can create tracks, using radar 'contacts'. This system can calculate the tracked object's course, speed and closest point of approach (CPA), thereby finding (mathematically) if there is a danger of collision with the other ship or landmass. ARPA radars started to emerge in the 1960s.



Global Positioning System (GPS)

Global Positioning System (GPS) is a satellite-based navigation system, developed by the US Department of Defense. The first experimental satellite was launched in 1978. Subsequently, it launched a chain of 24 satellites, which circle the globe. There are now 30 actively broadcasting satellites in the GPS system.

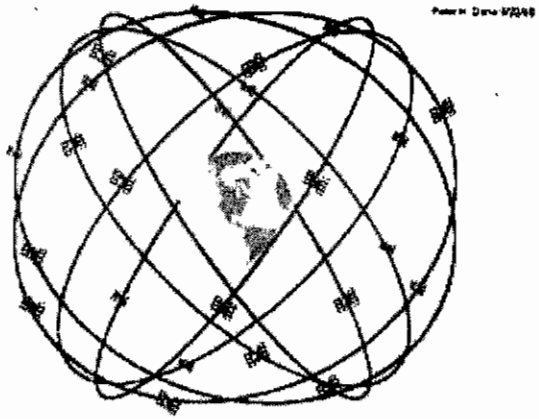
The ship carries a GPS Receiver and calculates its position by measuring the distance between itself and three or more GPS satellites, which are transmitting signals. The GPS receiver thus mathematically determines it's location, speed and direction.

Even though the GPS system can give an accuracy + 3 m under clear conditions, atmospheric conditions can cause a degradation of the accuracy, which may not be acceptable in certain applications. To correct this, the DGPS system was introduced.

Differential Global Positioning System (DGPS)

Differential Global Positioning System (DGPS) is an enhancement to the original Global Positionlng System, wherein a network of fixed (ground-based) Reference Stations are included, to calibrate the signals from each satellite and broadcast the difference (error), between the positions indicated by the satellite systems and the actual position.

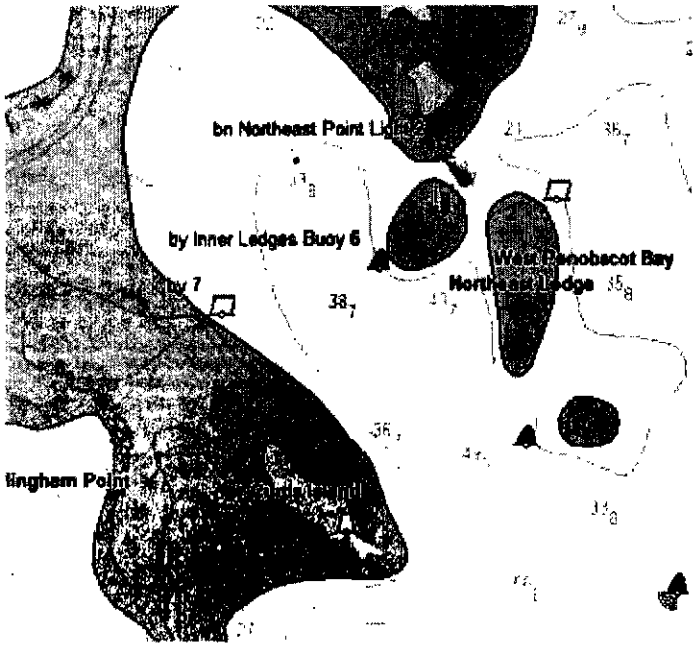
In effect, the Ground Station works backwards, i.e. knowing it's own geographical position (which is fixed and obviously cannot change !) it calculates how accurate the signal actually is. It then calculates the error-correction signal and sends this to all the (moving) GPS Receivers. Since it does not know which satellite each Receiver could be using, the signals from all Satellites are being continuously monitored and corrected by Ground Stations at different geographical locations.



GPS Nominal Constellation
24 Satellites in 6 Orbital Planes
4 Satellites in each Plane
20,200 km Altitudes, 55 Degree Inclination

Electronic Charts Display and Information System (ECDIS)

An Electronic Chart Display and Information System (ECDIS) is a computer-based navigation information system that can be used in lieu of 'paper' navigation charts. The system displays information from electronic navigational charts (ENCs) and integrates position information from the Global Positioning System (GPS) and other navigational sensors, such as Radar, Fathometer and Automatic Identification Systems (AIS). It may also display additional navigation-related information, such as Sailing Directions.



There are two type of electronic chart data in use : **Raster** and **Vector.**

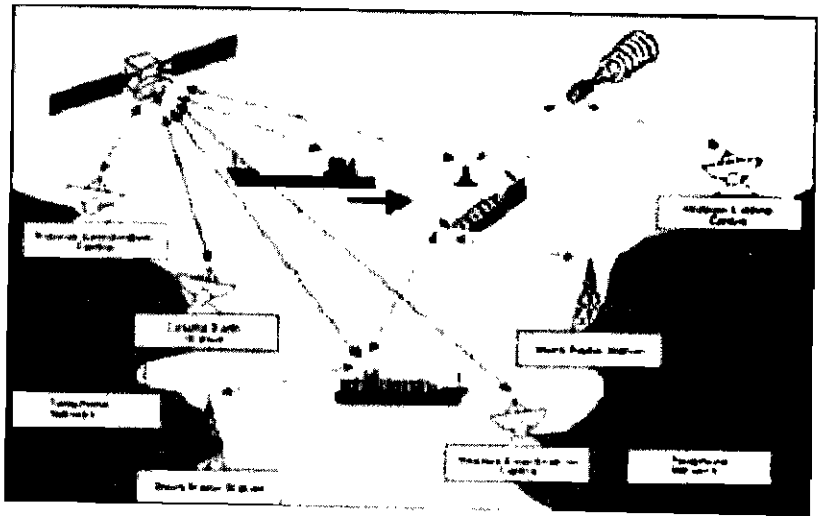
Although it is a computer-based system, a Raster chart is basically just a visual scan of a 'paper' chart, which uses charts issued by a hydrographic office, together with automatic continuous electronic positioning. Raster Nautical Charts, or RNC's, are charts that conform to IHO specifications.

A Vector chart is a more complex computerised model, wherein each point on the chart is digitally 'mapped', allowing the information to be used in a more sophisticated way. It also has additional features, such as 'clicking' on an item, to get additional details.

Global Maritime Distress and Safety System (GMDSS)

Global Maritime Distress and Safety System (GMDSS) is an automated ship to shore distress-alerting system, that relies on satellites, as well as terrestrial communication links. GMDSS is intended to perform various functions including :

- alerting (including position determination of the unit in distress),
- search and rescue coordination,
- locating (homing),
- maritime safety information broadcasts and communications.



Emergency Position-Indicating Radio Beacons (EPIRB)

Emergency Position-Indicating Radio Beacons (EPIRBs) are ship-based tracking transmitters, that operate as part of the Cospas-Sarsat satellite system. When the ship is in distress, the EPIRB floating in the vicinity of the vessel gets activated. The beacon sends out a distress signal, that allows it to be located by the satellite system as well as by Search And Rescue (SAR) aircraft / ships to locate the people, boats and aircraft needing rescue. They are a component of the Global Maritime Distress Safety System (GMDSS).

QUESTIONS ON AUTOMATION AND CONTROLS

1. What are the advantages of using automation for a marine system ? With reference to Unattended Machinery Spaces (UMS), explain some of the factors to be considered.
2. What are the advantages of using an electronic Indicator, as compared to a conventional Indicator using mechanical components ?
3. Sketch and describe any two sensors for measurement of :
   a) Pressure.
   b) Temperature.
   c) Level.
   d) Proximity.
4. What is Open Loop control and Closed Loop control, with reference to marine control system operation ? Describe where these are applied, with marine examples.
5. Sketch and describe a pneumatically operated control valve.
   a) Explain how the valve could be operated, if the control air supply fails and any fail-safe arrangements provided.
   b) Describe the routine maintenance required.
   c) Explain what might happen, if the valve were to be inserted the wrong way.
6. In case of Cascade control :
   a) Explain the operation.
   b) State why Cascade control is used in preference to other methods.
   c) Explain how the settings of the main loop differ from those of the sub loop, in the Cascade control system.
7. Controllers are provided with Three term control.
   a) Explain the circumstances which dictate the use of one or more of these terms.

   b) State how the optimum settings for the Controller may be achieved.
8. Sketch diagrammatically, a device for controlling the viscosity of the fuel oil to the injectors.
9. How is the vessel's speed measured ? Describe any device used on board to get the vessel's speed.
10. What is a Tachometer ? What parameter does it measure ? Describe, with the aid of a suitable sketch, any two types of Tachometers, explaining the fundamental principle of operation.
11. With reference to Pneumatic systems, what are the typical problems encountered in operation ? Why is cleanliness of such importance in the running of a pneumatic / hydraulic system ? Explain how you would remove impurities. What is a 'Flapper and Nozzle' set, with reference to Pneumatic controllers ?
12. What are the various types of Driers used in control systems ? Explain, with the aid of a neat sketch, the working of any one type.
13. What are the different types of control valves in marine use ? Explain, with the aid of suitable sketches, different ways of operating control valves. What are Actuators ? What is ATC and ATO, with reference to pneumatic valve Positioners ?
14. Briefly describe the materials used for pneumatic or hydraulic components. What factors does the choice of materials depend upon ?
15. With reference to marine engine control systems, what are the different means of starting ? Which is most suitable for the Main propulsion engine and why ? What is meant by the Starting air period ?
16. Explain some of the safety features provided in marine Air Starting systems. What is meant by an Interlock ? Where are these used in marine engine control systems ?
17. Describe, with the aid of a sketch, the manoeuvring system of a typical slow speed, two stroke, main propulsion engine, explaining the sequence of operation and the various safety features employed to prevent hazardous situations.

18. With reference to a Governor, what are the means provided to sense the speed of the engine being controlled ? What is meant by Compensation ? What is the function of a Power piston in hydraulic governors ? What are the advantages of using electronic governors ?

19. What are the means provided to prevent over-speeding of an engine during operation ? How is redundancy ensured ? Can this device be part of the speed governing mechanism ?

20. With reference to Electronic governors, how is the TDC position ascertained by the microprocessor ? What are the advantages / disadvantages of using electronic indicators, as compared to conventional 'piston-in-cylinder' mechanical types ?

21. With reference to control system terminology, differentiate between :

    a) ON condition and RUN condition.

    b) OFF condition and STOP condition.

    c) Latching and Inching.

22. What are Programmable Logic Controllers and where are they used on ships ? What are the various types of sensors / transducers used for the Inputs of either relay-logic circuits or the newer PLC based systems ? What is meant by analog input and digital input ? What are the advantages / disadvantages of analog as compared to digital signals ?

23. What are the different types of Timers and Counters used in Control systems ? What are Boolean logic functions ? Explain, with suitable examples, some typical Boolean functions used in automation.

24. With reference to Programmable Logic Controllers (PLC), discuss the advantages / disadvantages of these vis-à-vis conventional Relay logic circuits. What is meant by a Ladder Diagram and where is it used in PLC operation ?

25. What are the different Inputs and Outputs, that can be connected to the PLC ? Describe, with the aid of neat sketches, either a typical Input circuit or an Output circuit, showing how signals are processed.

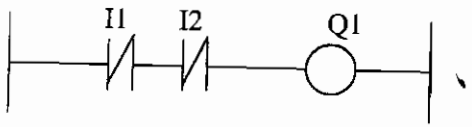
26. With reference to Programmable Logic Controllers (PLC), what are Normally Open (NO) and Normally Closed (NC) contacts ? Describe, with a neat sketch, a typical application, where a PLC could be substituted for a conventional relay circuit.

27. What is meant by an Opto-coupler, with reference to Programmable Logic Controller (PLC) operation ? Where are these used and why ? What is meant by Sinking and Sourcing, with reference to PLC I/O circuits ?

28. Can a PLC Input switch a Relay coil, to control a Motor ? Is AC or DC easier to interrupt ? What can happen if the Rated voltage on a device is exceeded ?

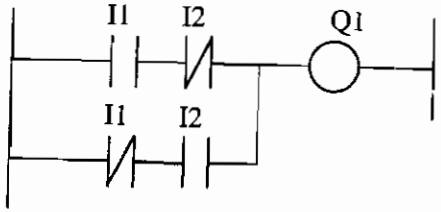
TASKS

Task 1:



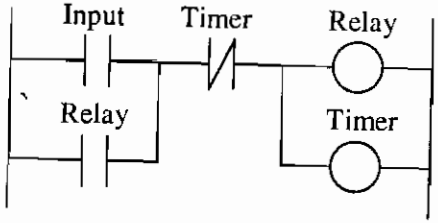
Is this NAND, NOR or neither ? _____

Task 2:



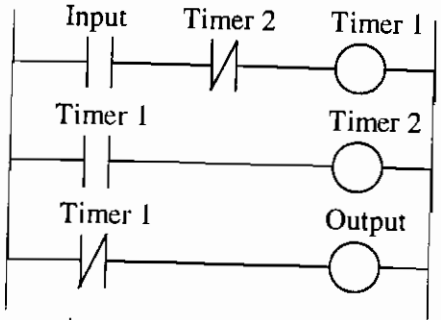
Is there a common name for the logic of this circuit ? _____

Task 3: Analyse this circuit. Explain the sequence of events, that would happen.



Task 4:

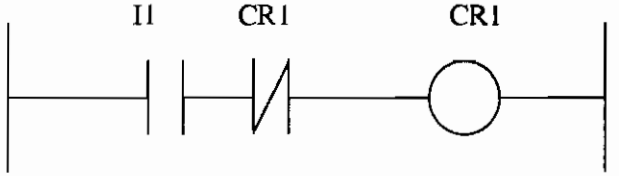
Describe the behaviour of this ladder logic.



What would be a good name for this circuit? _____

Task 5:

Can you predict how the system (below) will behave ?



What would be a good name for this circuit ? _____

Task 6:

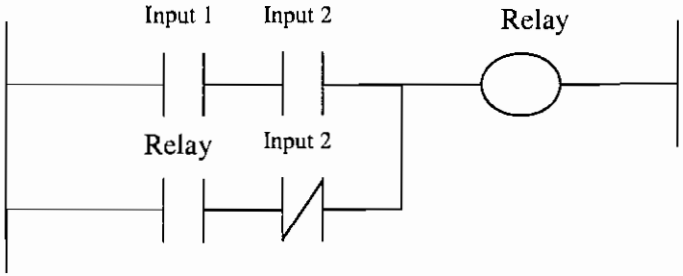


Figure out what this circuit does, explaining the logical sequence of operations.

Task 7:

Devise a circuit for a Conveyor belt. It is required, that the 3 phase AC Induction Motor must start with a delay of 10 seconds (after the switch is ON).

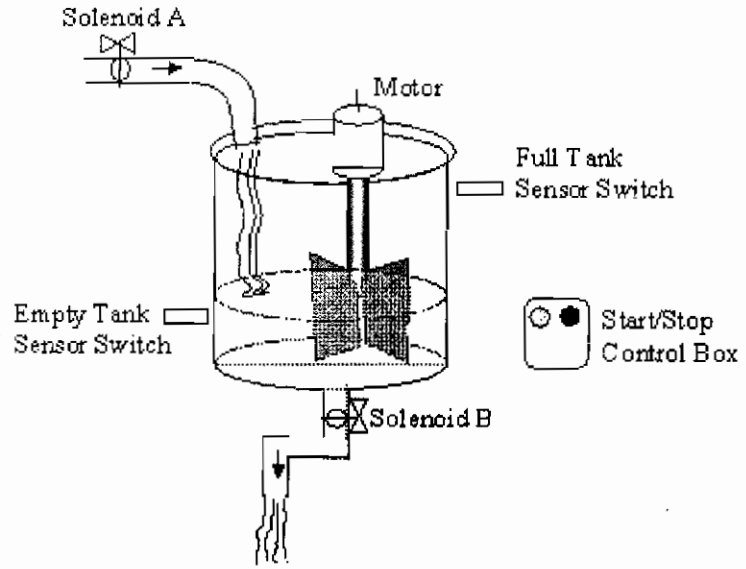
When the Motor is switched Off, there should be a delay of 10 s, before the system is switched off.

Count the number of packages, that pass along the conveyor belt. After a certain number of packages have passed, the system must shut-down, with the above mentioned 10 s delay period.

What are the inputs and outputs in the system ?

What components would be required to carry-out the timing and counting operations ?

What safety features / Interlocks would be required for such a system ?



With reference to the system shown above, create a ladder logic diagram to implement the PLC program to control the following sequence of operations :

a) Normally Open (NO) **Start** and Normally Closed (NC) **Stop** pushbuttons are to be used to Start and Stop the Process.

b) When the Start button is pressed, solenoid A energises, to start filling a Tank.

c) As the Tank fills, the empty level Normally Open sensor switch (NO) 'closes'.

d) When the tank is Full, the full level Normally Closed sensor switch (NC) 'opens'.

e) This causes the Solenoid A to de-energise and stop the filling. (Tank is full).

f) Now (after the Tank is Full), the Agitator Motor should start automatically and run for 3 minutes, to properly mix the liquid.

g) When the mixing is done, solenoid B is now **energised, to** empty the tank.

h) When the tank is nearly empty, the empty level **Normally** Open sensor switch (NO) 'opens', to de-energise **solenoid** B, which stops the emptying.

i) The Start button must now be pressed, to repeat the **above** sequence.

j) The Stop button may, however, be pressed at **anytime, to** halt the process.

requirements, relating to the time taken to fill the Main air Bottles. The system should be arranged to give three consecutive start attempts, failure of which should transfer controls to the Engine control room, where the fault can be rectified, and the Main engine started.

6] Remote control of the Main engine is to be possible from only one location at any time, and there should be an indication, e.g. a light, showing which location (Wheel-house, Engine control room or Emergency station) is in control, at any time. There should be a change-over system, to transfer control of the Main engine, from one location to another. It should be ensured that both the stations involved in transferring of controls (e.g. from Bridge to Engine control room), should show the same speed (fuel lever setting) and direction (Ahead or Astern), before the transfer of controls is possible. This ensures that there can be no conflicting instructions given, during any change -over of control, and prevent accidents. There must be an approved means of communication, e.g. telephone, at all three locations, such that it must be possible for all three stations to be in contact at all times. The local control station / Emergency station, at the Main engine, should be equipped with all the necessary Gauges and controls, to ensure safe running of the engine, in case of failure of the remote control system. If possible, it should be arranged that Speed and Direction (Ahead/ Astern) of the Main engine be kept constant, in case of remote control failure, till Local controls are taken over.

7] A comprehensive Alarm system is to be provided, for indication of machinery faults, which should be distinct from the Fire alarm system. This system should be capable of displaying (by means of a panel of lights) any abnormality of the machinery, both on the Bridge, as well as in the Accommodation spaces, including the Duty Engineer's cabin and the Chief Engineer's cabin, all public rooms. The power supply, for this alarm system, should have a back- up, from the Emergency switchboard, in case of a Black- out, and there should be an alarm, to indicate this condition.

The alarm system should be capable of indicating more than one alarm, at the same time.

For convenience, the alarms may be grouped, as e.g. Main engine, Aux. Machinery, Bilge levels , and so on. Non- acceptance of the Alarm, by the Duty Engineer, should further relay the alarm to the Chief Engineer's cabin.

8] Adequately designed safety systems are to be provided, for the automatic shut- down of Main engine, Aux. Engines or Boiler, in case of any serious malfunction. Aux. Engines used for power generation, must be capable of automatic start and coming on Load, in case of failure of the running machine. This can be done by means of a Stand- by feature.

In case of electrical power Generators, means should be provided for shedding excess load, e.g. a Preferential Trip, which will shed the load that is considered non- essential for the immediate running of the Main Engine.(e.g. Galley power, deck machinery power, accommodation Blowers and A/C).

All pumps, essential for the running of the Main engine, must be provided in duplicate, and a auto stand- by facility provided, so that the stand- by pump must be capable of automatically starting, on detection of failure of the running pump, e.g. by means of a pressure switch. Each pump must be capable of performing the duty independently.

9] Steering gear for U.M.S. operation must be of the type, such that it can automatically isolate and regain Steering control, in case of single failure, by automatic operation of the Isolating valves, e.g. safe-matic steering system. There should be two separate Steering gear pumps, which can independently steer the vessel, and one of them must have their power supply through the Emergency switch-board, such that it will continue to work, inspite of a Black- out, by drawing its power from the emergency supply.

10] There must be an approved system for ensuring safety of the personnel entering the Machinery spaces, during U.M.S. operation, for answering Alarms or carrying out essential repairs. E.g. Dead man alarm system.

This operates by means of a timer, which is activated, when the Duty engineer enters the Machinery space. It must be manually reset, after a pre-determined time interval, by means of reset switches located in the engine room.

In case of incapacitation of the Duty engineer, failure to reset the timer will sound the alarm on the Bridge, and someone can be deputed to investigate and take requisite action.

Remote control of machinery

Microprocessor controllers are the principle devices used to control various processes. Each controller is able to receive inputs from the engine room, perform processing and alarm checking, perform control algorithms, and output to motor controllers, valves and other types of actuators.

Besides direct monitoring and control of engine room systems, the microprocessor can perform trend analysis to give an early warning of possible break-downs.

One such system is the SIPWA system used to detect piston ring wear. Piston ring wear is one of the main reasons for having a unit overhaul every 8000 hours. Since the actual wear can be monitored, the Time Between Overhaul (TBO) can be safely extended, and this saves costs- spare parts, man hours as well as reduced down-time, allowing more time for other maintenance. This also helps to plan the intervals between unit overhauls, so that work can be easily carried out, at a convenient place, e.g. when the ship is alongside for a longer time.

The Engine Control Room usually contains various panels for monitoring and running of the machinery - Main engine, various pumps, compressors, boilers, alternators and other equipment.

Automated control rooms usually have a CRT display, which gives various parameters, mimic diagrams, bar- charts, performance graphs and other data. Engine combustion can easily be monitored by means of electronic indicator and draw cards. The advantage of using an electronic indicator is -

1. There is a consistancy of results, since many readings can be stored and the average taken.

2. The readings can be superimposed on each other, so that there can be a comparison of performance, eg readings taken on successive Loaded voyages can be compared on the screen / analysed by the microprocessor.

3. Since the mechanical indicator depended on a spring, it was only accurate for slow speed engines. The electronic indicator uses sensors like peizo-electric transducers, which can work with any speed engine - from 50 rpm to 2000 rpm, so that the same instrument can be used for the Main engine as well as the auxiliary engines.

4. The electronic indicator can be used on any engine having a standard indicator cock and is also portable.

An on-line printer can print out the log 'on-demand' or at regular intervals. An Alarm printer will provide the time, type and numerical values of various alarm conditions.

The control systems for Main engine, Boilers and Alternators contain numerous interlocks, to safeguard against possible emergency conditions, including remote shut-downs / trips and indications.

There is an alarm panel provided, which is duplicated in various remote locations, viz. Navigation Bridge, Chief Engineer's and Duty Engineer's cabins, and public spaces in the Accomodation.

With the ship in UMS mode, the Navigating watchkeeper can control the speed and direction from the Bridge by means of the remote control system, and the Main engine control system is able to automatically bypass the Critical Speed range.

Fig. 1

